Canonical GHZ configurations - cluster P1

```
species: electron
polarization cluster: P1
source cluster configurations: 111
canonical configurations collected: 72

canonical display criterion: exactly 2 retained final-state components
retained-component cutoff: fraction >= 2.0%
normalized amplitude display: bar length = |A_tilde|, bar color/text = phase

cluster retained-component count distribution:
  2 components: 72 configurations
  3 components: 1 configurations
  4 components: 9 configurations
  5 components: 7 configurations
  6 components: 18 configurations
  7 components: 4 configurations

CSV index:
  Output\GradientPhaseSpaceScan\electron\Data\GHZ\dghz\polarization_cluster_01\combined\
  canonical_dghz_polarization_cluster_01_configurations_electron.csv

Each following page is one complete momentum, kinematic-summary, and normalized-amplitude configuration.
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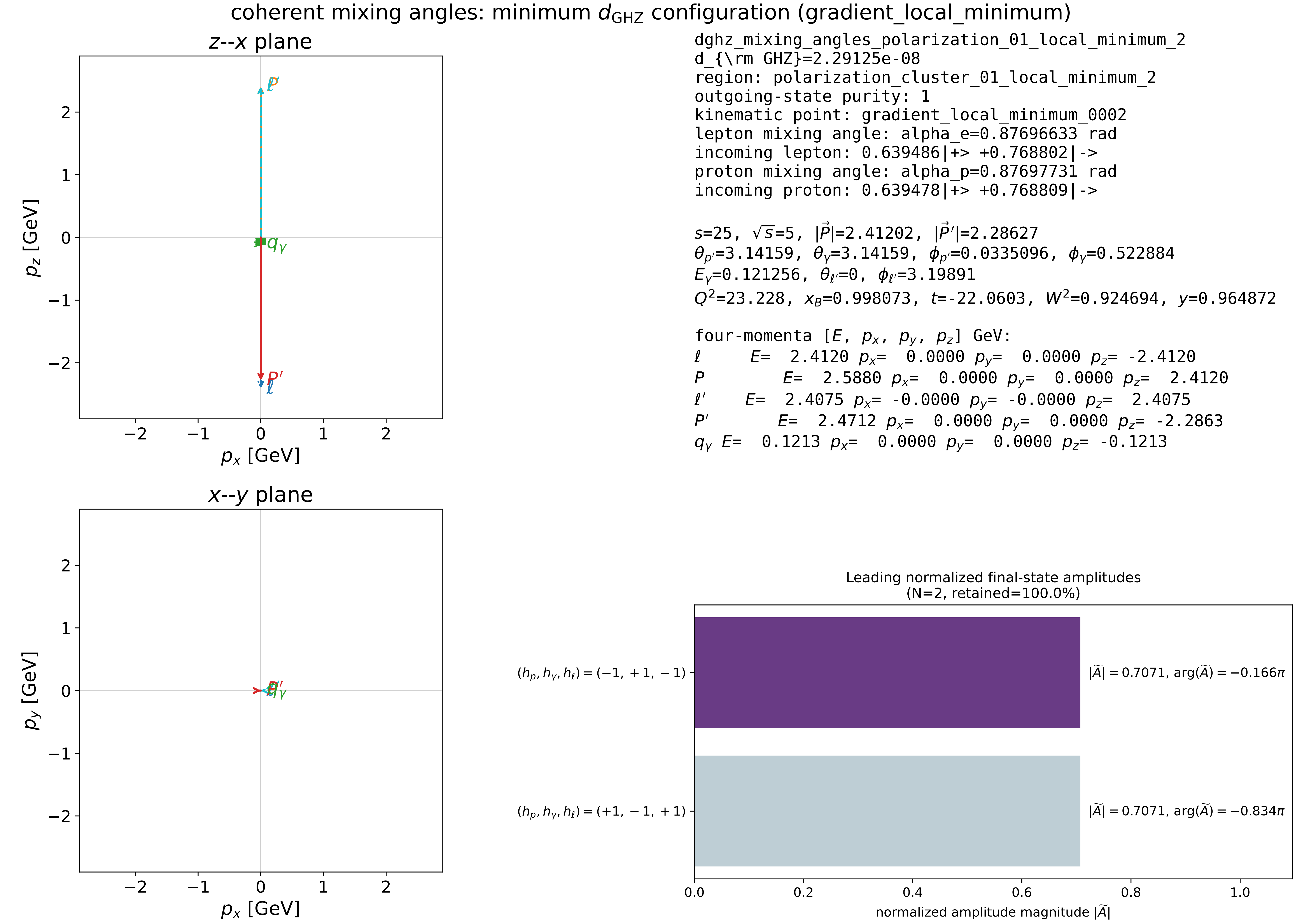
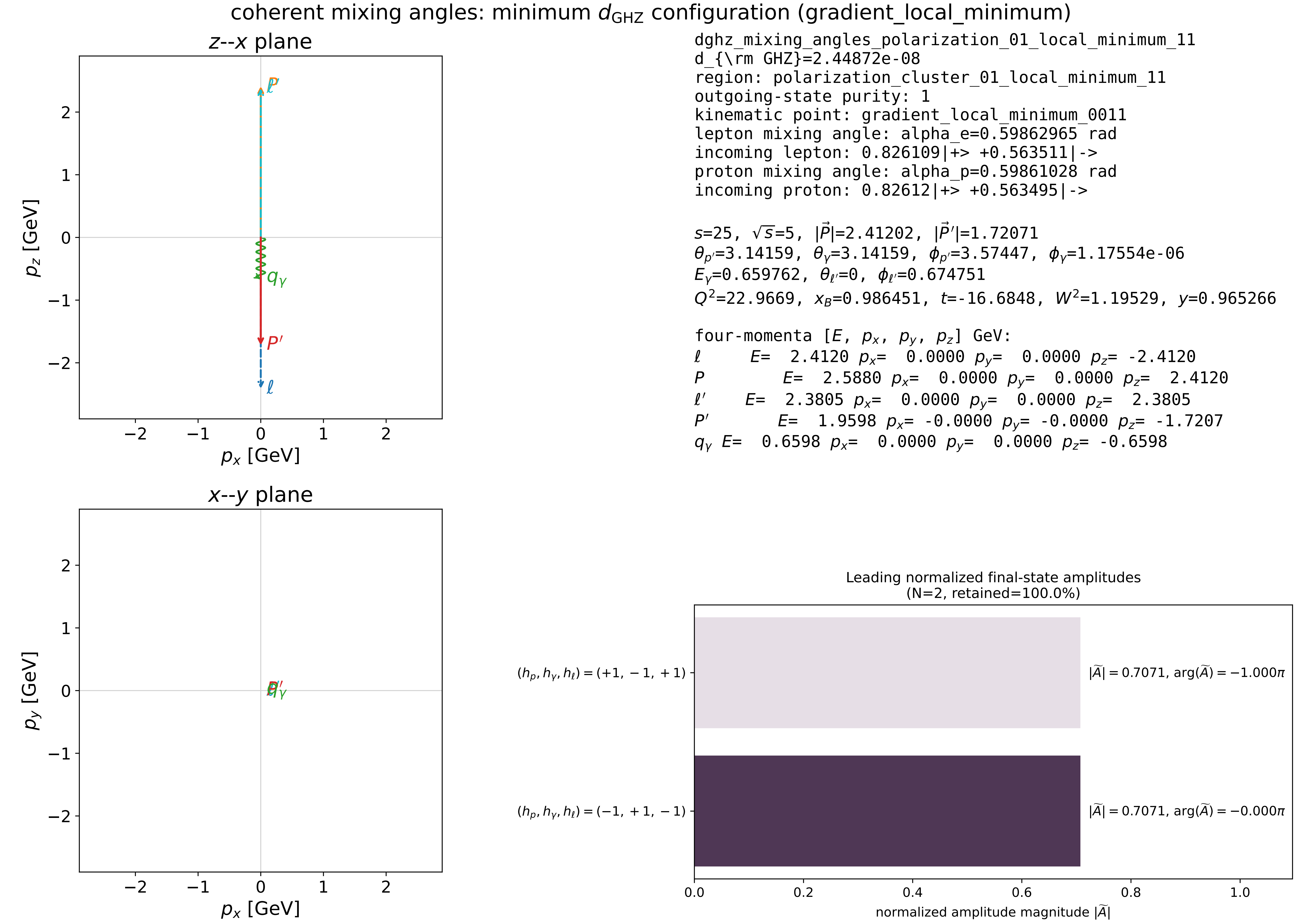
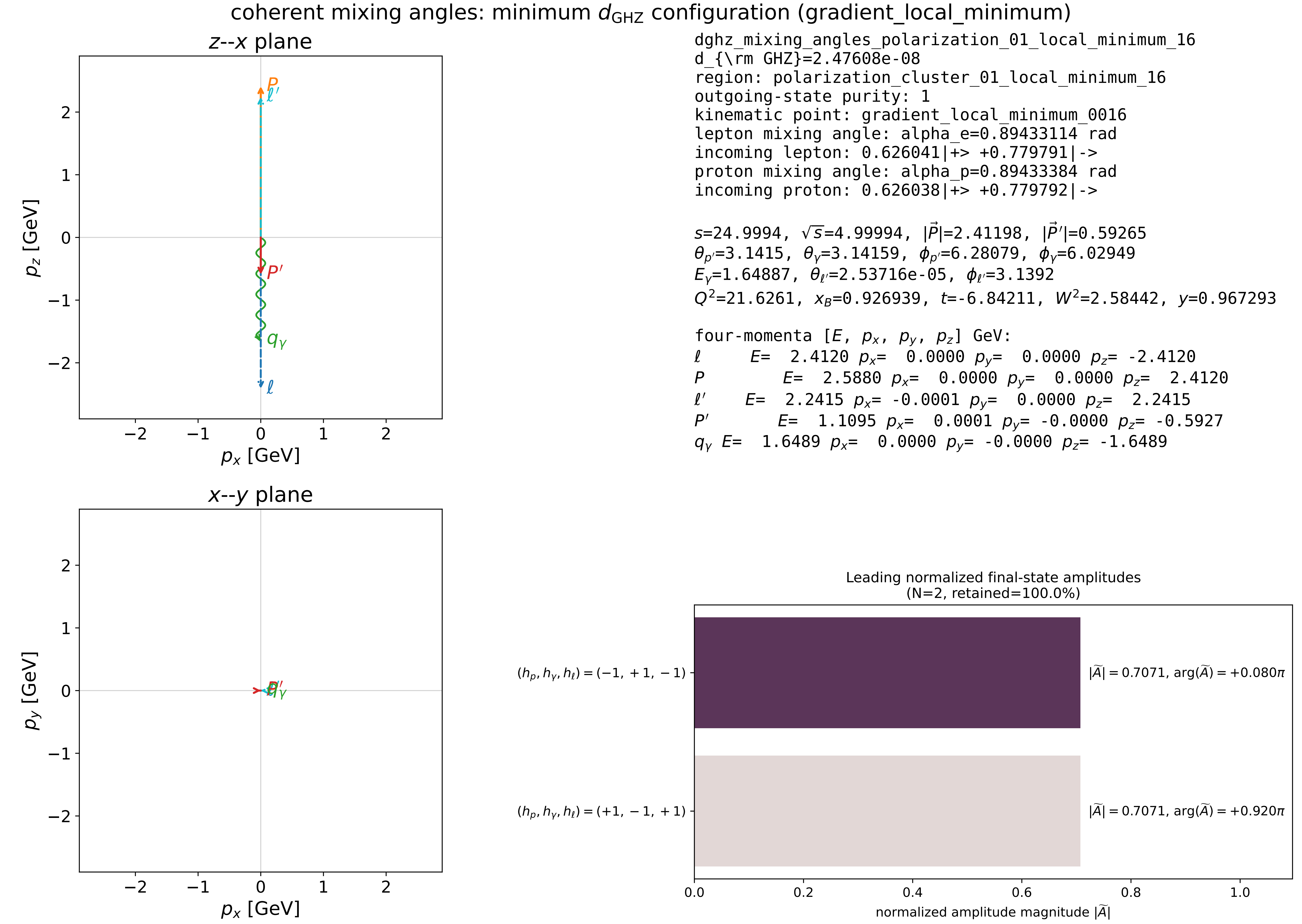
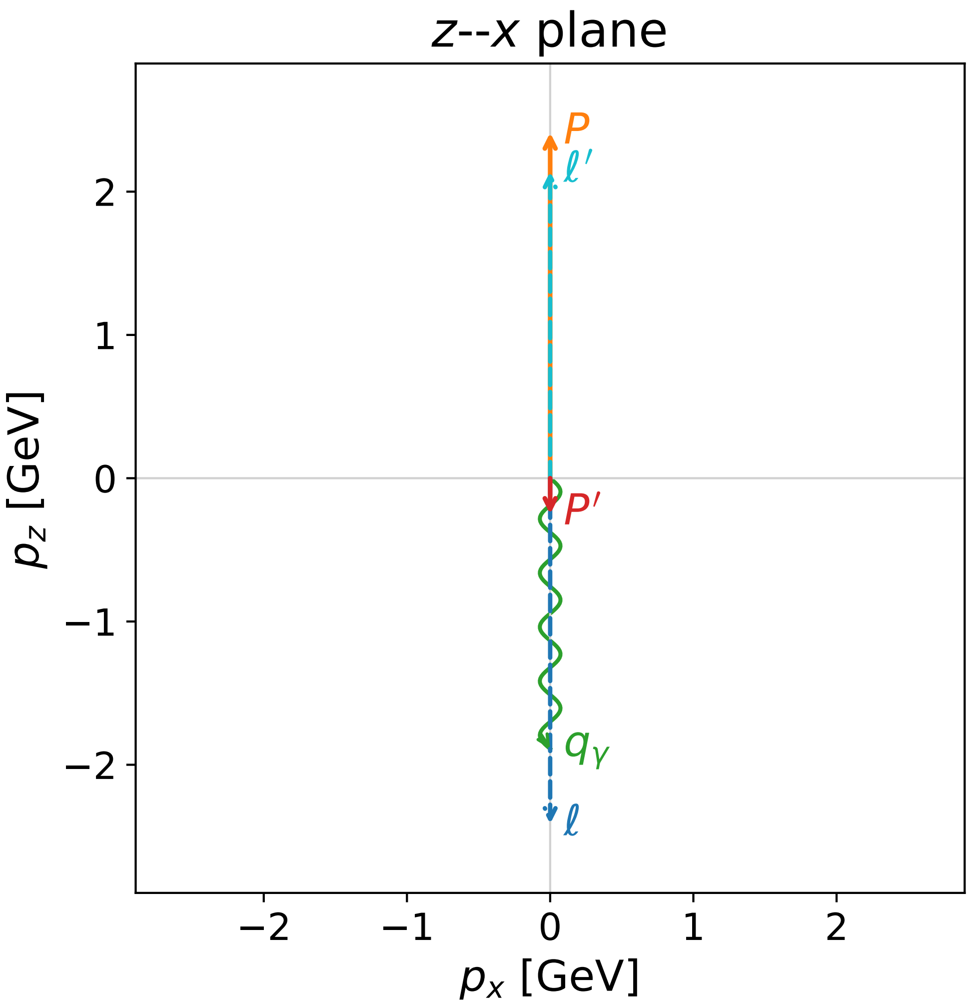

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_01_local_minimum_2
d_{\rm GHZ}=2.29125e-08
region: polarization_cluster_01_local_minimum_2
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0002
lepton mixing angle: alpha_e=0.87696633 rad
incoming lepton: 0.639486|+> +0.768802|->
proton mixing angle: alpha_p=0.87697731 rad
incoming proton: 0.639478|+> +0.768809|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.28627
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=0.0335096, ϕ_γ=0.522884
E_γ=0.121256, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.19891
Q^2=23.228, x_B=0.998073, t=-22.0603, W^2=0.924694, y=0.964872
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.4075 p_x= -0.0000 p_y= -0.0000 p_z= 2.4075
P' E= 2.4712 p_x= 0.0000 p_y= 0.0000 p_z= -2.2863
q_γ E= 0.1213 p_x= 0.0000 p_y= 0.0000 p_z= -0.1213
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.166\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.834\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_11
d_{\rm GHZ}=2.44872e-08
region: polarization_cluster_01_local_minimum_11
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0011
lepton mixing angle: alpha_e=0.59862965 rad
incoming lepton: 0.826109|+> +0.563511|->
proton mixing angle: alpha_p=0.59861028 rad
incoming proton: 0.82612|+> +0.563495|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=1.72071
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.57447, ϕ_γ=1.17554e-06
E_γ=0.659762, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.674751
Q^2=22.9669, x_B=0.986451, t=-16.6848, W^2=1.19529, y=0.965266
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3805 p_x= 0.0000 p_y= 0.0000 p_z= 2.3805
P' E= 1.9598 p_x= -0.0000 p_y= -0.0000 p_z= -1.7207
q_γ E= 0.6598 p_x= 0.0000 p_y= 0.0000 p_z= -0.6598
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_16
d_{\rm GHZ}=2.47608e-08
region: polarization_cluster_01_local_minimum_16
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0016
lepton mixing angle: alpha_e=0.89433114 rad
incoming lepton: 0.626041|+> +0.779791|->
proton mixing angle: alpha_p=0.89433384 rad
incoming proton: 0.626038|+> +0.779792|->
s=24.9994, $\sqrt{s}$=4.99994, $|\vec{P}|$=2.41198, $|\vec{P}'|$=0.59265
$\theta_{p'}$=3.1415, θ_γ=3.14159, $\phi_{p'}$=6.28079, ϕ_γ=6.02949
E_γ=1.64887, $\theta_{\ell'}$=2.53716e-05, $\phi_{\ell'}$=3.1392
Q^2=21.6261, x_B=0.926939, t=-6.84211, W^2=2.58442, y=0.967293
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.2415 p_x= -0.0001 p_y= 0.0000 p_z= 2.2415
P' E= 1.1095 p_x= 0.0001 p_y= -0.0000 p_z= -0.5927
q_γ E= 1.6489 p_x= 0.0000 p_y= -0.0000 p_z= -1.6489
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.080\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.920\pi$
normalized amplitude magnitude $|\widetilde{A}|$

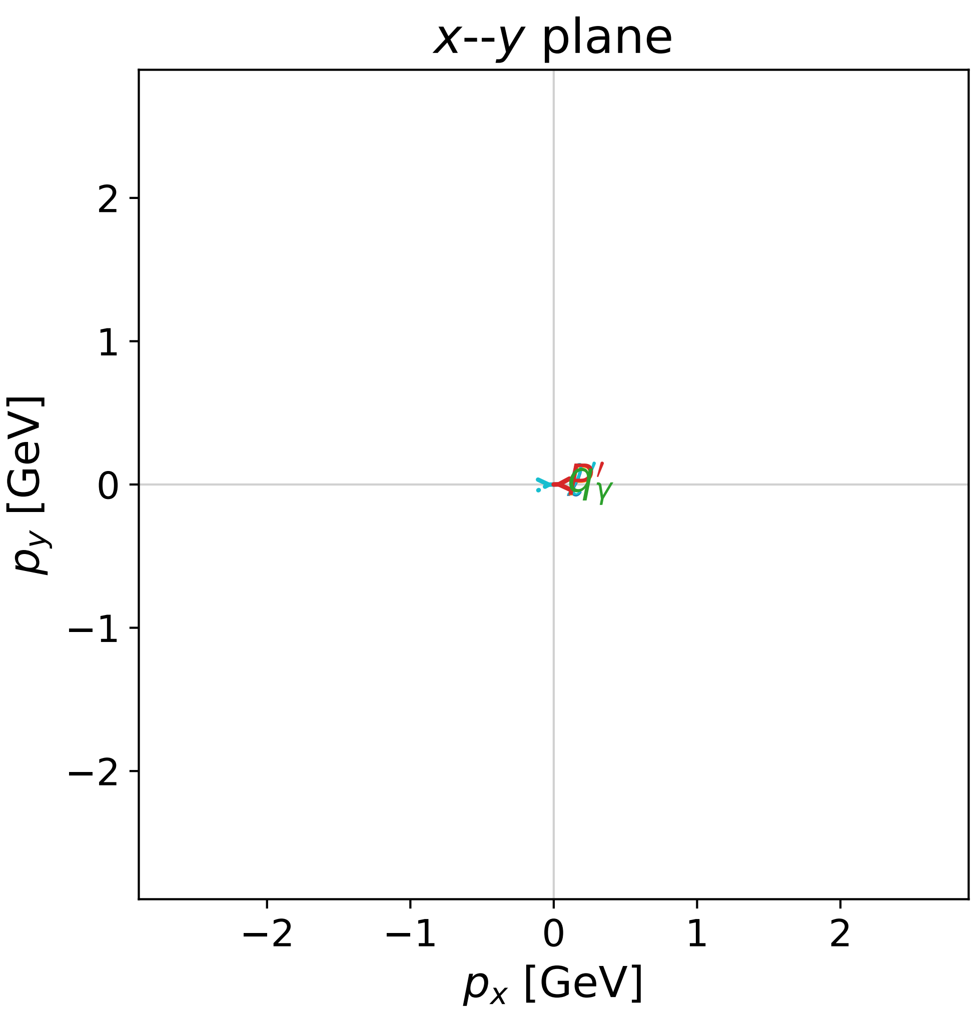
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

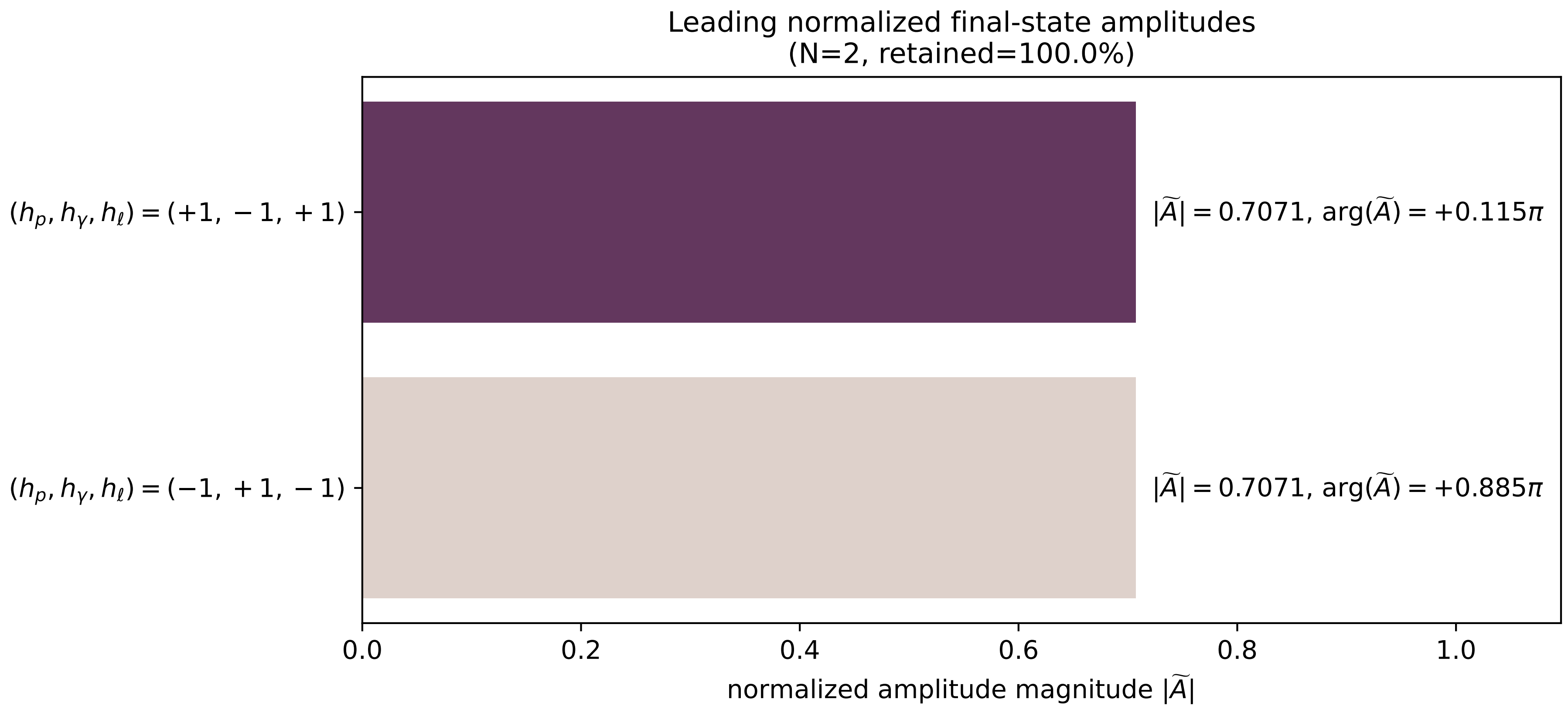


dghz_mixing_angles_polarization_01_local_minimum_19
d_{\rm GHZ}=2.53539e-08
region: polarization_cluster_01_local_minimum_19
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0019
lepton mixing angle: alpha_e=0.78519375 rad
incoming lepton: 0.707251|+> +0.706962|->
proton mixing angle: alpha_p=0.7851987 rad
incoming proton: 0.707248|+> +0.706966|->

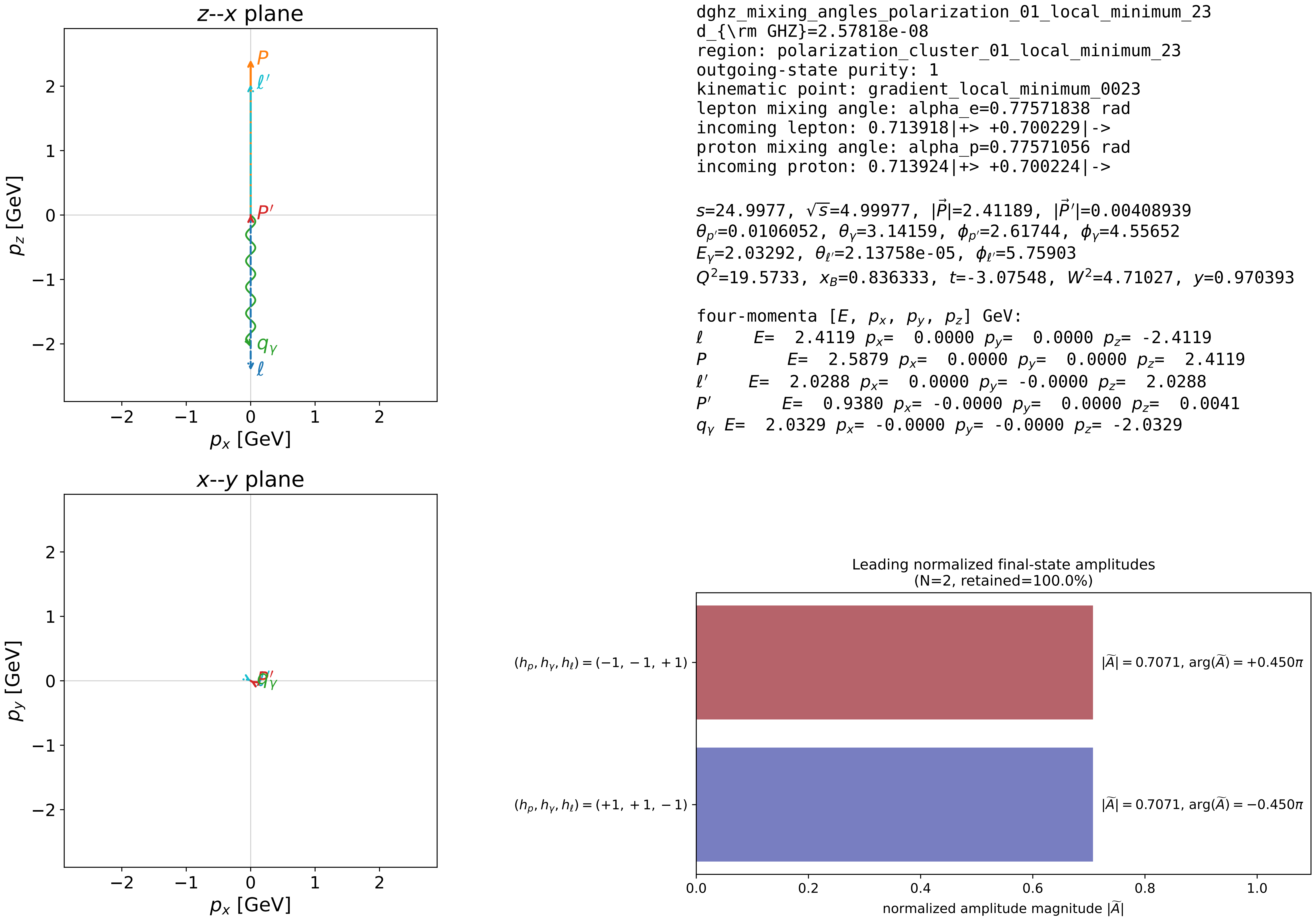
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.251318
$\theta_{p'}$=3.14131, θ_γ=3.14159, $\phi_{p'}$=3.17031, ϕ_γ=0.388985
E_γ=1.8888, $\theta_{\ell'}$=3.33062e-05, $\phi_{\ell'}$=0.0287211
Q^2=20.648, x_B=0.88364, t=-4.47898, W^2=3.59883, y=0.968774

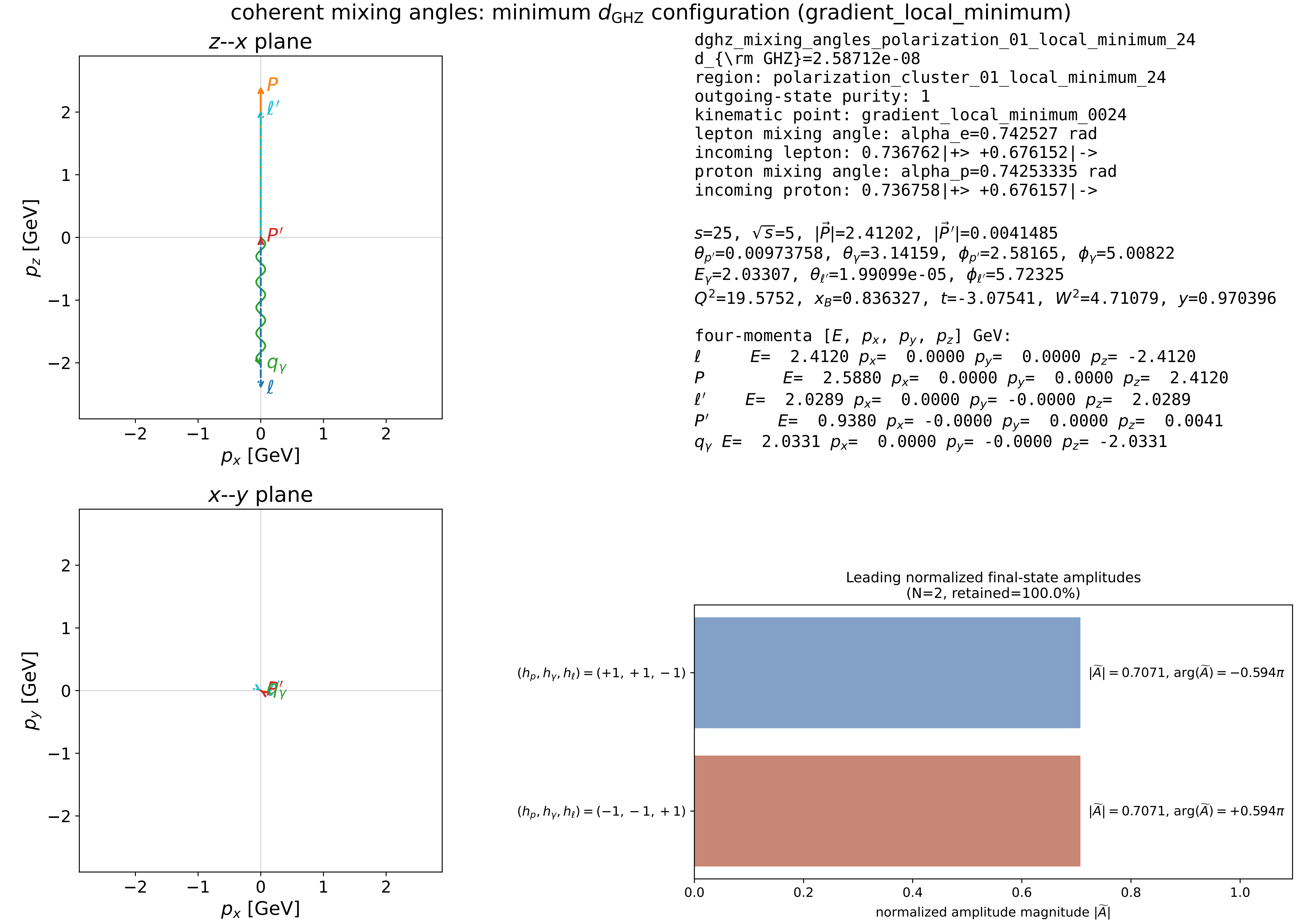
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.1401 p_x= 0.0001 p_y= 0.0000 p_z= 2.1401
P' E= 0.9711 p_x= -0.0001 p_y= -0.0000 p_z= -0.2513
q_γ E= 1.8888 p_x= 0.0000 p_y= 0.0000 p_z= -1.8888

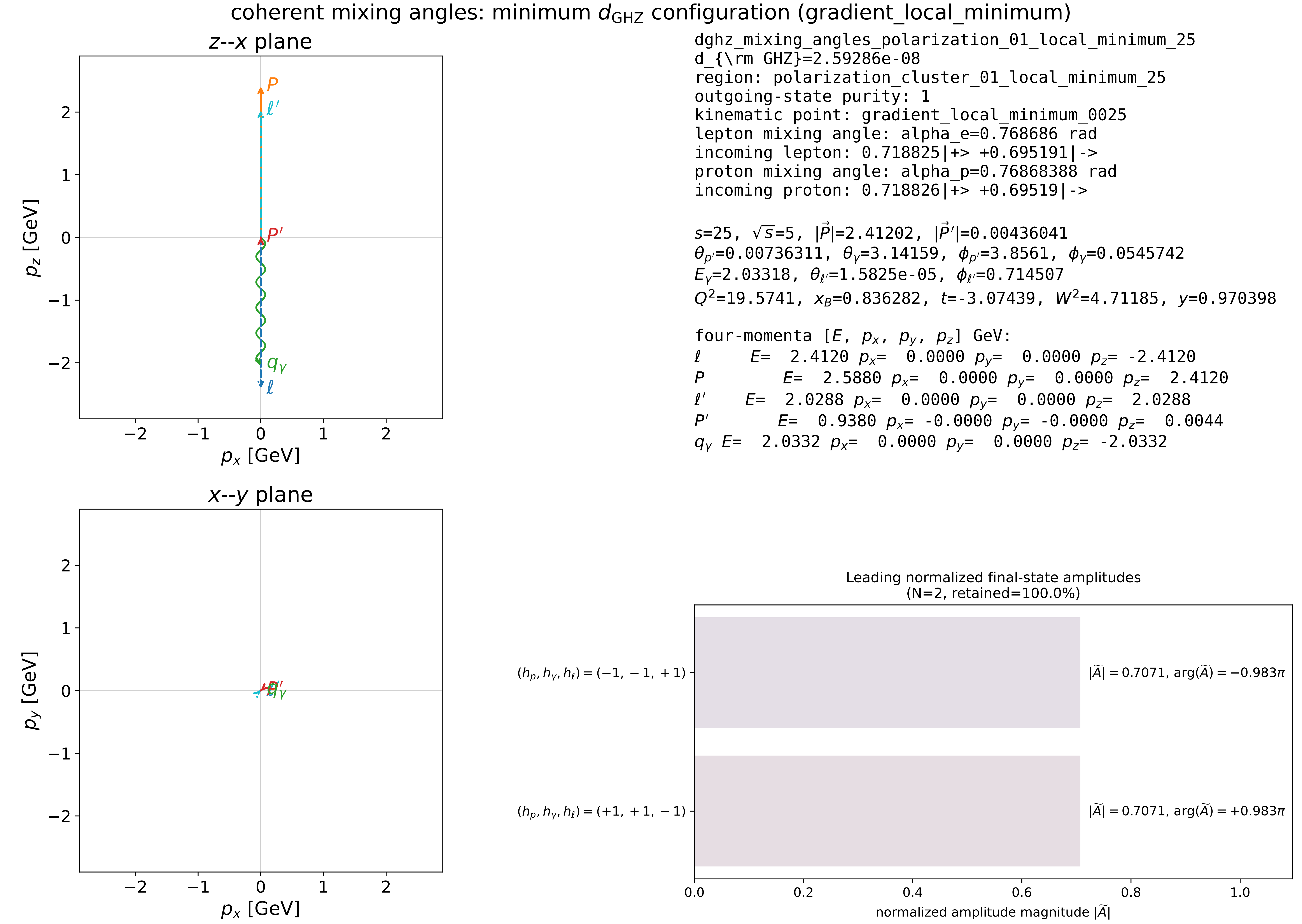




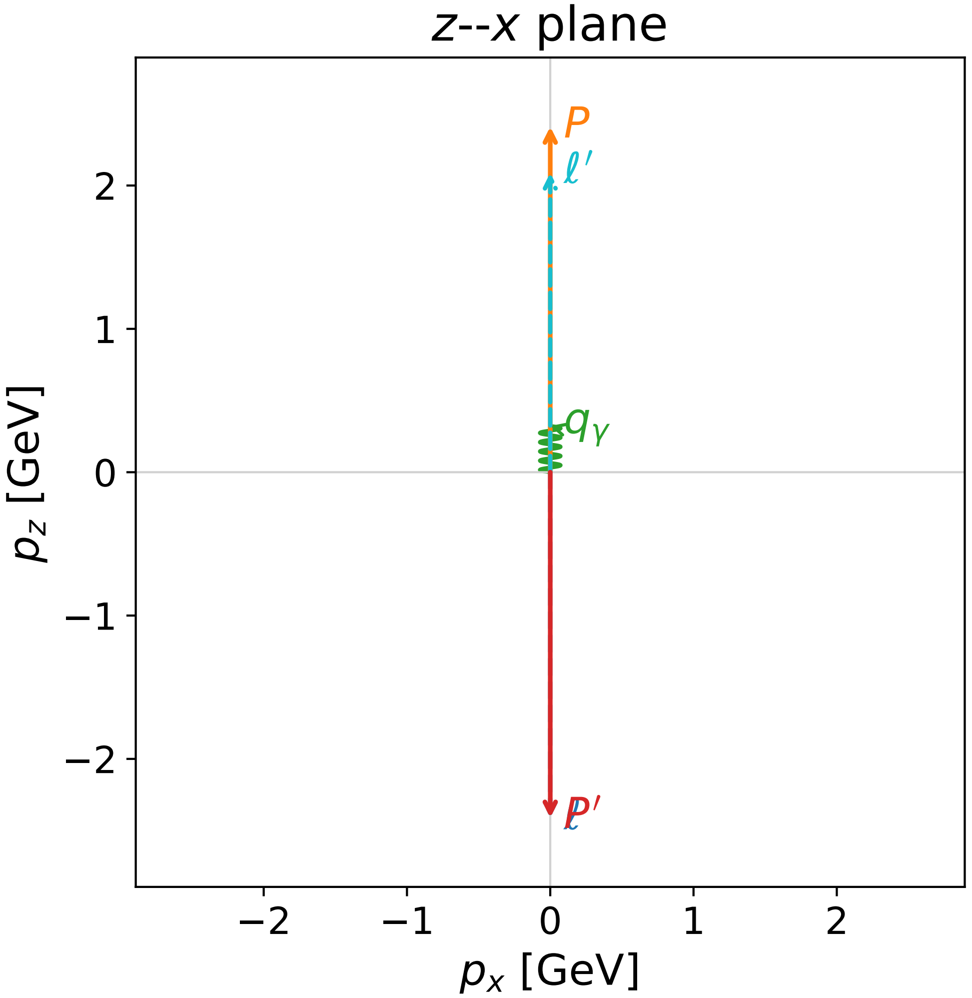
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_24
d_{\rm GHZ}=2.58712e-08
region: polarization_cluster_01_local_minimum_24
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0024
lepton mixing angle: alpha_e=0.742527 rad
incoming lepton: 0.736762|+> +0.676152|->
proton mixing angle: alpha_p=0.74253335 rad
incoming proton: 0.736758|+> +0.676157|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0041485
$\theta_{p'}$=0.00973758, θ_γ=3.14159, $\phi_{p'}$=2.58165, ϕ_γ=5.00822
E_γ=2.03307, $\theta_{\ell'}$=1.99099e-05, $\phi_{\ell'}$=5.72325
Q^2=19.5752, x_B=0.836327, t=-3.07541, W^2=4.71079, y=0.970396
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0289 p_x= 0.0000 p_y= -0.0000 p_z= 2.0289
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0041
q_γ E= 2.0331 p_x= 0.0000 p_y= -0.0000 p_z= -2.0331
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.594\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.594\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_25
d_{\rm GHZ}=2.59286e-08
region: polarization_cluster_01_local_minimum_25
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0025
lepton mixing angle: alpha_e=0.768686 rad
incoming lepton: 0.718825|+> +0.695191|->
proton mixing angle: alpha_p=0.76868388 rad
incoming proton: 0.718826|+> +0.69519|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.00436041
$\theta_{p'}$=0.00736311, θ_γ=3.14159, $\phi_{p'}$=3.8561, ϕ_γ=0.0545742
E_γ=2.03318, $\theta_{\ell'}$=1.5825e-05, $\phi_{\ell'}$=0.714507
Q^2=19.5741, x_B=0.836282, t=-3.07439, W^2=4.71185, y=0.970398
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0288 p_x= 0.0000 p_y= 0.0000 p_z= 2.0288
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0044
q_γ E= 2.0332 p_x= 0.0000 p_y= 0.0000 p_z= -2.0332
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.983\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.983\pi$
normalized amplitude magnitude $|\widetilde{A}|$

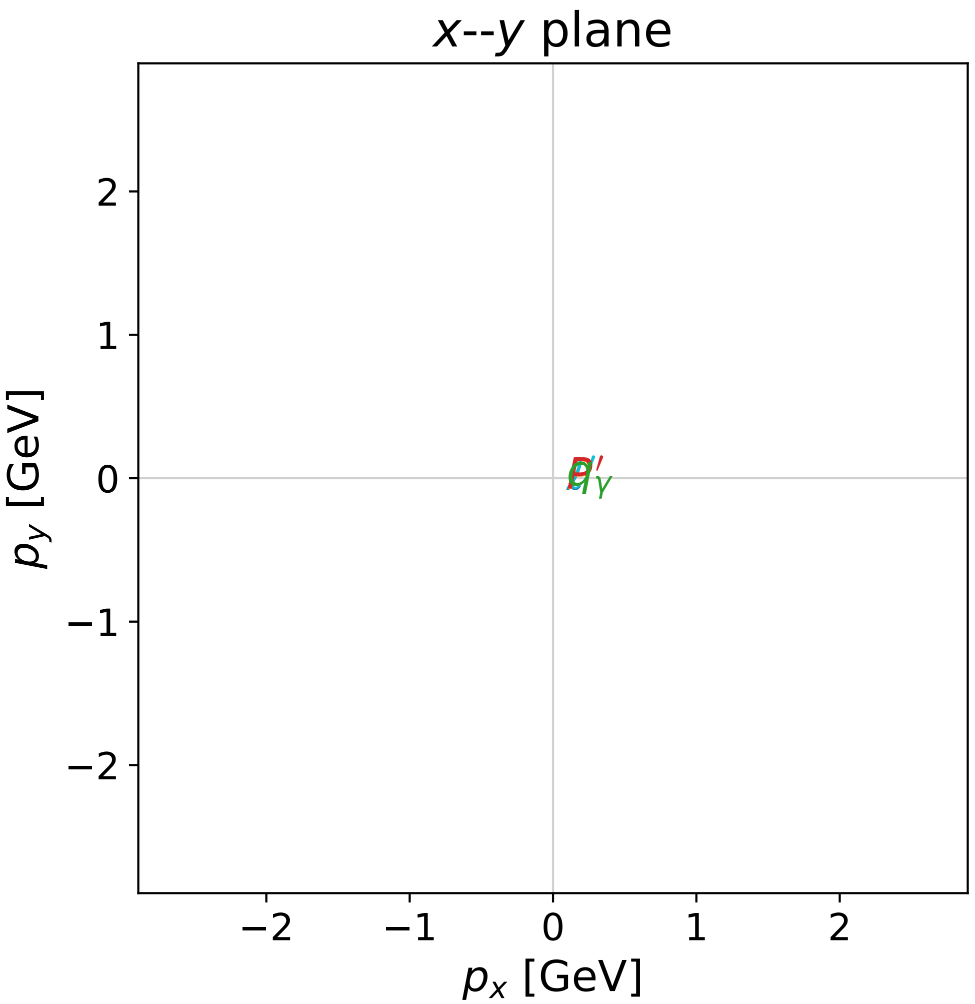
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

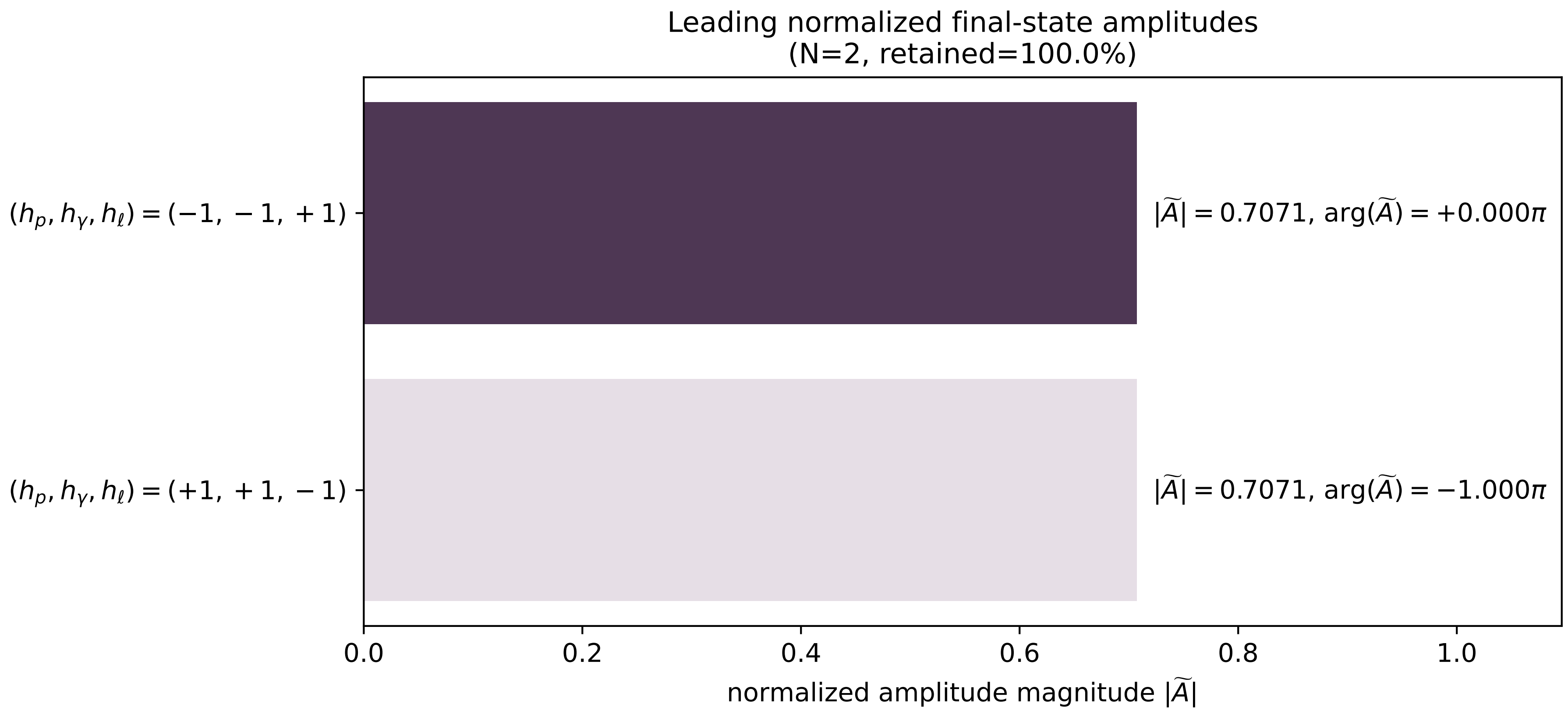


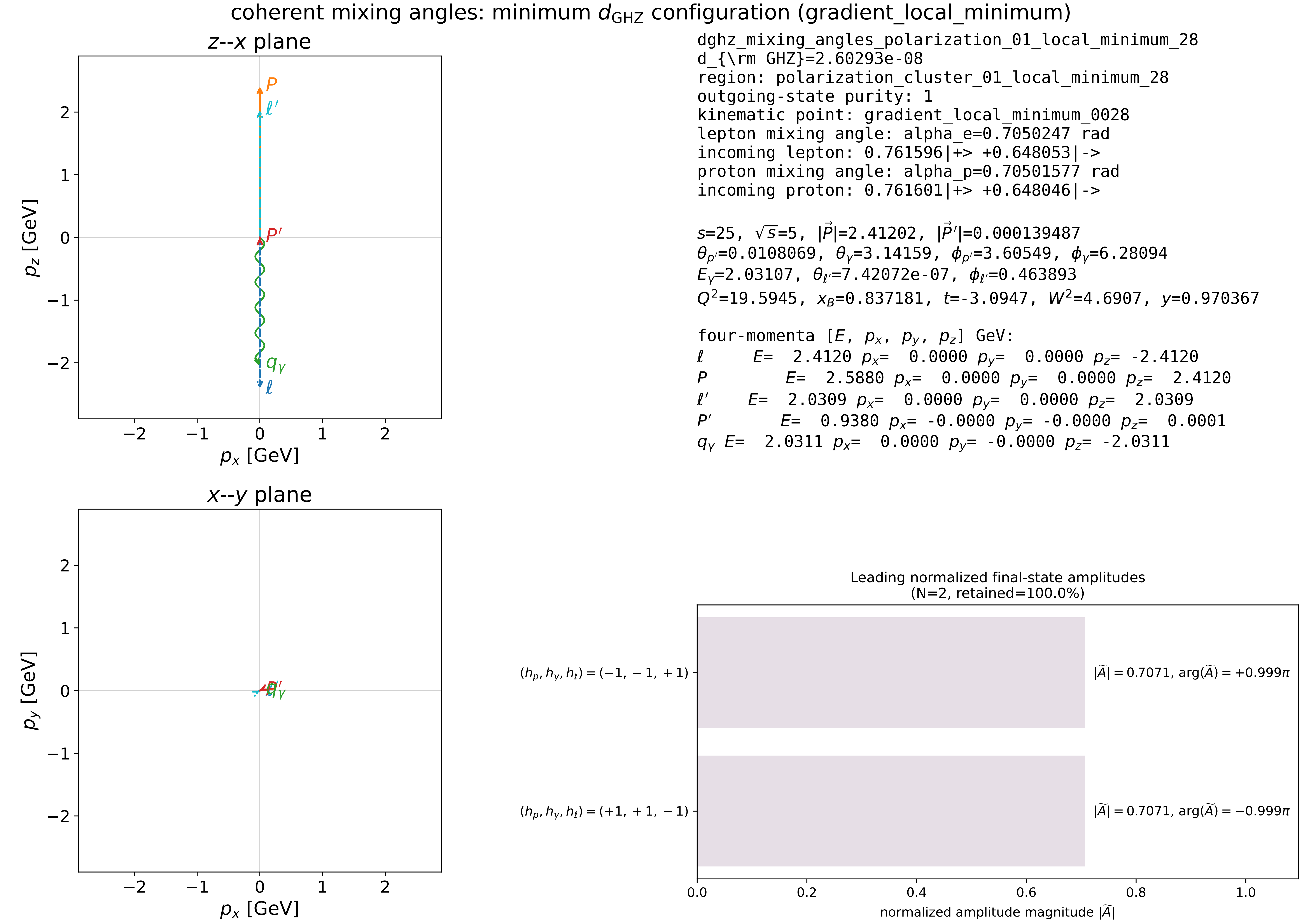
dghz_mixing_angles_polarization_01_local_minimum_27
d_{\rm GHZ}=2.59874e-08
region: polarization_cluster_01_local_minimum_27
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0027
lepton mixing angle: alpha_e=0.75349531 rad
incoming lepton: 0.729302|+> +0.684192|->
proton mixing angle: alpha_p=0.81730854 rad
incoming proton: 0.684187|+> +0.729307|->

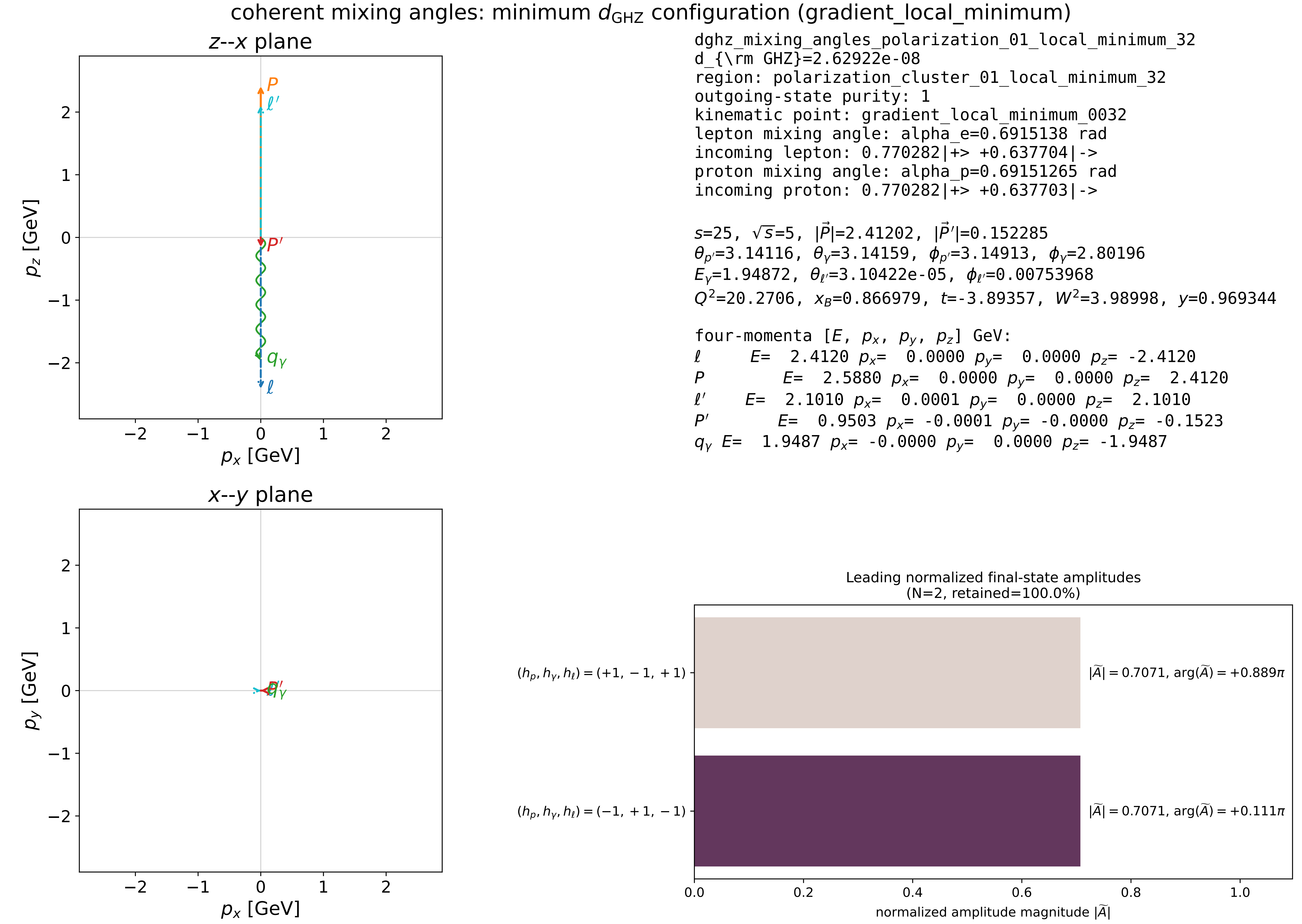
s=24.9779, $\sqrt{s}$=4.99779, $|\vec{P}|$=2.41087, $|\vec{P}'|$=2.41087
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=0.0939447, ϕ_γ=0.966166
E_γ=0.322487, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.23554
Q^2=20.1393, x_B=0.862026, t=-23.2492, W^2=4.10329, y=0.969487

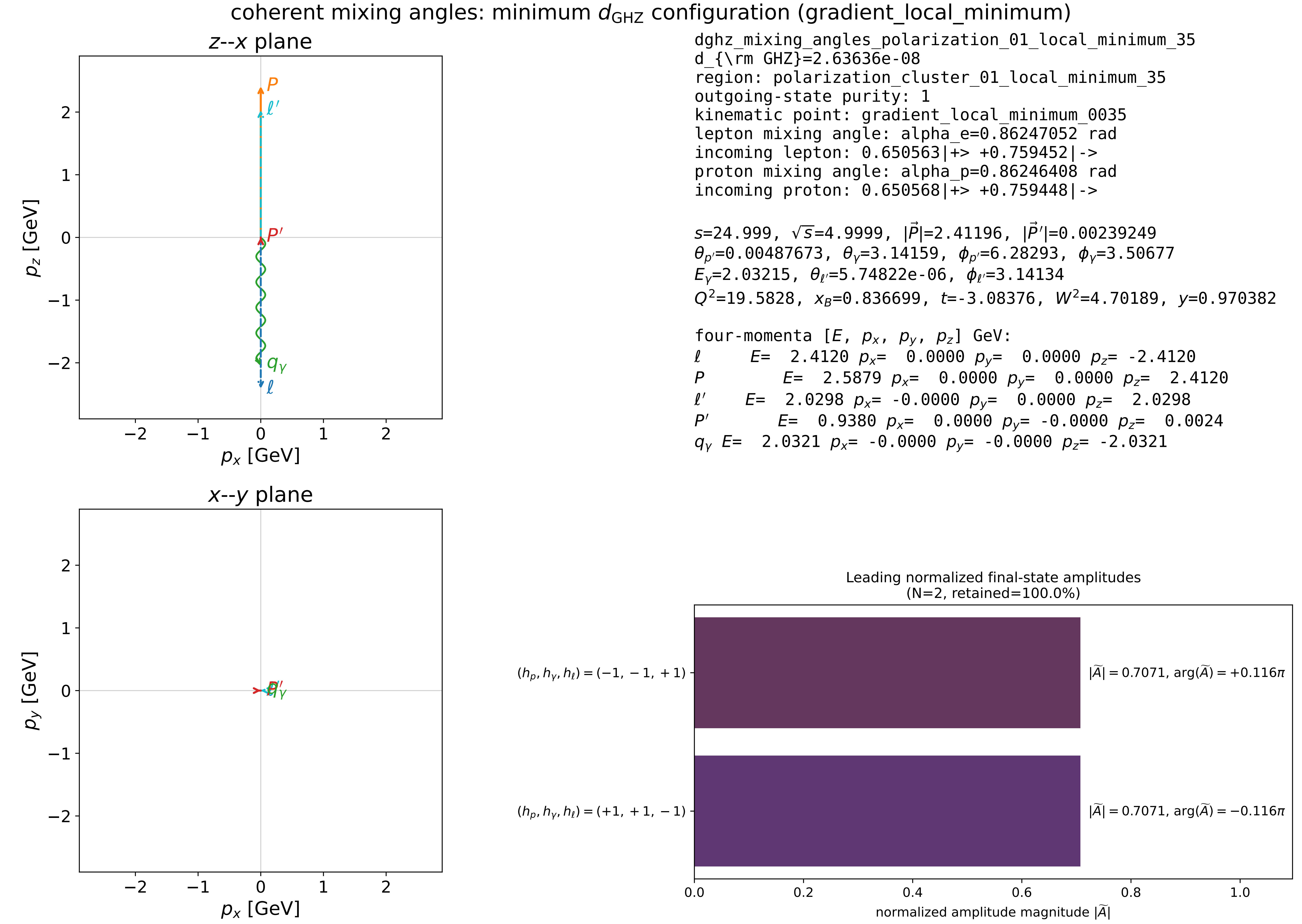
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4109 p_x= 0.0000 p_y= 0.0000 p_z= -2.4109
P E= 2.5869 p_x= 0.0000 p_y= 0.0000 p_z= 2.4109
ℓ' E= 2.0884 p_x= -0.0000 p_y= -0.0000 p_z= 2.0884
P' E= 2.5869 p_x= 0.0000 p_y= 0.0000 p_z= -2.4109
q_γ E= 0.3225 p_x= 0.0000 p_y= 0.0000 p_z= 0.3225



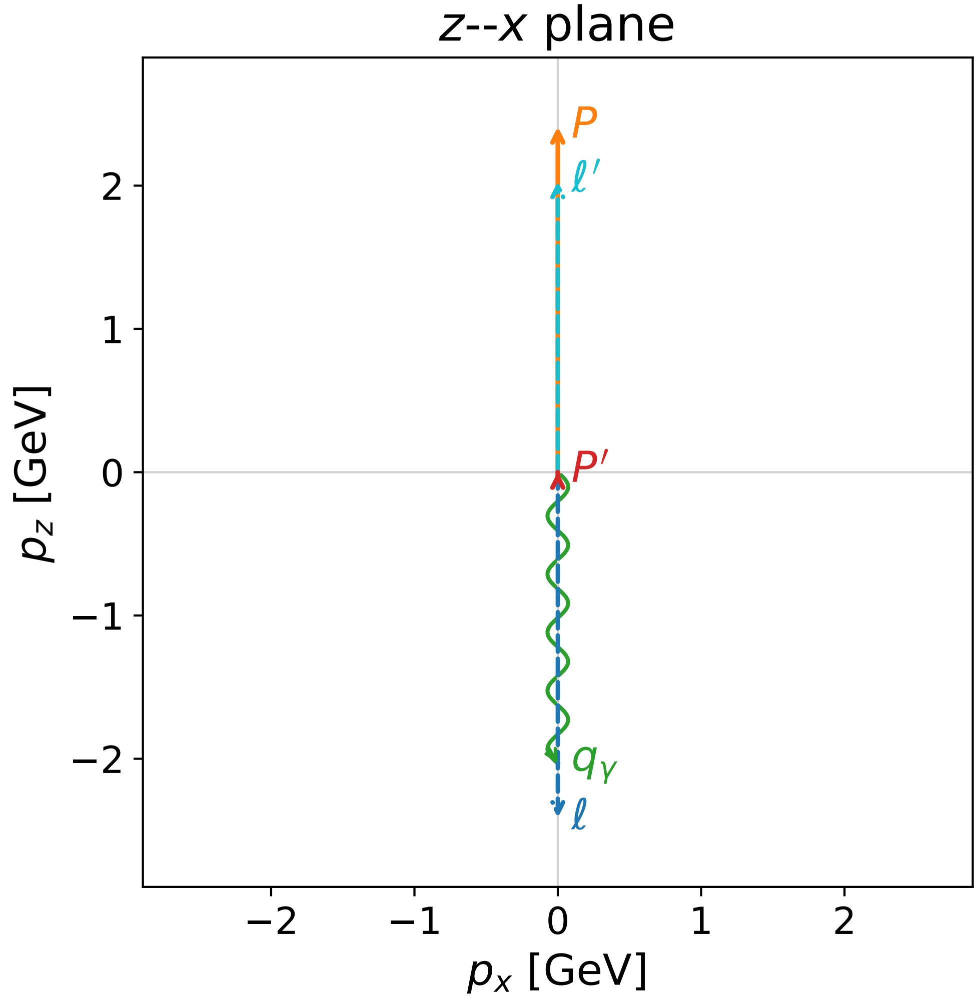



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_28
d_{\rm GHZ}=2.60293e-08
region: polarization_cluster_01_local_minimum_28
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0028
lepton mixing angle: alpha_e=0.7050247 rad
incoming lepton: 0.761596|+> +0.648053|->
proton mixing angle: alpha_p=0.70501577 rad
incoming proton: 0.761601|+> +0.648046|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.000139487
$\theta_{p'}$=0.0108069, θ_γ=3.14159, $\phi_{p'}$=3.60549, ϕ_γ=6.28094
E_γ=2.03107, $\theta_{\ell'}$=7.42072e-07, $\phi_{\ell'}$=0.463893
Q^2=19.5945, x_B=0.837181, t=-3.0947, W^2=4.6907, y=0.970367
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0309 p_x= 0.0000 p_y= 0.0000 p_z= 2.0309
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0001
q_γ E= 2.0311 p_x= 0.0000 p_y= -0.0000 p_z= -2.0311
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.999\pi$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.999\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_32
d_{\rm GHZ}=2.62922e-08
region: polarization_cluster_01_local_minimum_32
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0032
lepton mixing angle: alpha_e=0.6915138 rad
incoming lepton: 0.770282|+> +0.637704|->
proton mixing angle: alpha_p=0.69151265 rad
incoming proton: 0.770282|+> +0.637703|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.152285
$\theta_{p'}$=3.14116, θ_γ=3.14159, $\phi_{p'}$=3.14913, ϕ_γ=2.80196
E_γ=1.94872, $\theta_{\ell'}$=3.10422e-05, $\phi_{\ell'}$=0.00753968
Q^2=20.2706, x_B=0.866979, t=-3.89357, W^2=3.98998, y=0.969344
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.1010 p_x= 0.0001 p_y= 0.0000 p_z= 2.1010
P' E= 0.9503 p_x= -0.0001 p_y= -0.0000 p_z= -0.1523
q_γ E= 1.9487 p_x= -0.0000 p_y= 0.0000 p_z= -1.9487
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.889\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.111\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_35
d_{\rm GHZ}=2.63636e-08
region: polarization_cluster_01_local_minimum_35
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0035
lepton mixing angle: alpha_e=0.86247052 rad
incoming lepton: 0.650563|+> +0.759452|->
proton mixing angle: alpha_p=0.86246408 rad
incoming proton: 0.650568|+> +0.759448|->
s=24.999, $\sqrt{s}$=4.9999, $|\vec{P}|$=2.41196, $|\vec{P}'|$=0.00239249
$\theta_{p'}$=0.00487673, θ_γ=3.14159, $\phi_{p'}$=6.28293, ϕ_γ=3.50677
E_γ=2.03215, $\theta_{\ell'}$=5.74822e-06, $\phi_{\ell'}$=3.14134
Q^2=19.5828, x_B=0.836699, t=-3.08376, W^2=4.70189, y=0.970382
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5879 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0298 p_x= -0.0000 p_y= 0.0000 p_z= 2.0298
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0024
q_γ E= 2.0321 p_x= -0.0000 p_y= -0.0000 p_z= -2.0321
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.116\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.116\pi$
normalized amplitude magnitude $|\widetilde{A}|$

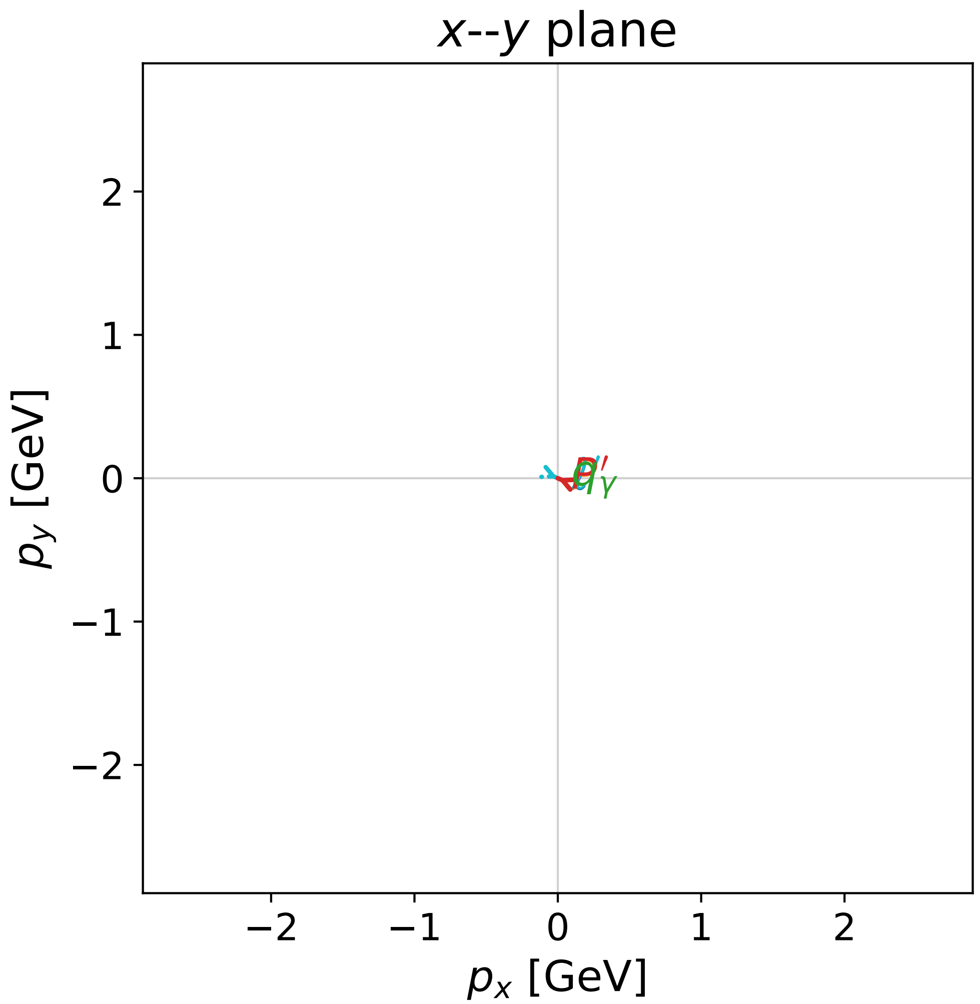
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

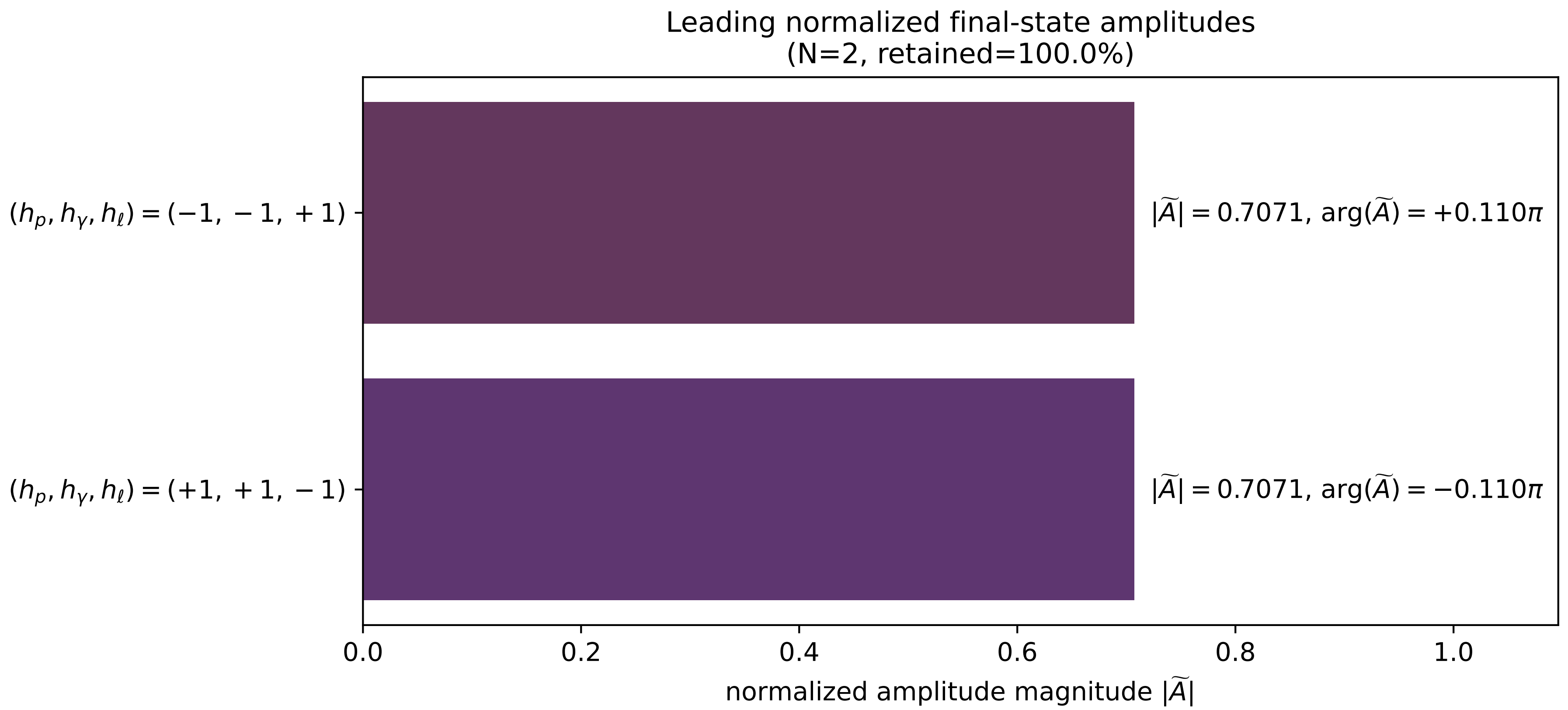


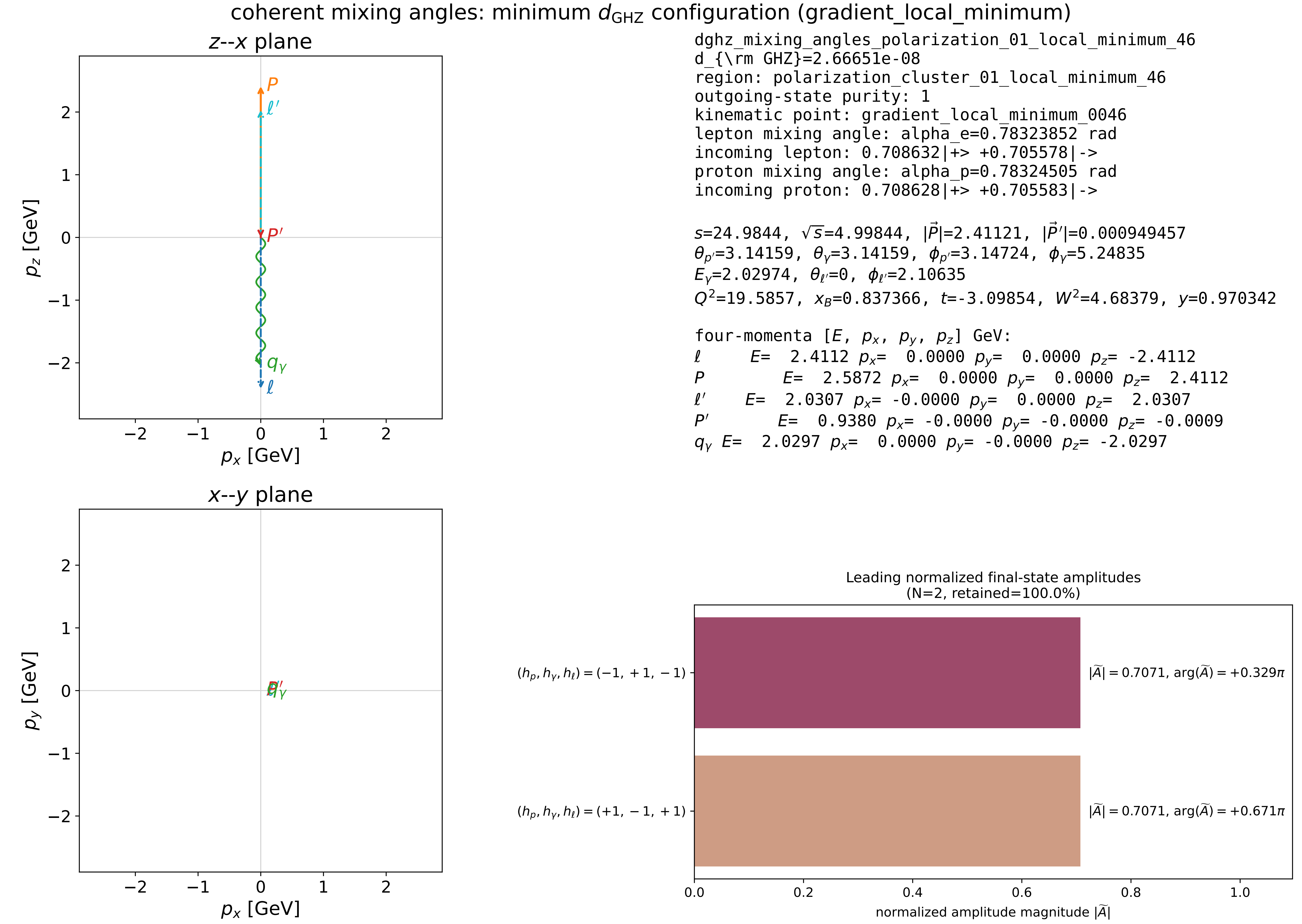

```
dghz_mixing_angles_polarization_01_local_minimum_44
d_{\rm GHZ}=2.65428e-08
region: polarization_cluster_01_local_minimum_44
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0044
lepton mixing angle: alpha_e=0.70477583 rad
incoming lepton: 0.761757|+> +0.647863|->
proton mixing angle: alpha_p=0.70475776 rad
incoming proton: 0.761769|+> +0.647849|->
```

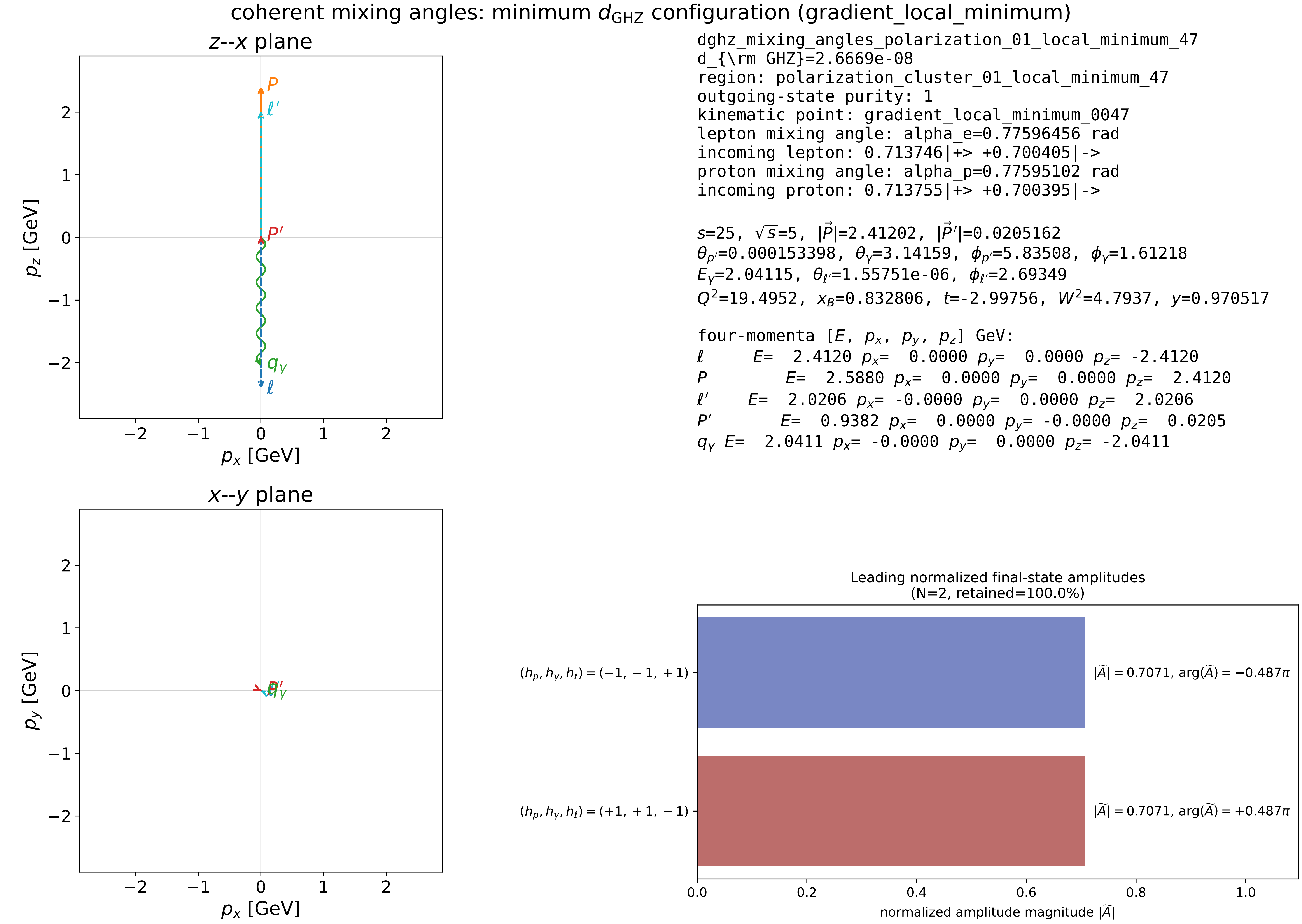
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.00358496
$\theta_{p'}$=2.38484e-05, θ_γ=3.14159, $\phi_{p'}$=2.72855, ϕ_γ=3.48858
E_γ=2.03279, $\theta_{\ell'}$=4.21468e-08, $\phi_{\ell'}$=5.87014
Q^2=19.5779, x_B=0.836447, t=-3.07811, W^2=4.70796, y=0.970392

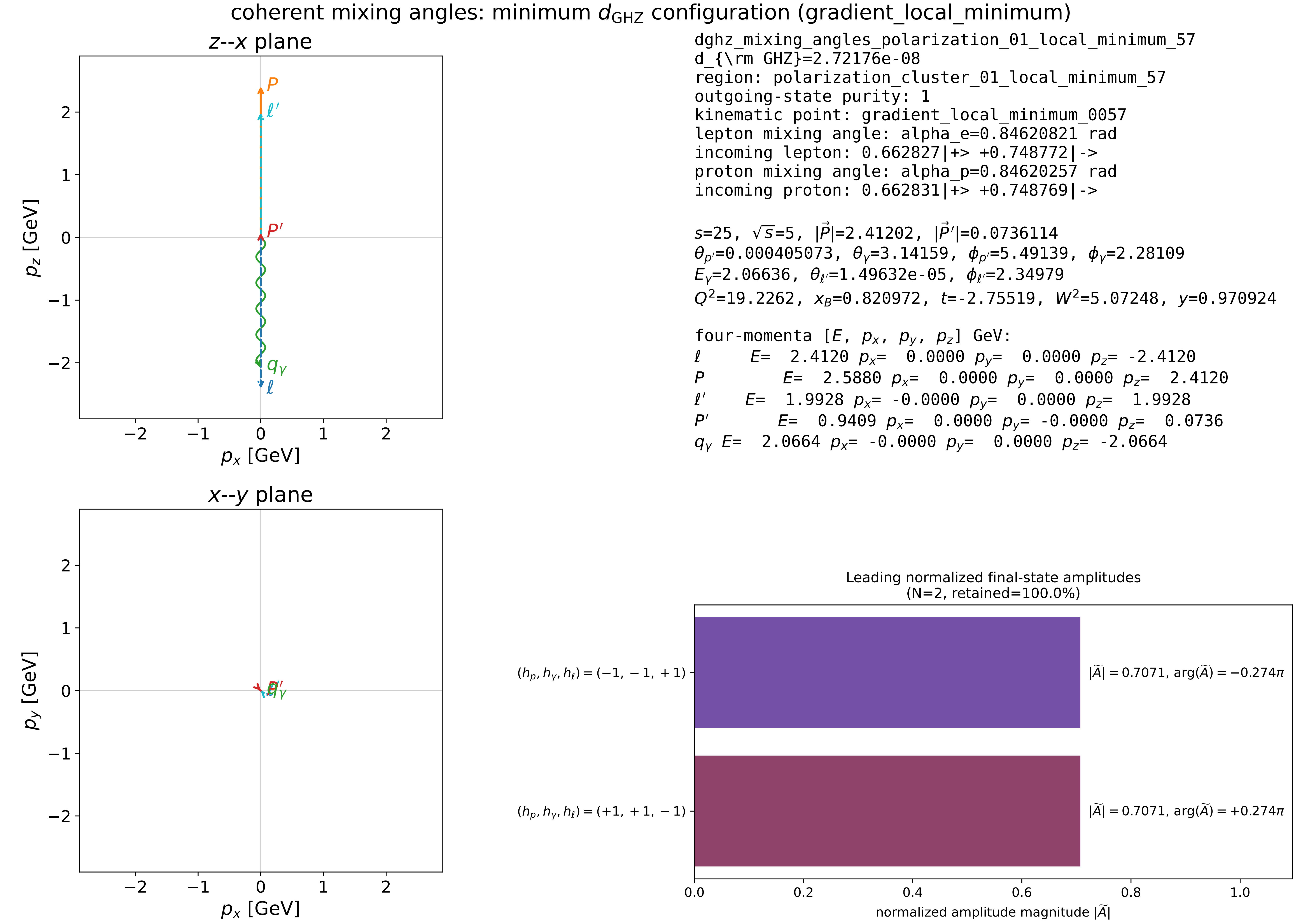
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0292 p_x= 0.0000 p_y= -0.0000 p_z= 2.0292
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0036
q_γ E= 2.0328 p_x= -0.0000 p_y= -0.0000 p_z= -2.0328

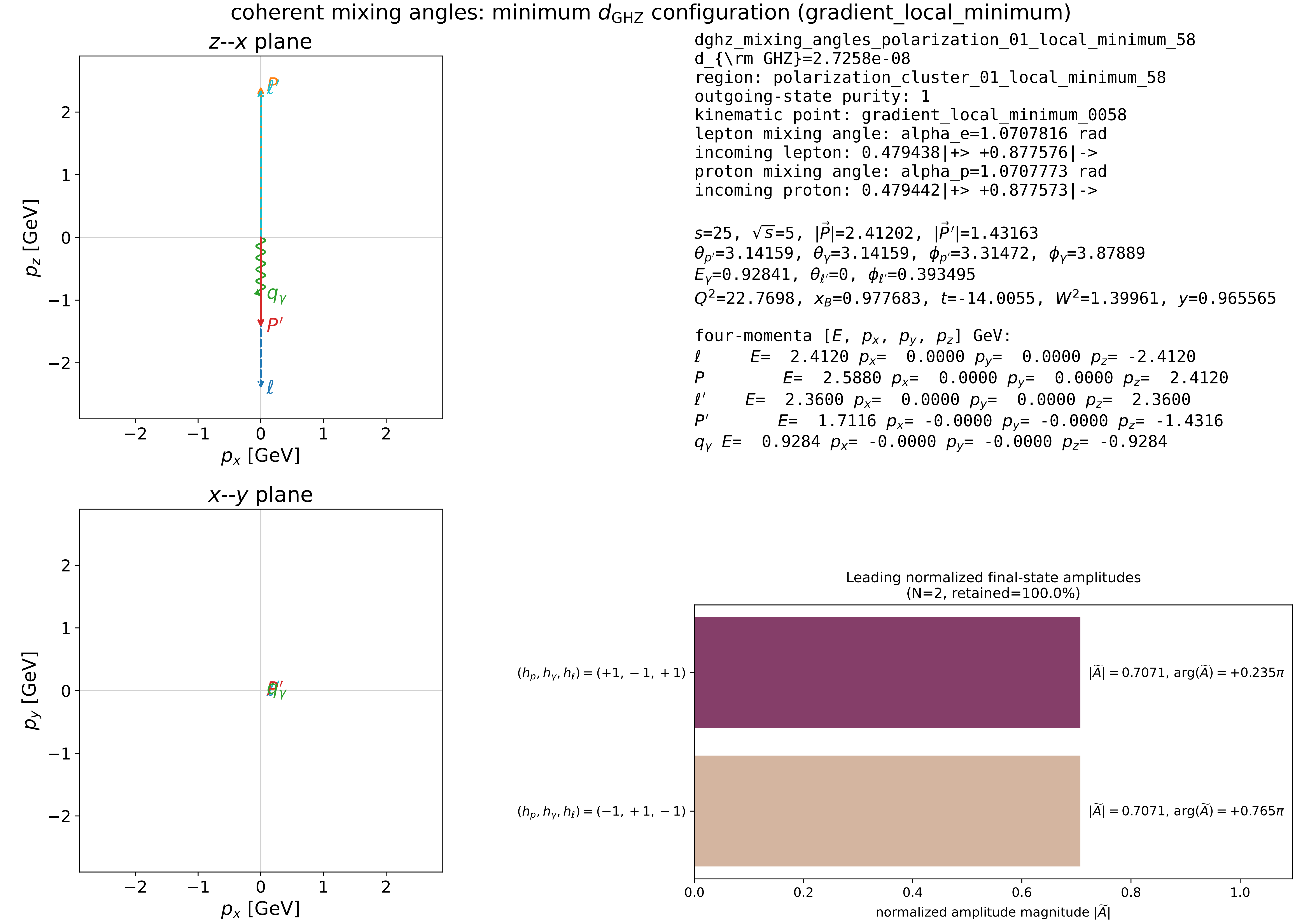


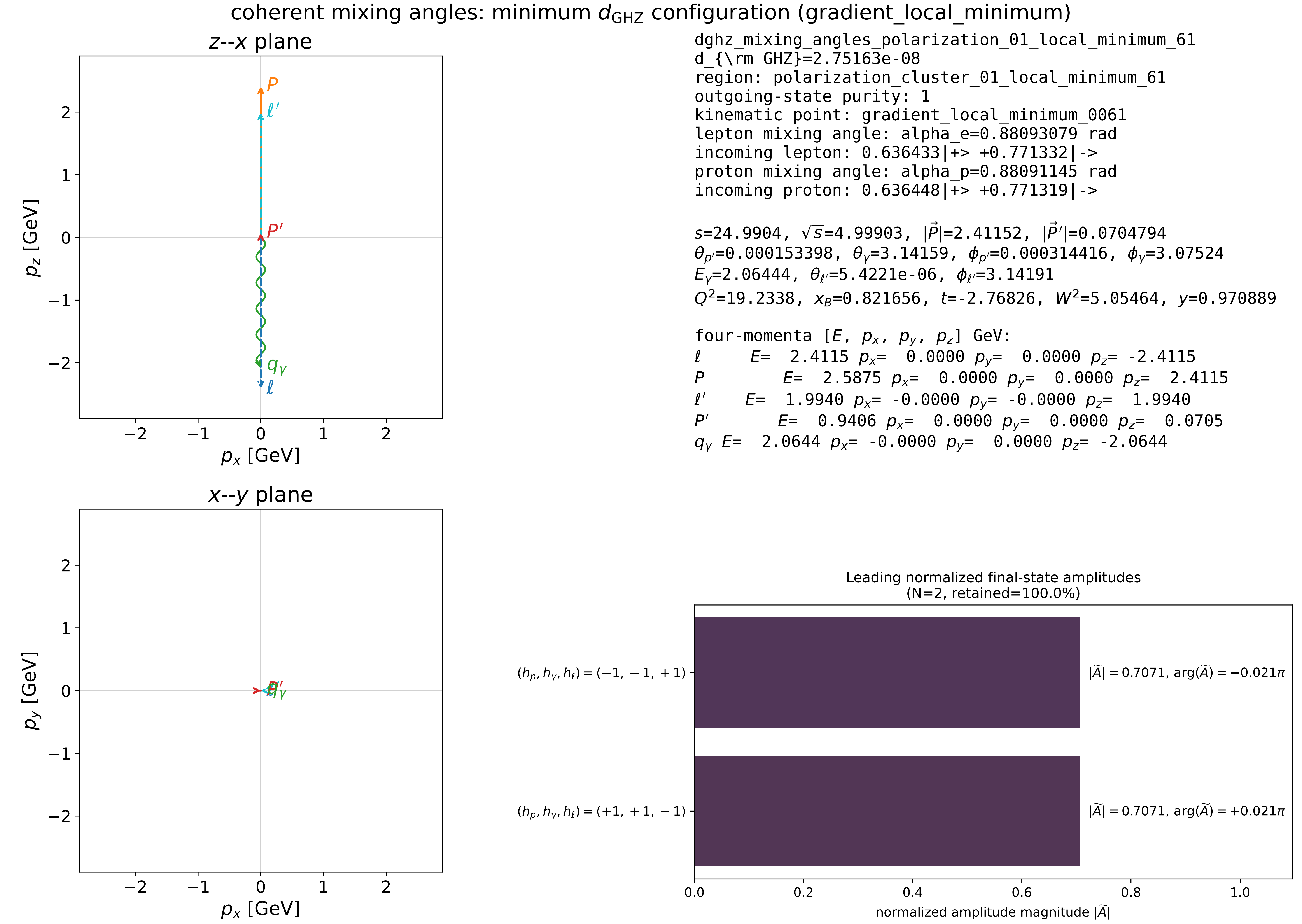


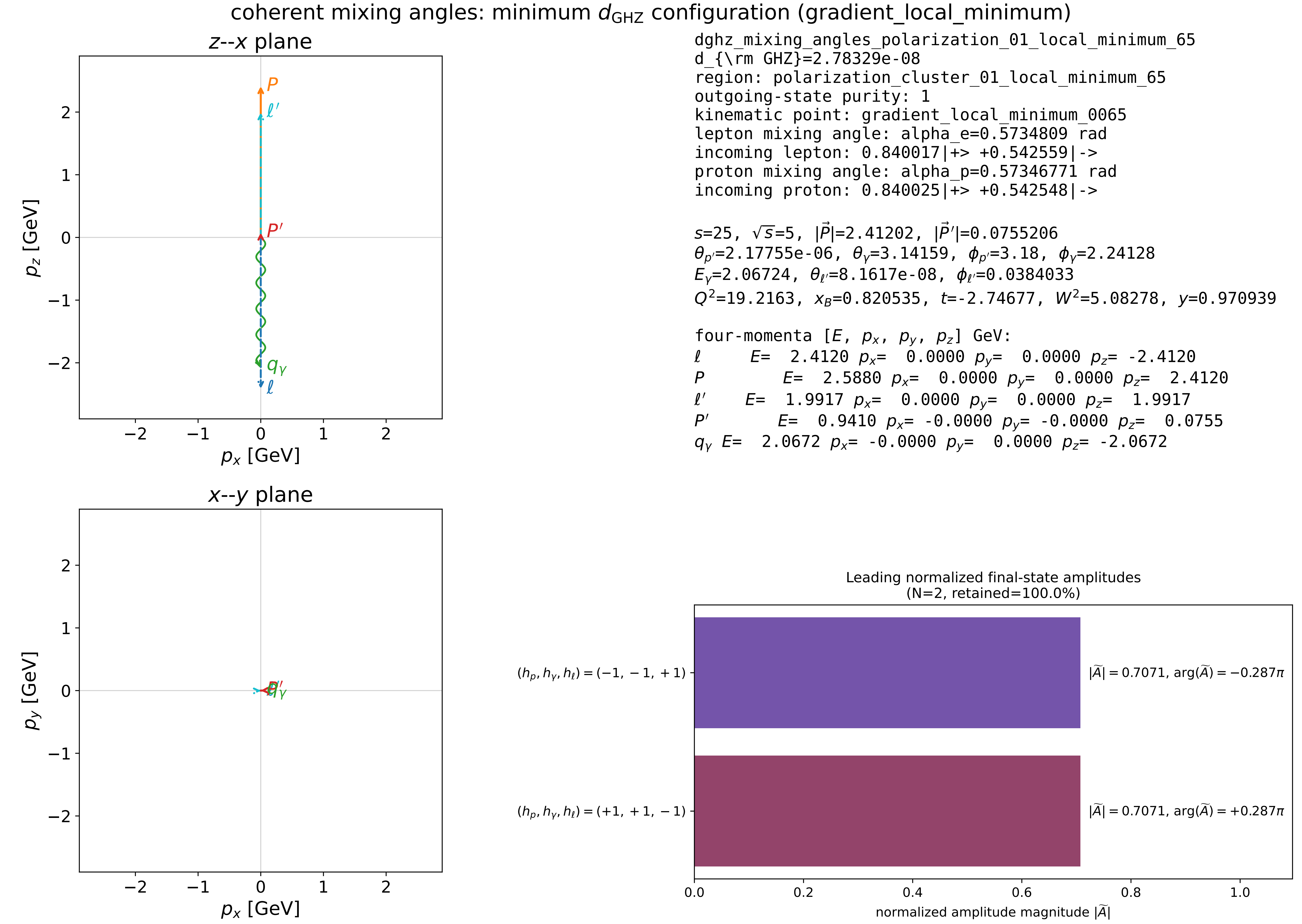

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_46
d_{\rm GHZ}=2.66651e-08
region: polarization_cluster_01_local_minimum_46
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0046
lepton mixing angle: alpha_e=0.78323852 rad
incoming lepton: 0.708632|+> +0.705578|->
proton mixing angle: alpha_p=0.78324505 rad
incoming proton: 0.708628|+> +0.705583|->
s=24.9844, $\sqrt{s}$=4.99844, $|\vec{P}|$=2.41121, $|\vec{P}'|$=0.000949457
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.14724, ϕ_γ=5.24835
E_γ=2.02974, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.10635
Q^2=19.5857, x_B=0.837366, t=-3.09854, W^2=4.68379, y=0.970342
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4112 p_x= 0.0000 p_y= 0.0000 p_z= -2.4112
P E= 2.5872 p_x= 0.0000 p_y= 0.0000 p_z= 2.4112
ℓ' E= 2.0307 p_x= -0.0000 p_y= 0.0000 p_z= 2.0307
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0009
q_γ E= 2.0297 p_x= 0.0000 p_y= -0.0000 p_z= -2.0297
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.329\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.671\pi$
normalized amplitude magnitude $|\widetilde{A}|$

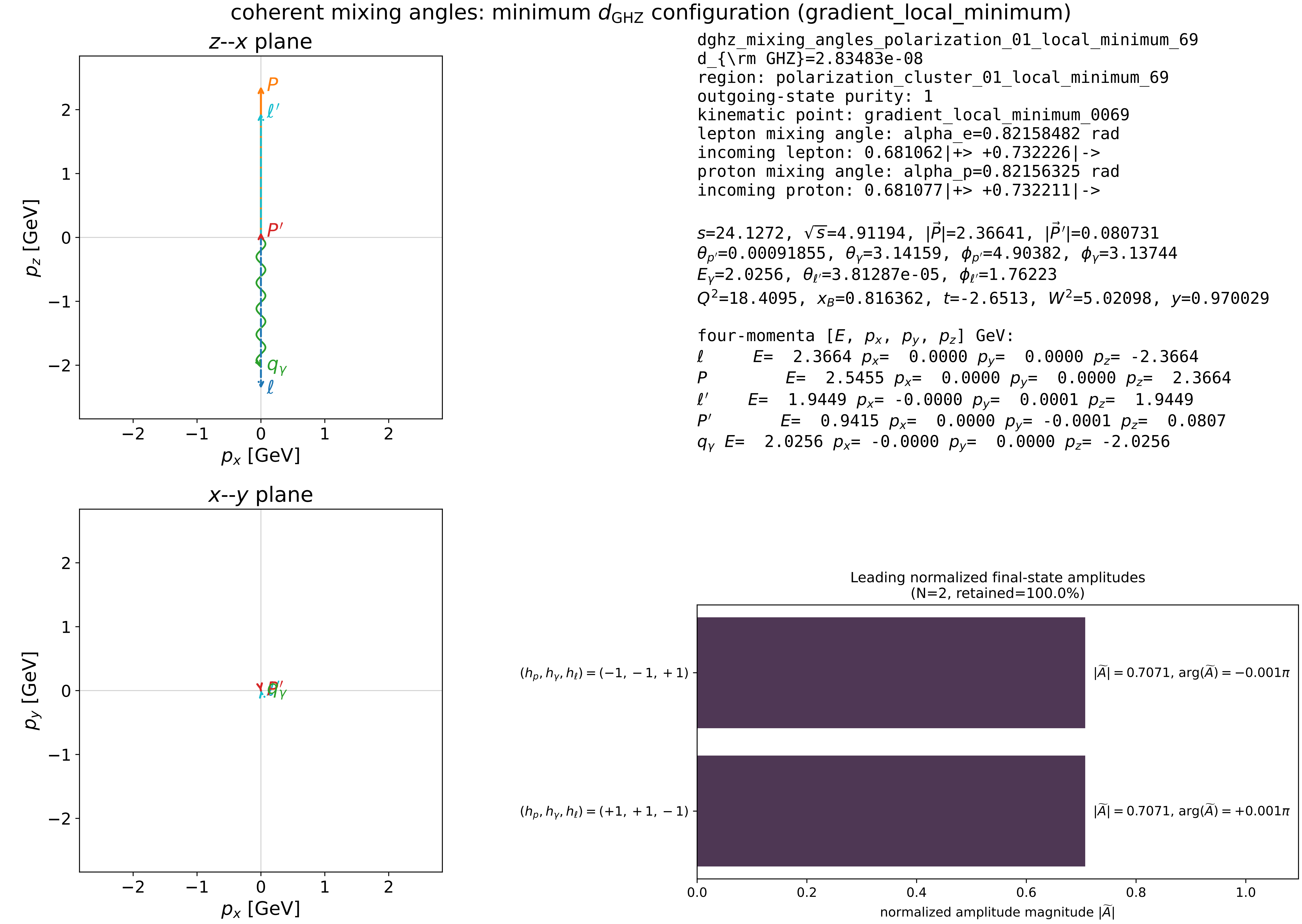

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_47
d_{\rm GHZ}=2.6669e-08
region: polarization_cluster_01_local_minimum_47
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0047
lepton mixing angle: alpha_e=0.77596456 rad
incoming lepton: 0.713746|+> +0.700405|->
proton mixing angle: alpha_p=0.77595102 rad
incoming proton: 0.713755|+> +0.700395|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0205162
$\theta_{p'}$=0.000153398, θ_γ=3.14159, $\phi_{p'}$=5.83508, ϕ_γ=1.61218
E_γ=2.04115, $\theta_{\ell'}$=1.55751e-06, $\phi_{\ell'}$=2.69349
Q^2=19.4952, x_B=0.832806, t=-2.99756, W^2=4.7937, y=0.970517
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0206 p_x= -0.0000 p_y= 0.0000 p_z= 2.0206
P' E= 0.9382 p_x= 0.0000 p_y= -0.0000 p_z= 0.0205
q_γ E= 2.0411 p_x= -0.0000 p_y= 0.0000 p_z= -2.0411
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.487\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.487\pi$
normalized amplitude magnitude $|\widetilde{A}|$

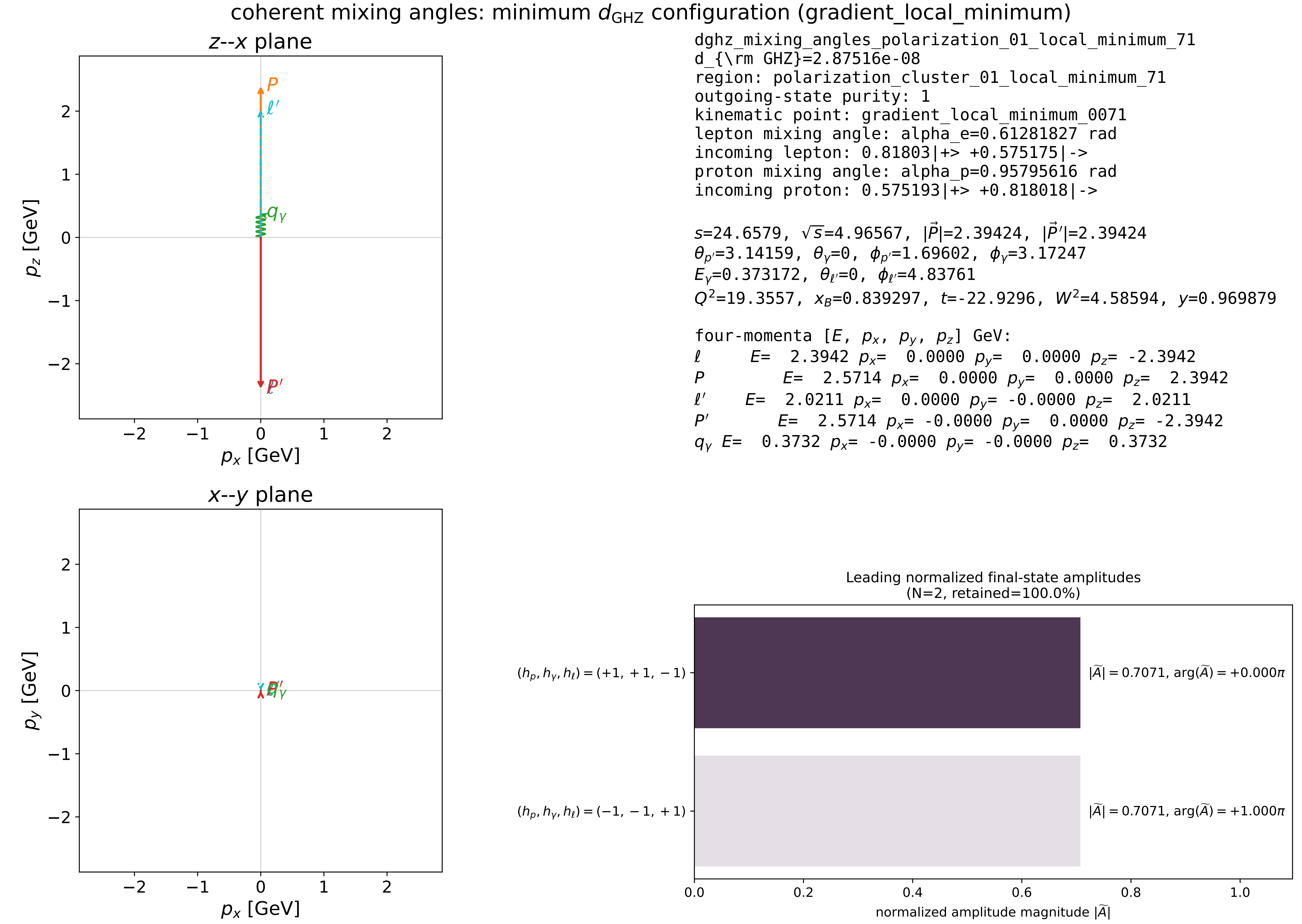

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_57
d_{\rm GHZ}=2.72176e-08
region: polarization_cluster_01_local_minimum_57
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0057
lepton mixing angle: alpha_e=0.84620821 rad
incoming lepton: 0.662827|+> +0.748772|->
proton mixing angle: alpha_p=0.84620257 rad
incoming proton: 0.662831|+> +0.748769|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0736114
$\theta_{p'}$=0.000405073, θ_γ=3.14159, $\phi_{p'}$=5.49139, ϕ_γ=2.28109
E_γ=2.06636, $\theta_{\ell'}$=1.49632e-05, $\phi_{\ell'}$=2.34979
Q^2=19.2262, x_B=0.820972, t=-2.75519, W^2=5.07248, y=0.970924
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 1.9928 p_x= -0.0000 p_y= 0.0000 p_z= 1.9928
P' E= 0.9409 p_x= 0.0000 p_y= -0.0000 p_z= 0.0736
q_γ E= 2.0664 p_x= -0.0000 p_y= 0.0000 p_z= -2.0664
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.274\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.274\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_01_local_minimum_58
d_{\rm GHZ}=2.7258e-08
region: polarization_cluster_01_local_minimum_58
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0058
lepton mixing angle: alpha_e=1.0707816 rad
incoming lepton: 0.479438|+> +0.877576|->
proton mixing angle: alpha_p=1.0707773 rad
incoming proton: 0.479442|+> +0.877573|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=1.43163
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.31472, ϕ_γ=3.87889
E_γ=0.92841, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.393495
Q^2=22.7698, x_B=0.977683, t=-14.0055, W^2=1.39961, y=0.965565
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3600 p_x= 0.0000 p_y= 0.0000 p_z= 2.3600
P' E= 1.7116 p_x= -0.0000 p_y= -0.0000 p_z= -1.4316
q_γ E= 0.9284 p_x= -0.0000 p_y= -0.0000 p_z= -0.9284
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.235\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.765\pi$
normalized amplitude magnitude $|\widetilde{A}|$

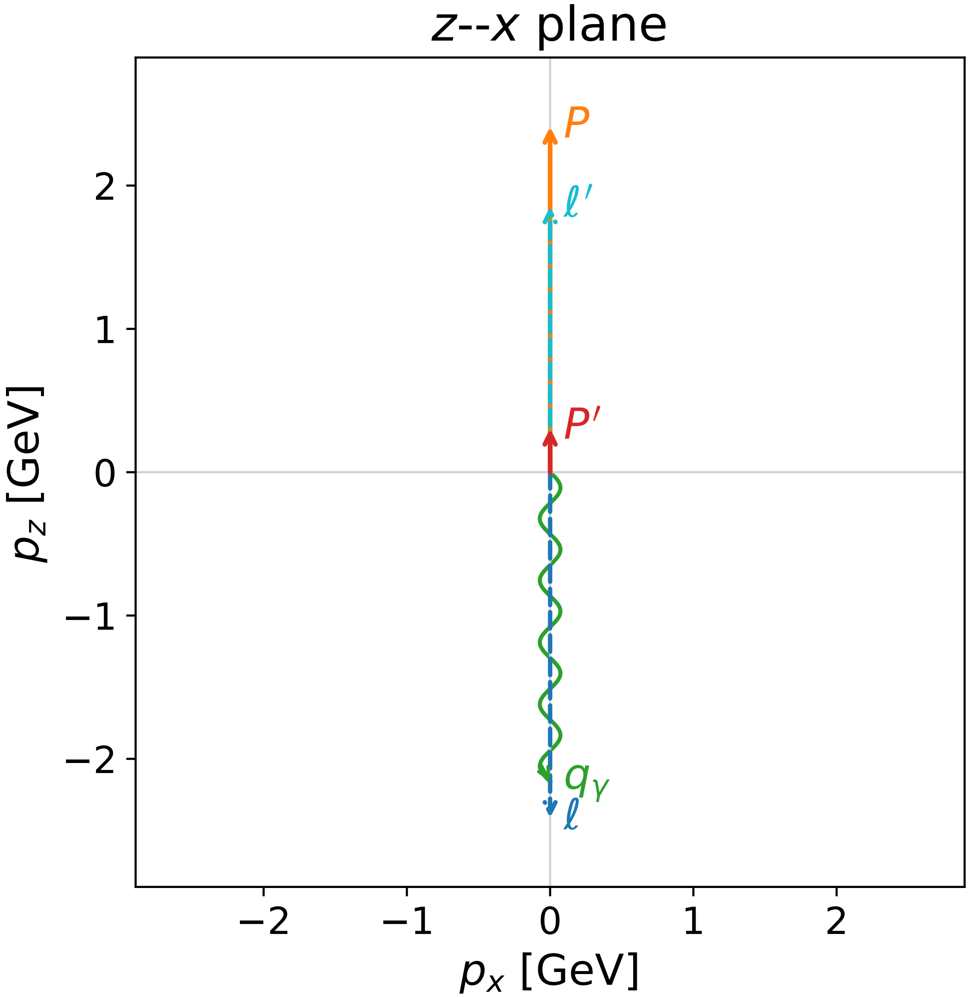

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_61
d_{\rm GHZ}=2.75163e-08
region: polarization_cluster_01_local_minimum_61
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0061
lepton mixing angle: alpha_e=0.88093079 rad
incoming lepton: 0.636433|+> +0.771332|->
proton mixing angle: alpha_p=0.88091145 rad
incoming proton: 0.636448|+> +0.771319|->
s=24.9904, $\sqrt{s}$=4.99903, $|\vec{P}|$=2.41152, $|\vec{P}'|$=0.0704794
$\theta_{p'}$=0.000153398, θ_γ=3.14159, $\phi_{p'}$=0.000314416, ϕ_γ=3.07524
E_γ=2.06444, $\theta_{\ell'}$=5.4221e-06, $\phi_{\ell'}$=3.14191
Q^2=19.2338, x_B=0.821656, t=-2.76826, W^2=5.05464, y=0.970889
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4115 p_x= 0.0000 p_y= 0.0000 p_z= -2.4115
P E= 2.5875 p_x= 0.0000 p_y= 0.0000 p_z= 2.4115
ℓ' E= 1.9940 p_x= -0.0000 p_y= -0.0000 p_z= 1.9940
P' E= 0.9406 p_x= 0.0000 p_y= 0.0000 p_z= 0.0705
q_γ E= 2.0644 p_x= -0.0000 p_y= 0.0000 p_z= -2.0644
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.021\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.021\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_65
d_{\rm GHZ}=2.78329e-08
region: polarization_cluster_01_local_minimum_65
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0065
lepton mixing angle: alpha_e=0.5734809 rad
incoming lepton: 0.840017|+> +0.542559|->
proton mixing angle: alpha_p=0.57346771 rad
incoming proton: 0.840025|+> +0.542548|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.0755206
$\theta_{p'}$=2.17755e-06, θ_γ=3.14159, $\phi_{p'}$=3.18, ϕ_γ=2.24128
E_γ=2.06724, $\theta_{\ell'}$=8.1617e-08, $\phi_{\ell'}$=0.0384033
Q^2=19.2163, x_B=0.820535, t=-2.74677, W^2=5.08278, y=0.970939
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 1.9917 p_x= 0.0000 p_y= 0.0000 p_z= 1.9917
P' E= 0.9410 p_x= -0.0000 p_y= -0.0000 p_z= 0.0755
q_γ E= 2.0672 p_x= -0.0000 p_y= 0.0000 p_z= -2.0672
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.287\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.287\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_69
d_{\rm GHZ}=2.83483e-08
region: polarization_cluster_01_local_minimum_69
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0069
lepton mixing angle: alpha_e=0.82158482 rad
incoming lepton: 0.681062|+> +0.732226|->
proton mixing angle: alpha_p=0.82156325 rad
incoming proton: 0.681077|+> +0.732211|->
s=24.1272, $\sqrt{s}$=4.91194, $|\vec{P}|$=2.36641, $|\vec{P}'|$=0.080731
$\theta_{p'}$=0.00091855, θ_γ=3.14159, $\phi_{p'}$=4.90382, ϕ_γ=3.13744
E_γ=2.0256, $\theta_{\ell'}$=3.81287e-05, $\phi_{\ell'}$=1.76223
Q^2=18.4095, x_B=0.816362, t=-2.6513, W^2=5.02098, y=0.970029
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3664 p_x= 0.0000 p_y= 0.0000 p_z= -2.3664
P E= 2.5455 p_x= 0.0000 p_y= 0.0000 p_z= 2.3664
ℓ' E= 1.9449 p_x= -0.0000 p_y= 0.0001 p_z= 1.9449
P' E= 0.9415 p_x= 0.0000 p_y= -0.0001 p_z= 0.0807
q_γ E= 2.0256 p_x= -0.0000 p_y= 0.0000 p_z= -2.0256
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.001\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.001\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
q_γ
P'
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_71
d_{\rm GHZ}=2.87516e-08
region: polarization_cluster_01_local_minimum_71
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0071
lepton mixing angle: alpha_e=0.61281827 rad
incoming lepton: 0.81803|+> +0.575175|->
proton mixing angle: alpha_p=0.95795616 rad
incoming proton: 0.575193|+> +0.818018|->
s=24.6579, $\sqrt{s}$=4.96567, $|\vec{P}|$=2.39424, $|\vec{P}'|$=2.39424
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.69602, ϕ_γ=3.17247
E_γ=0.373172, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.83761
Q^2=19.3557, x_B=0.839297, t=-22.9296, W^2=4.58594, y=0.969879
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3942 p_x= 0.0000 p_y= 0.0000 p_z= -2.3942
P E= 2.5714 p_x= 0.0000 p_y= 0.0000 p_z= 2.3942
ℓ' E= 2.0211 p_x= 0.0000 p_y= -0.0000 p_z= 2.0211
P' E= 2.5714 p_x= -0.0000 p_y= 0.0000 p_z= -2.3942
q_γ E= 0.3732 p_x= -0.0000 p_y= -0.0000 p_z= 0.3732
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

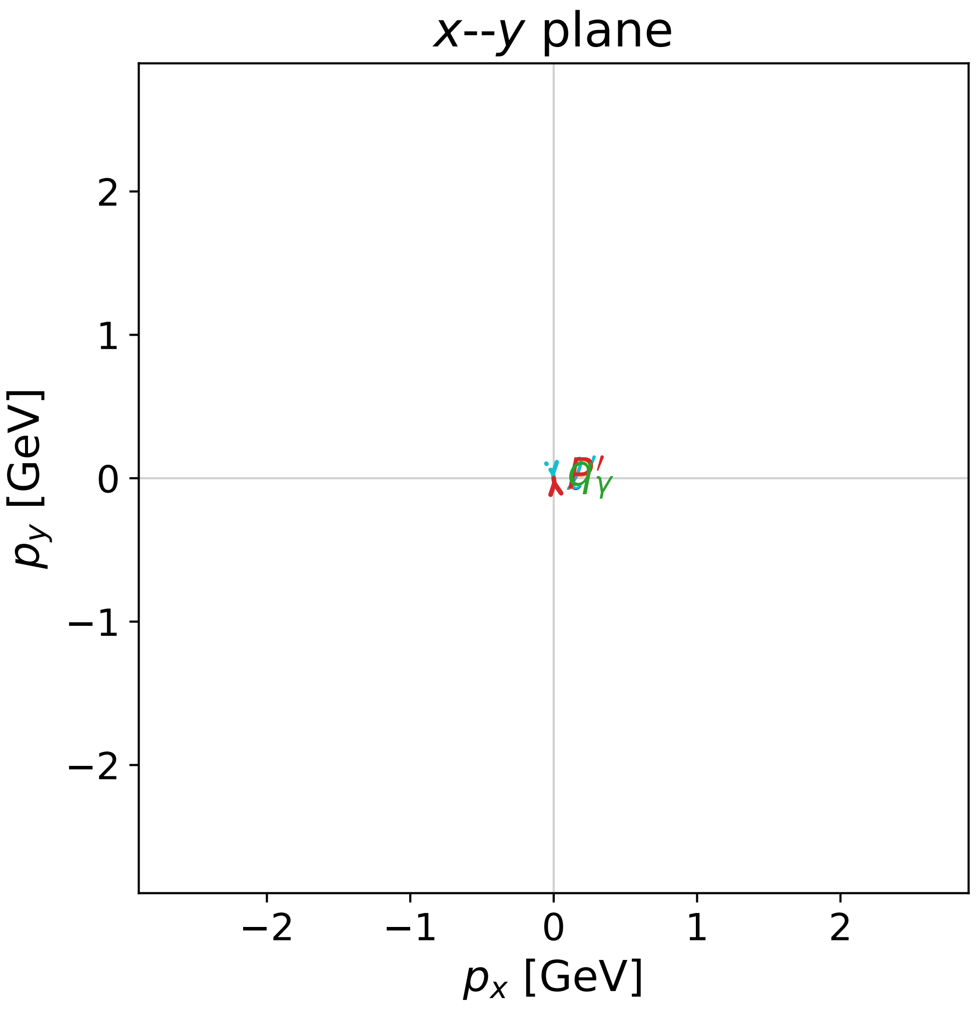
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

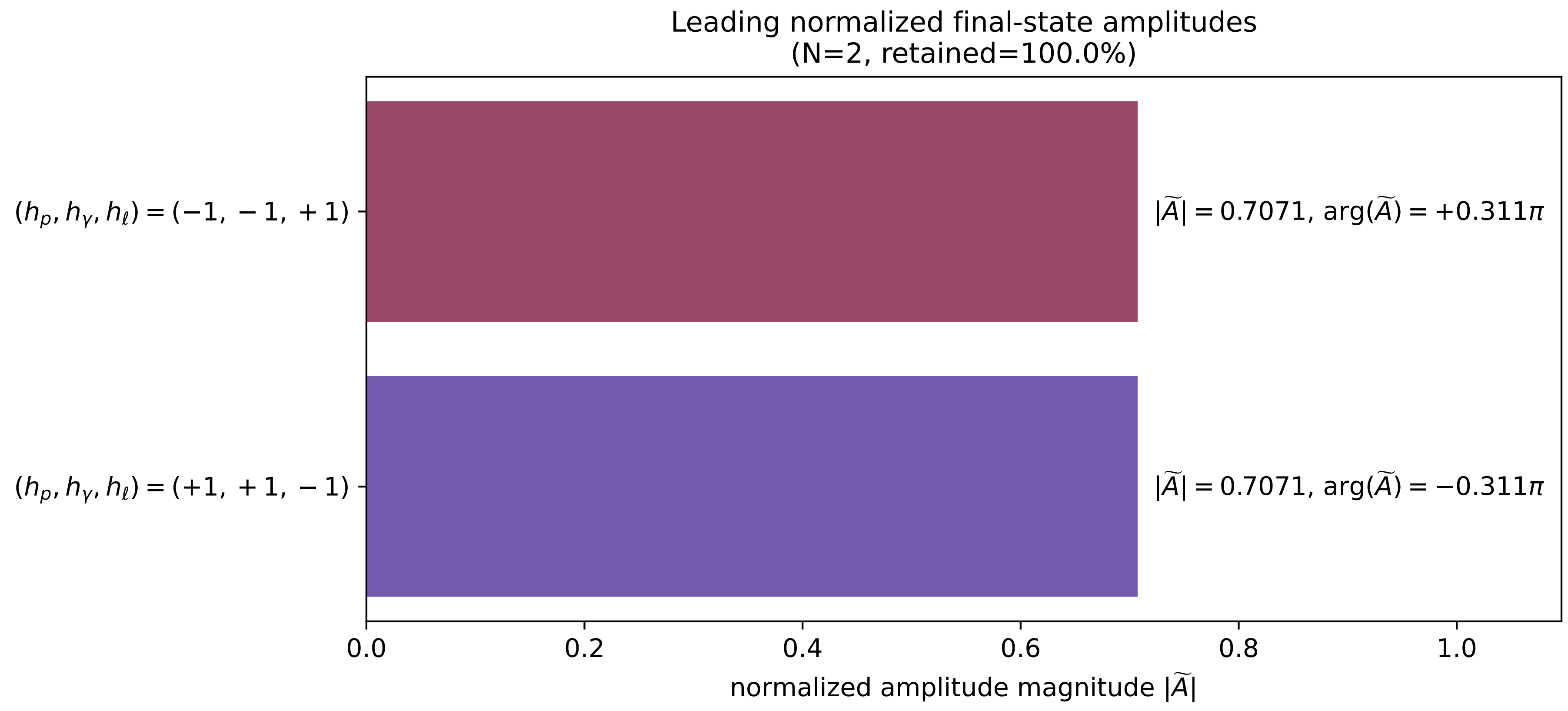


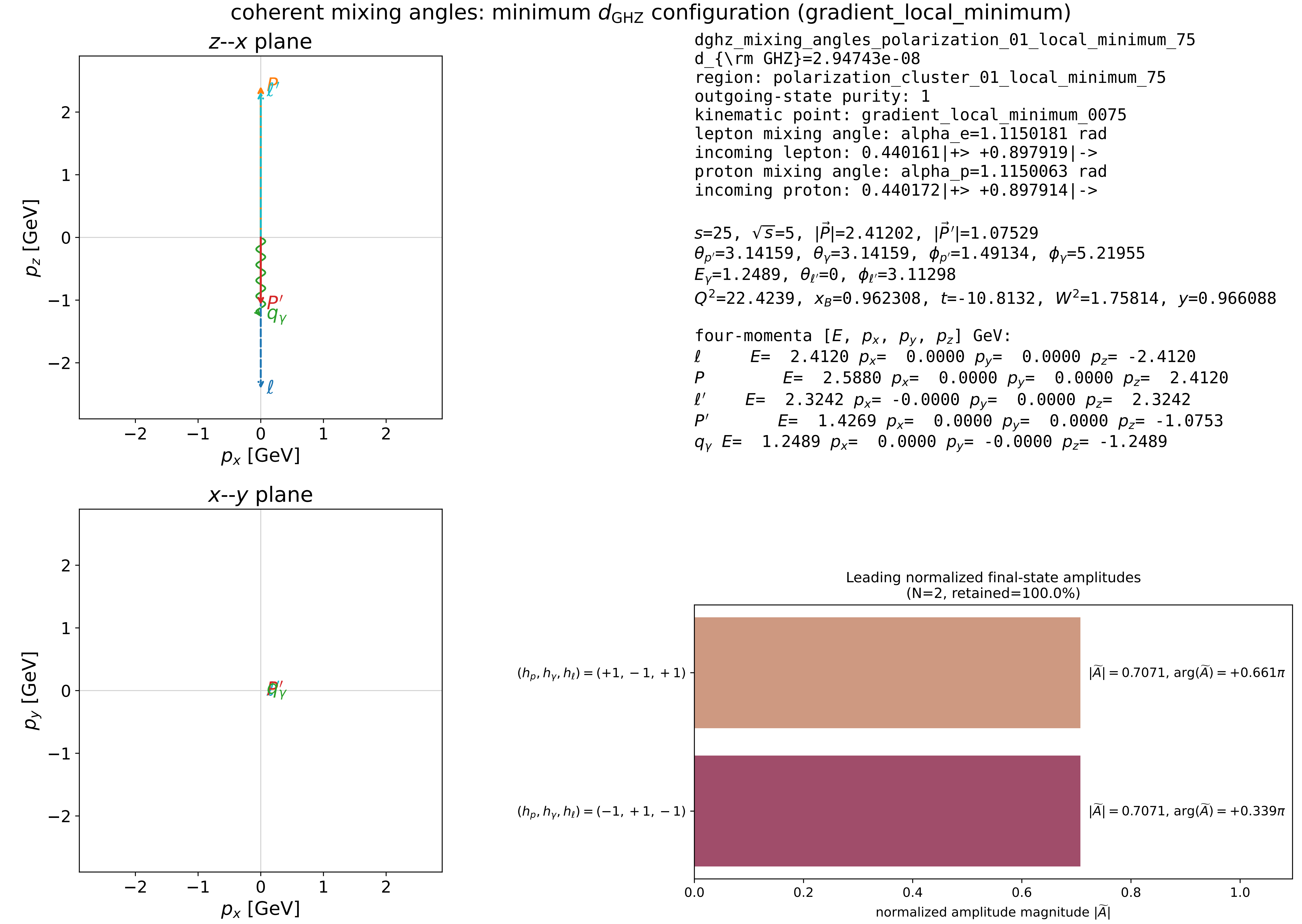

```
dghz_mixing_angles_polarization_01_local_minimum_72
d_{\rm GHZ}=2.90848e-08
region: polarization_cluster_01_local_minimum_72
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0072
lepton mixing angle: alpha_e=0.79108284 rad
incoming lepton: 0.703076|+> +0.711115|->
proton mixing angle: alpha_p=0.7910723 rad
incoming proton: 0.703083|+> +0.711108|->
```

s=24.9857, $\sqrt{s}$=4.99857, $|\vec{P}|$=2.41128, $|\vec{P}'|$=0.305039
$\theta_{p'}$=0.000101732, θ_γ=3.14159, $\phi_{p'}$=1.70875, ϕ_γ=4.11801
E_γ=2.15863, $\theta_{\ell'}$=1.67417e-05, $\phi_{\ell'}$=4.85035
Q^2=17.8781, x_B=0.762282, t=-1.87322, W^2=6.45512, y=0.97293

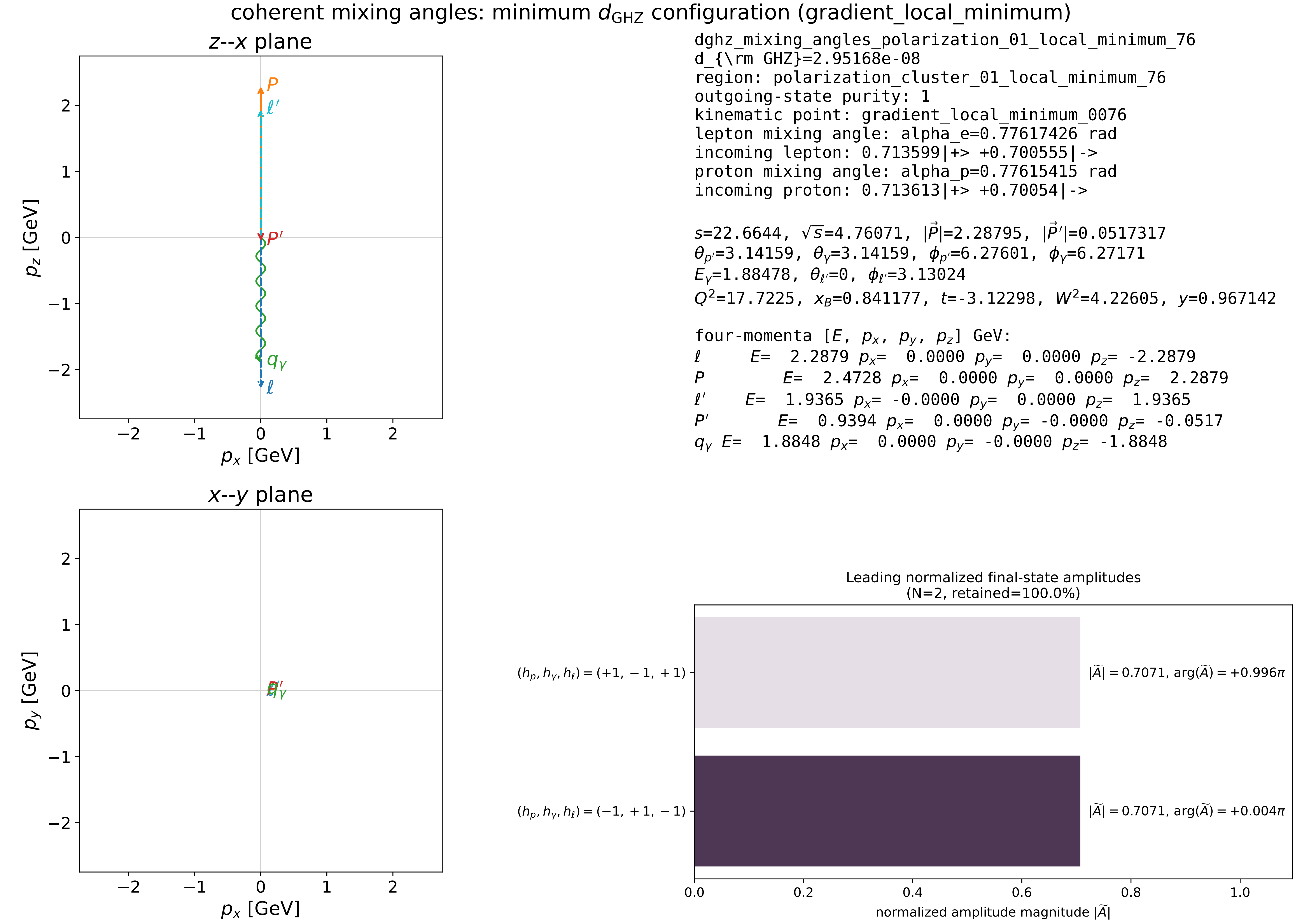
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4113 p_x= 0.0000 p_y= 0.0000 p_z= -2.4113
P E= 2.5873 p_x= 0.0000 p_y= 0.0000 p_z= 2.4113
ℓ' E= 1.8536 p_x= 0.0000 p_y= -0.0000 p_z= 1.8536
P' E= 0.9864 p_x= -0.0000 p_y= 0.0000 p_z= 0.3050
q_γ E= 2.1586 p_x= -0.0000 p_y= -0.0000 p_z= -2.1586

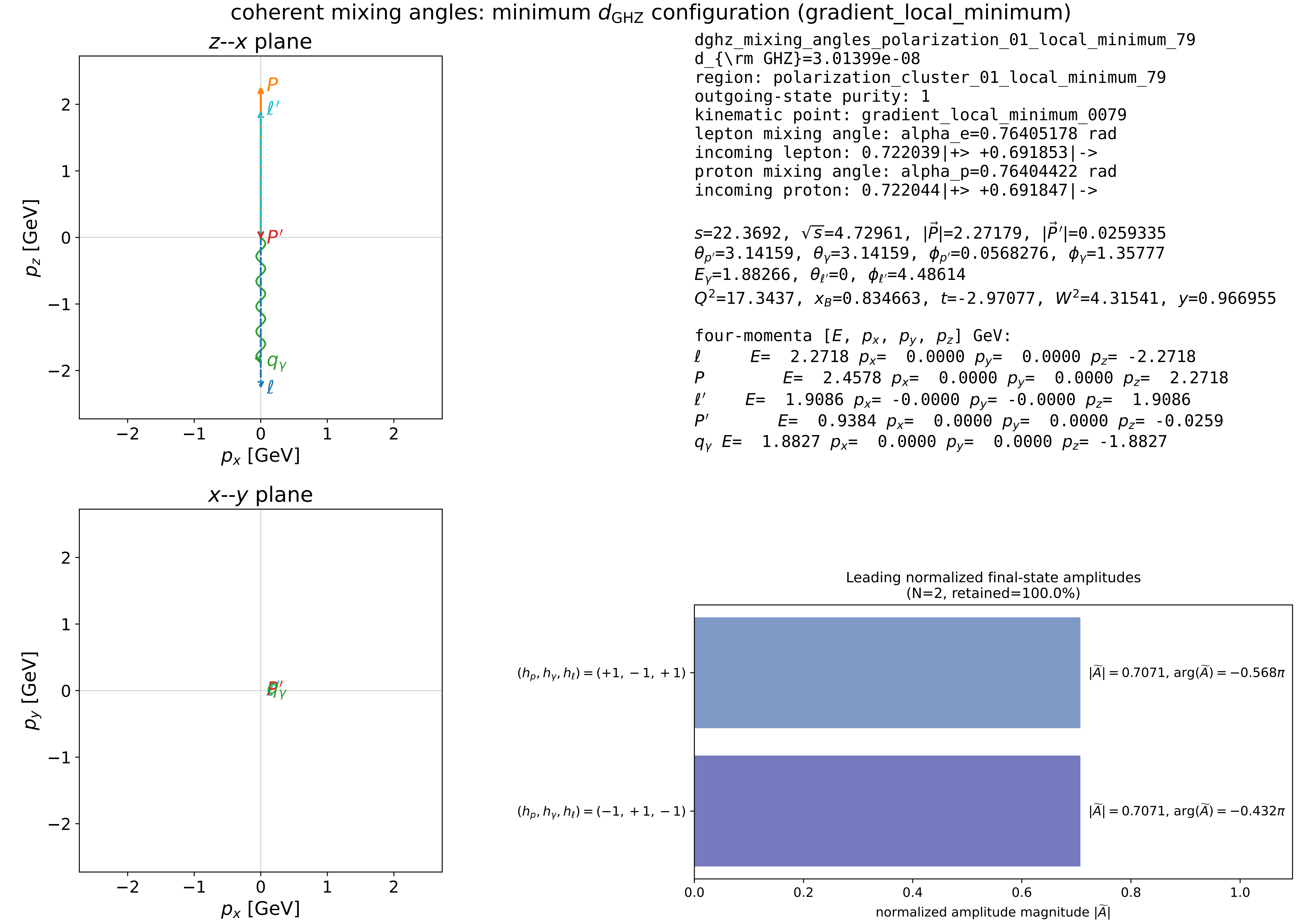




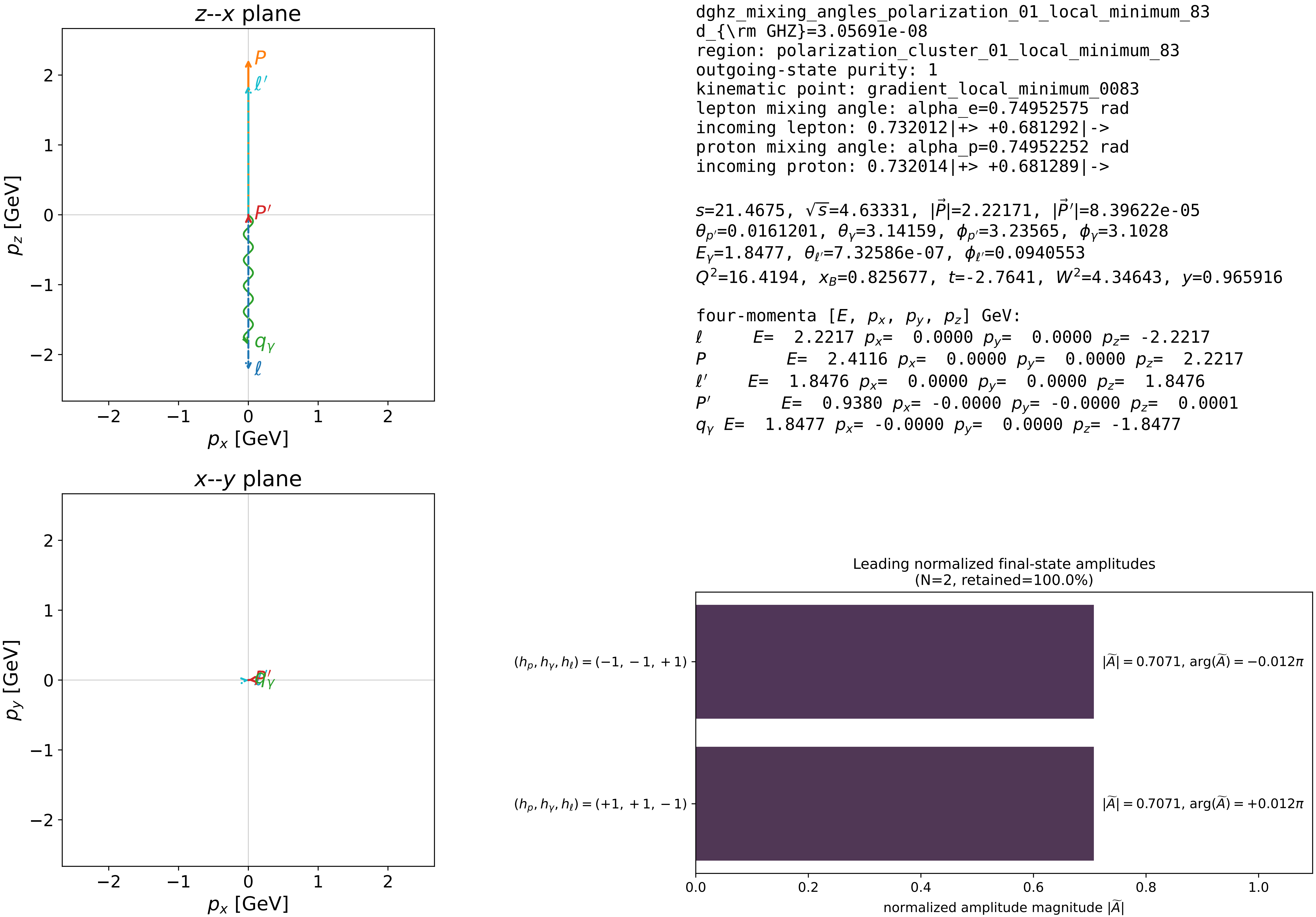


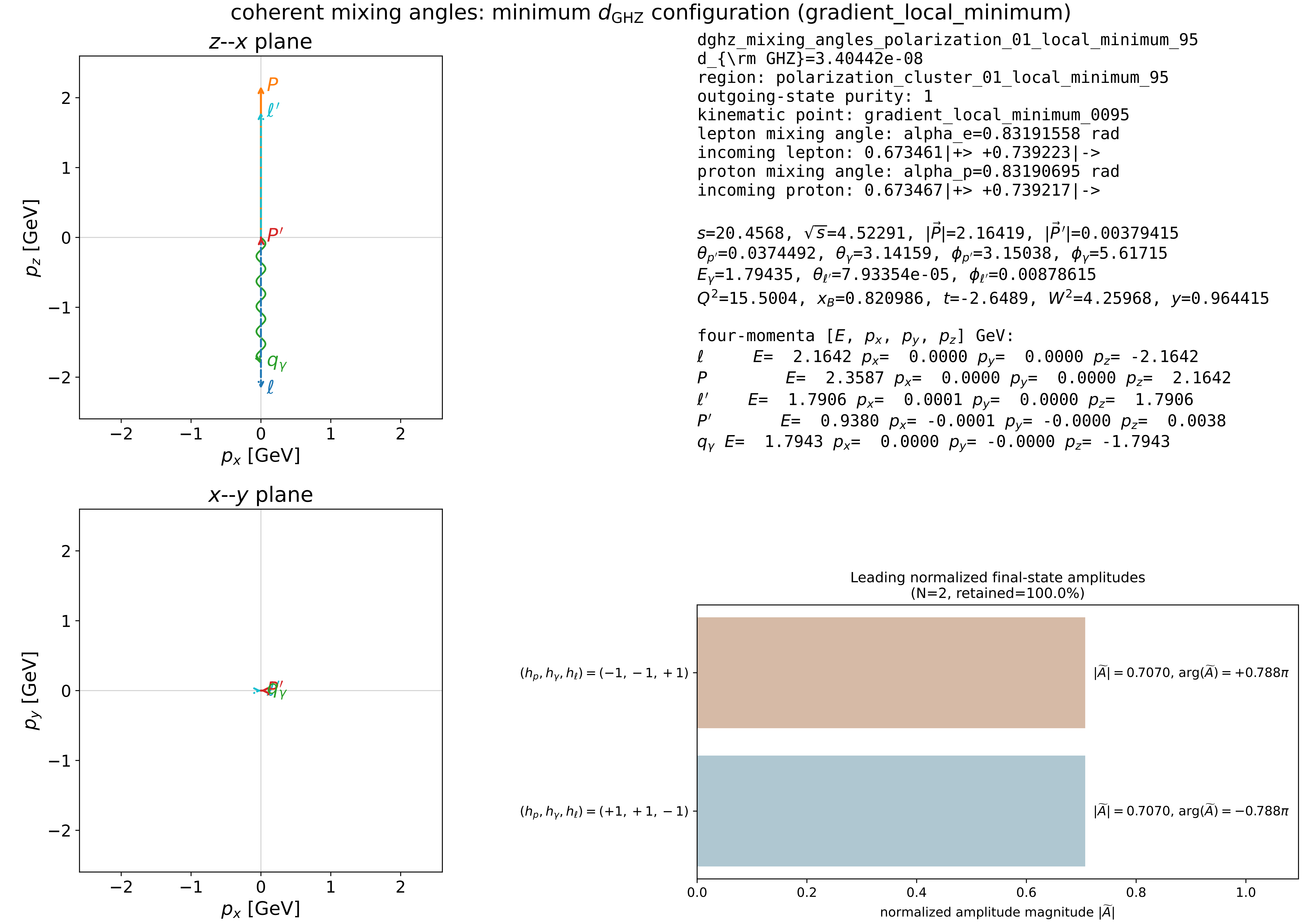
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_75
d_{\rm GHZ}=2.94743e-08
region: polarization_cluster_01_local_minimum_75
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0075
lepton mixing angle: alpha_e=1.1150181 rad
incoming lepton: 0.440161|+> +0.897919|->
proton mixing angle: alpha_p=1.1150063 rad
incoming proton: 0.440172|+> +0.897914|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=1.07529
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=1.49134, ϕ_γ=5.21955
E_γ=1.2489, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.11298
Q^2=22.4239, x_B=0.962308, t=-10.8132, W^2=1.75814, y=0.966088
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.3242 p_x= -0.0000 p_y= 0.0000 p_z= 2.3242
P' E= 1.4269 p_x= 0.0000 p_y= 0.0000 p_z= -1.0753
q_γ E= 1.2489 p_x= 0.0000 p_y= -0.0000 p_z= -1.2489
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.661\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.339\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_76
d_{\rm GHZ}=2.95168e-08
region: polarization_cluster_01_local_minimum_76
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0076
lepton mixing angle: alpha_e=0.77617426 rad
incoming lepton: 0.713599|+> +0.700555|->
proton mixing angle: alpha_p=0.77615415 rad
incoming proton: 0.713613|+> +0.70054|->
s=22.6644, $\sqrt{s}$=4.76071, $|\vec{P}|$=2.28795, $|\vec{P}'|$=0.0517317
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=6.27601, ϕ_γ=6.27171
E_γ=1.88478, $\theta_{\ell'}$=0, $\phi_{\ell'}$=3.13024
Q^2=17.7225, x_B=0.841177, t=-3.12298, W^2=4.22605, y=0.967142
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2879 p_x= 0.0000 p_y= 0.0000 p_z= -2.2879
P E= 2.4728 p_x= 0.0000 p_y= 0.0000 p_z= 2.2879
ℓ' E= 1.9365 p_x= -0.0000 p_y= 0.0000 p_z= 1.9365
P' E= 0.9394 p_x= 0.0000 p_y= -0.0000 p_z= -0.0517
q_γ E= 1.8848 p_x= 0.0000 p_y= -0.0000 p_z= -1.8848
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.996\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.004\pi$
normalized amplitude magnitude $|\widetilde{A}|$

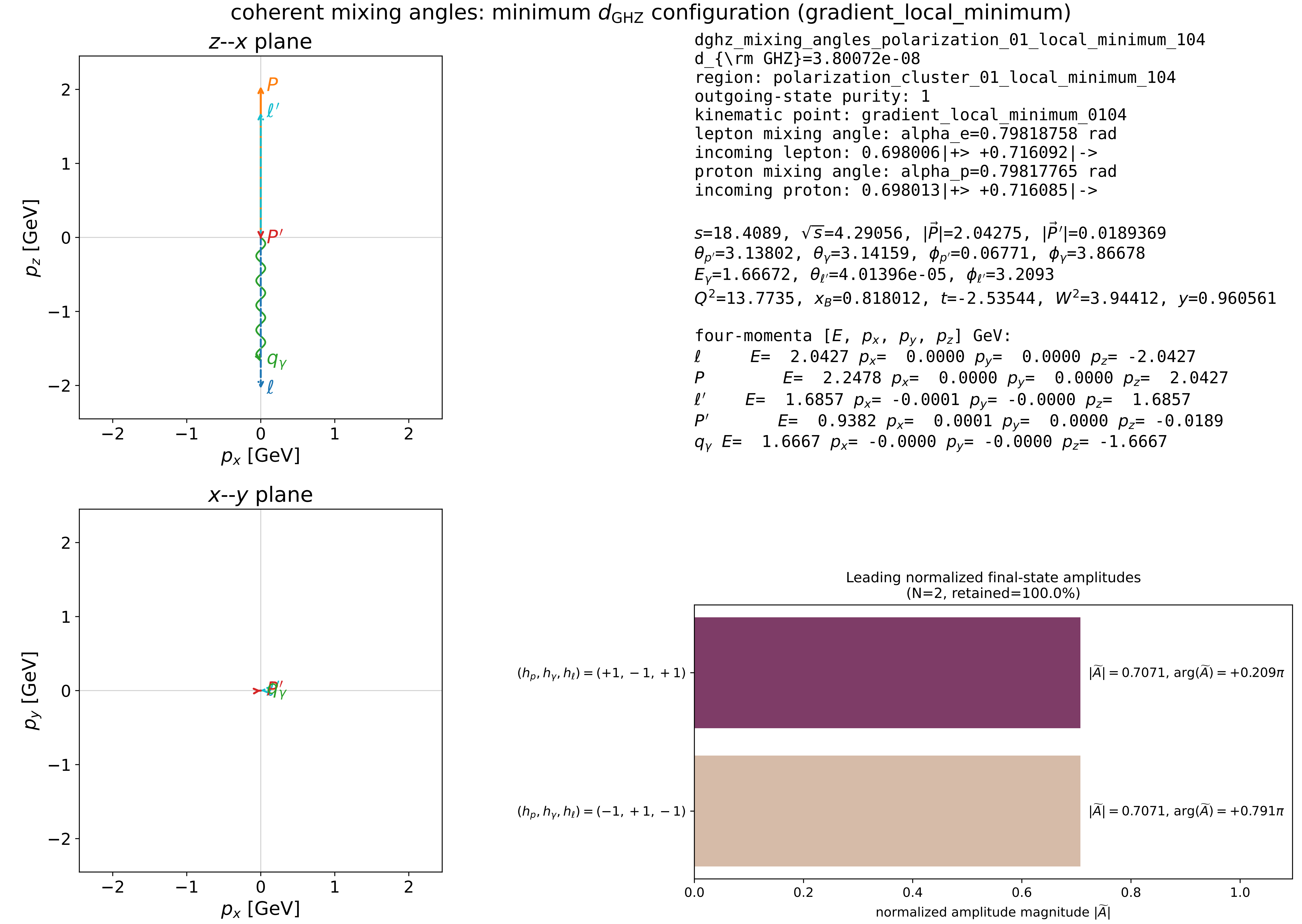

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_79
d_{\rm GHZ}=3.01399e-08
region: polarization_cluster_01_local_minimum_79
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0079
lepton mixing angle: alpha_e=0.76405178 rad
incoming lepton: 0.722039|+> +0.691853|->
proton mixing angle: alpha_p=0.76404422 rad
incoming proton: 0.722044|+> +0.691847|->
s=22.3692, $\sqrt{s}$=4.72961, $|\vec{P}|$=2.27179, $|\vec{P}'|$=0.0259335
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=0.0568276, ϕ_γ=1.35777
E_γ=1.88266, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.48614
Q^2=17.3437, x_B=0.834663, t=-2.97077, W^2=4.31541, y=0.966955
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2718 p_x= 0.0000 p_y= 0.0000 p_z= -2.2718
P E= 2.4578 p_x= 0.0000 p_y= 0.0000 p_z= 2.2718
ℓ' E= 1.9086 p_x= -0.0000 p_y= -0.0000 p_z= 1.9086
P' E= 0.9384 p_x= 0.0000 p_y= 0.0000 p_z= -0.0259
q_γ E= 1.8827 p_x= 0.0000 p_y= 0.0000 p_z= -1.8827
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.568\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.432\pi$
normalized amplitude magnitude $|\widetilde{A}|$

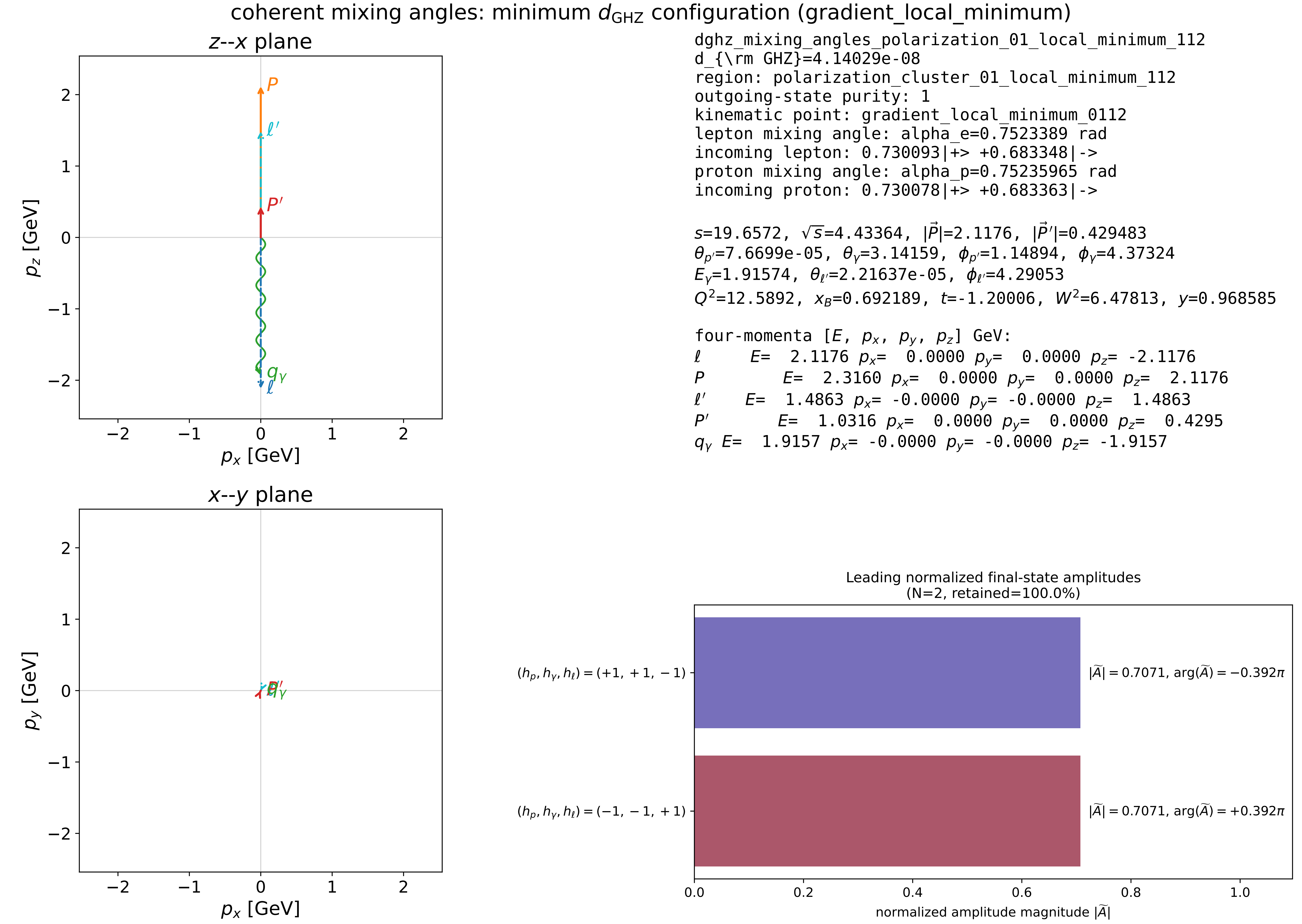
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



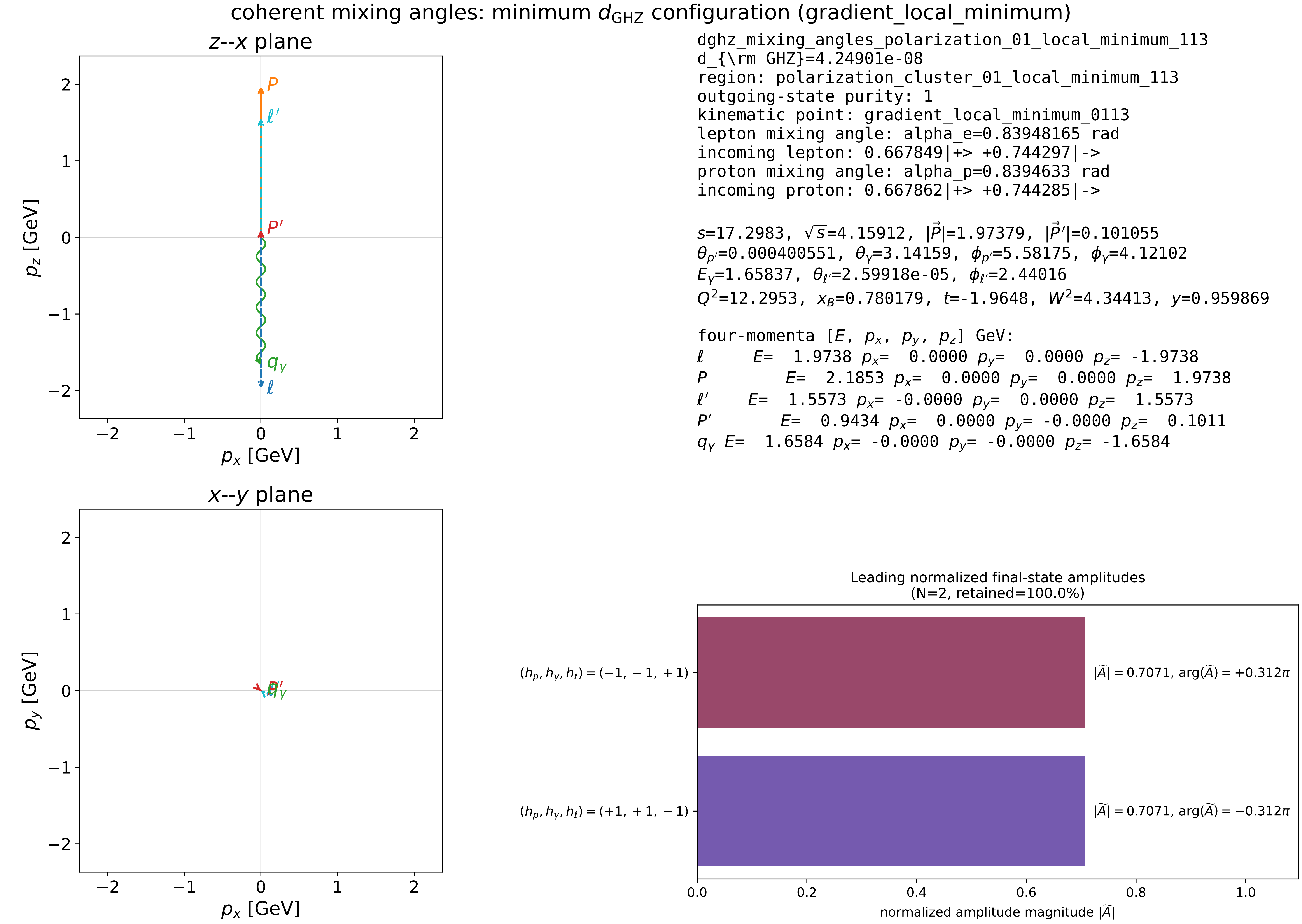


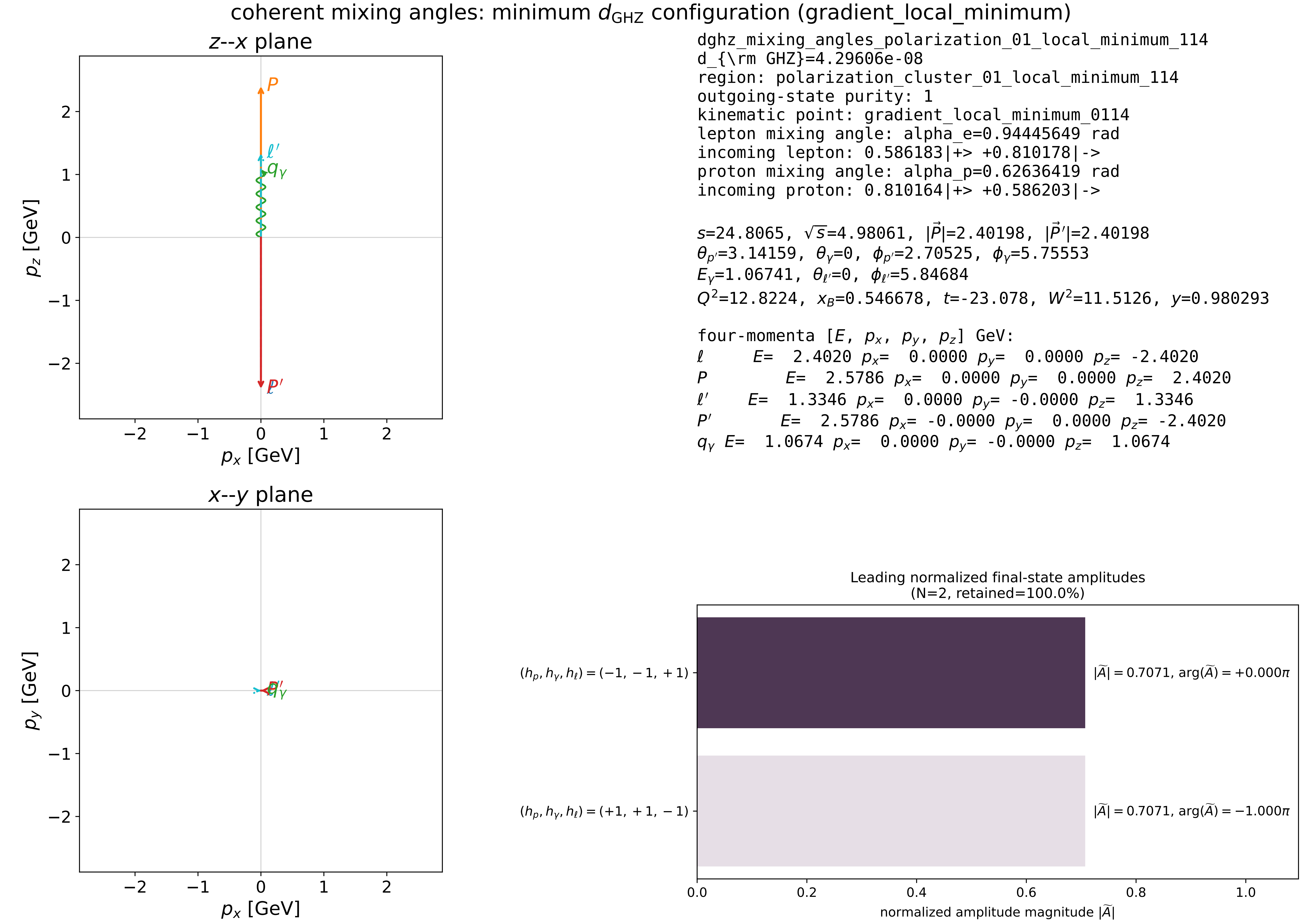
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_95
d_{\rm GHZ}=3.40442e-08
region: polarization_cluster_01_local_minimum_95
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0095
lepton mixing angle: alpha_e=0.83191558 rad
incoming lepton: 0.673461|+> +0.739223|->
proton mixing angle: alpha_p=0.83190695 rad
incoming proton: 0.673467|+> +0.739217|->
s=20.4568, $\sqrt{s}$=4.52291, $|\vec{P}|$=2.16419, $|\vec{P}'|$=0.00379415
$\theta_{p'}$=0.0374492, θ_γ=3.14159, $\phi_{p'}$=3.15038, ϕ_γ=5.61715
E_γ=1.79435, $\theta_{\ell'}$=7.93354e-05, $\phi_{\ell'}$=0.00878615
Q^2=15.5004, x_B=0.820986, t=-2.6489, W^2=4.25968, y=0.964415
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1642 p_x= 0.0000 p_y= 0.0000 p_z= -2.1642
P E= 2.3587 p_x= 0.0000 p_y= 0.0000 p_z= 2.1642
ℓ' E= 1.7906 p_x= 0.0001 p_y= 0.0000 p_z= 1.7906
P' E= 0.9380 p_x= -0.0001 p_y= -0.0000 p_z= 0.0038
q_γ E= 1.7943 p_x= 0.0000 p_y= -0.0000 p_z= -1.7943
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = +0.788\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = -0.788\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_104
d_{\rm GHZ}=3.80072e-08
region: polarization_cluster_01_local_minimum_104
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0104
lepton mixing angle: alpha_e=0.79818758 rad
incoming lepton: 0.698006|+> +0.716092|->
proton mixing angle: alpha_p=0.79817765 rad
incoming proton: 0.698013|+> +0.716085|->
s=18.4089, $\sqrt{s}$=4.29056, $|\vec{P}|$=2.04275, $|\vec{P}'|$=0.0189369
$\theta_{p'}$=3.13802, θ_γ=3.14159, $\phi_{p'}$=0.06771, ϕ_γ=3.86678
E_γ=1.66672, $\theta_{\ell'}$=4.01396e-05, $\phi_{\ell'}$=3.2093
Q^2=13.7735, x_B=0.818012, t=-2.53544, W^2=3.94412, y=0.960561
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.0427 p_x= 0.0000 p_y= 0.0000 p_z= -2.0427
P E= 2.2478 p_x= 0.0000 p_y= 0.0000 p_z= 2.0427
ℓ' E= 1.6857 p_x= -0.0001 p_y= -0.0000 p_z= 1.6857
P' E= 0.9382 p_x= 0.0001 p_y= 0.0000 p_z= -0.0189
q_γ E= 1.6667 p_x= -0.0000 p_y= -0.0000 p_z= -1.6667
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.209\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.791\pi$
normalized amplitude magnitude $|\widetilde{A}|$

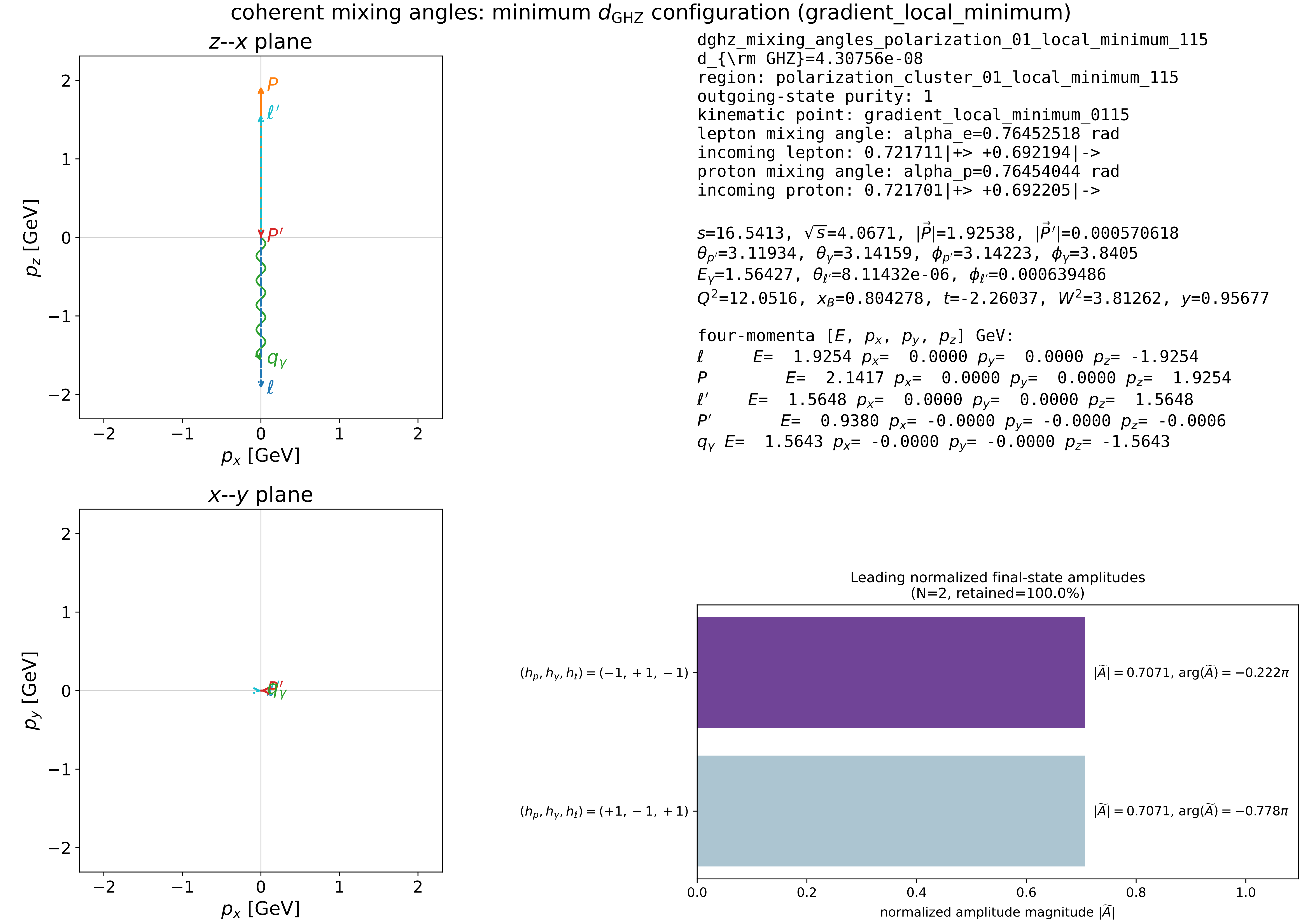


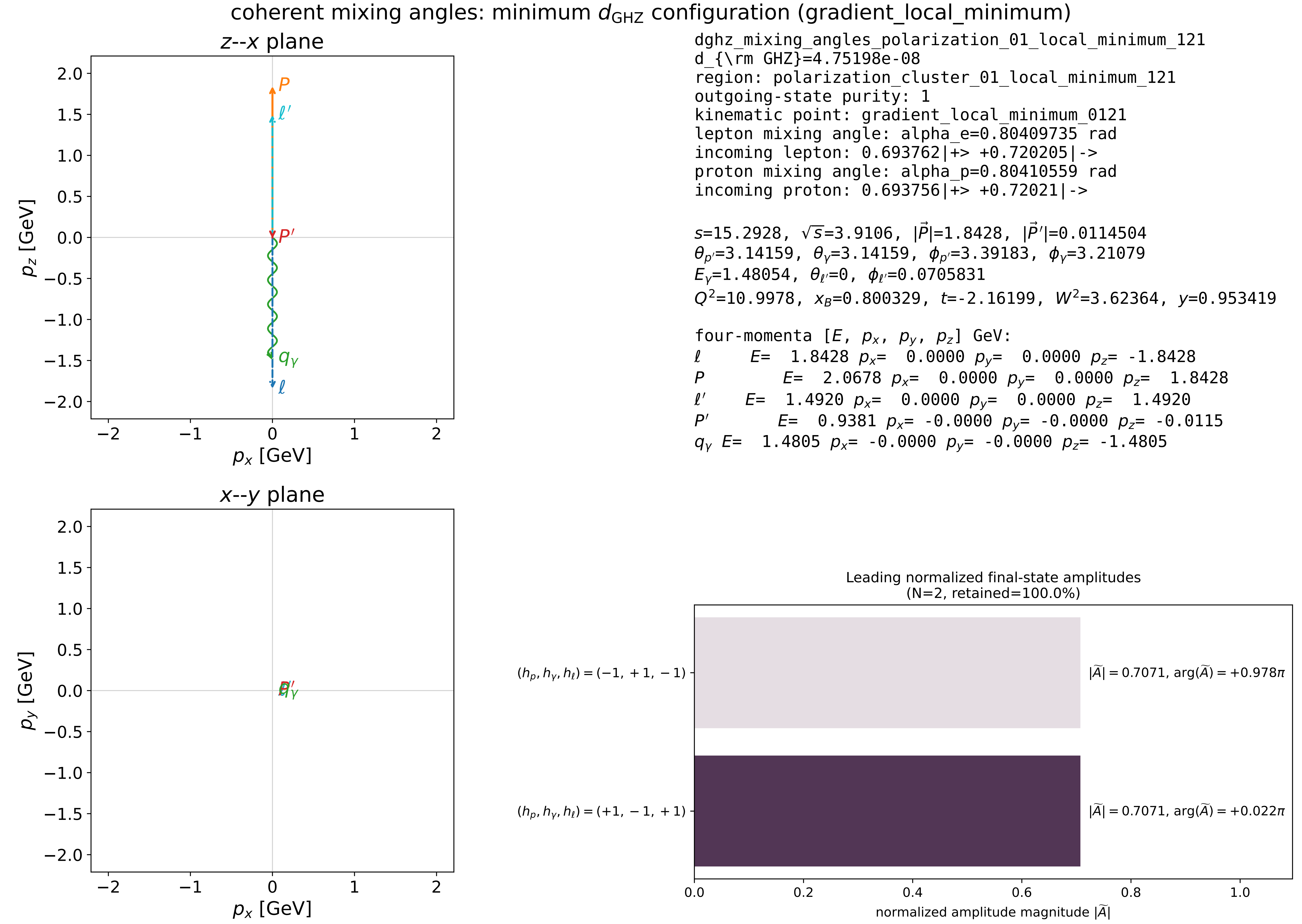
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_112
d_{\rm GHZ}=4.14029e-08
region: polarization_cluster_01_local_minimum_112
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0112
lepton mixing angle: alpha_e=0.7523389 rad
incoming lepton: 0.730093|+> +0.683348|->
proton mixing angle: alpha_p=0.75235965 rad
incoming proton: 0.730078|+> +0.683363|->
s=19.6572, $\sqrt{s}$=4.43364, $|\vec{P}|$=2.1176, $|\vec{P}'|$=0.429483
$\theta_{p'}$=7.6699e-05, θ_γ=3.14159, $\phi_{p'}$=1.14894, ϕ_γ=4.37324
E_γ=1.91574, $\theta_{\ell'}$=2.21637e-05, $\phi_{\ell'}$=4.29053
Q^2=12.5892, x_B=0.692189, t=-1.20006, W^2=6.47813, y=0.968585
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1176 p_x= 0.0000 p_y= 0.0000 p_z= -2.1176
P E= 2.3160 p_x= 0.0000 p_y= 0.0000 p_z= 2.1176
ℓ' E= 1.4863 p_x= -0.0000 p_y= -0.0000 p_z= 1.4863
P' E= 1.0316 p_x= 0.0000 p_y= 0.0000 p_z= 0.4295
q_γ E= 1.9157 p_x= -0.0000 p_y= -0.0000 p_z= -1.9157
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.392\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.392\pi$
normalized amplitude magnitude $|\widetilde{A}|$

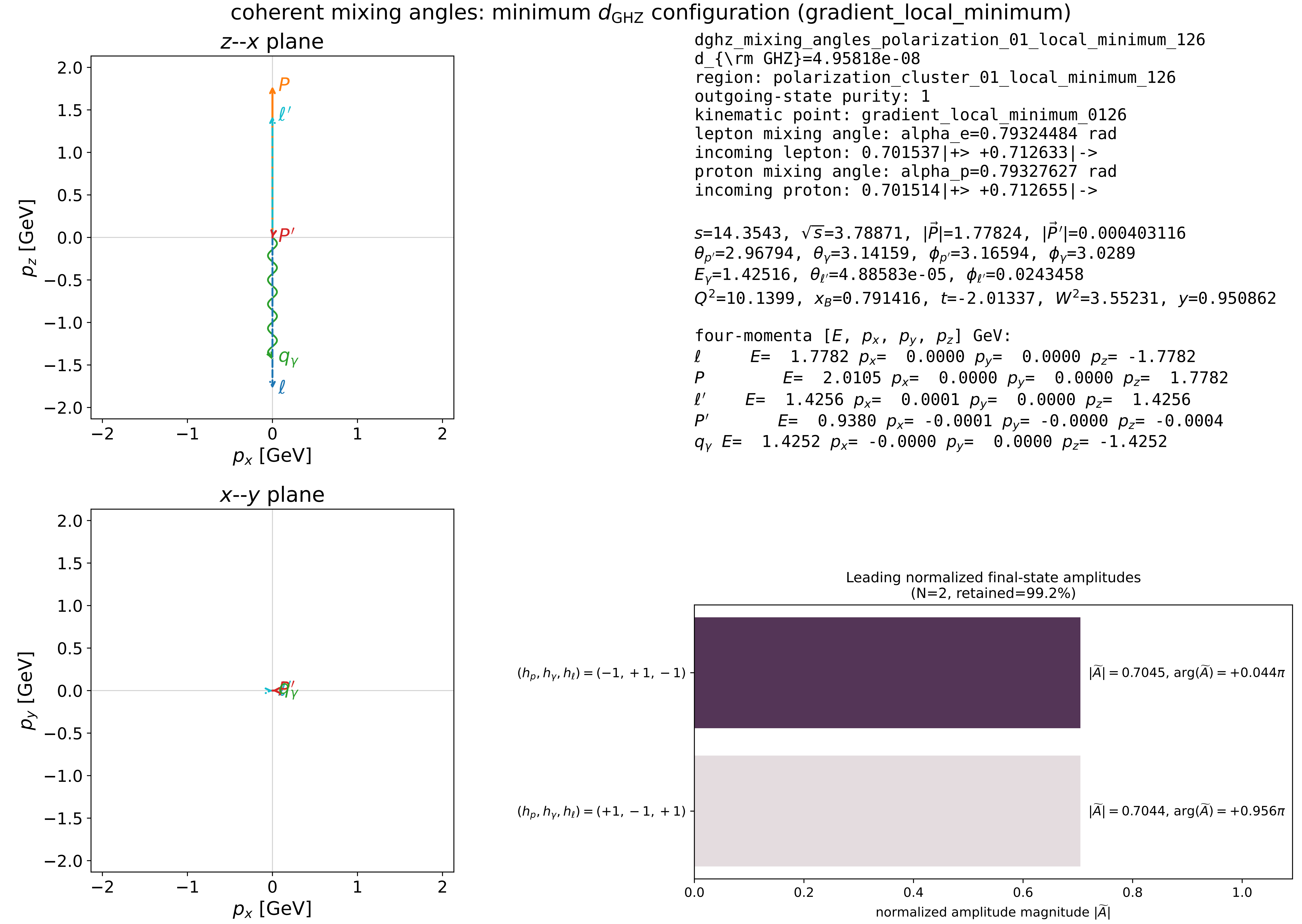

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_113
d_{\rm GHZ}=4.24901e-08
region: polarization_cluster_01_local_minimum_113
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0113
lepton mixing angle: alpha_e=0.83948165 rad
incoming lepton: 0.667849|+> +0.744297|->
proton mixing angle: alpha_p=0.8394633 rad
incoming proton: 0.667862|+> +0.744285|->
s=17.2983, $\sqrt{s}$=4.15912, $|\vec{P}|$=1.97379, $|\vec{P}'|$=0.101055
$\theta_{p'}$=0.000400551, θ_γ=3.14159, $\phi_{p'}$=5.58175, ϕ_γ=4.12102
E_γ=1.65837, $\theta_{\ell'}$=2.59918e-05, $\phi_{\ell'}$=2.44016
Q^2=12.2953, x_B=0.780179, t=-1.9648, W^2=4.34413, y=0.959869
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9738 p_x= 0.0000 p_y= 0.0000 p_z= -1.9738
P E= 2.1853 p_x= 0.0000 p_y= 0.0000 p_z= 1.9738
ℓ' E= 1.5573 p_x= -0.0000 p_y= 0.0000 p_z= 1.5573
P' E= 0.9434 p_x= 0.0000 p_y= -0.0000 p_z= 0.1011
q_γ E= 1.6584 p_x= -0.0000 p_y= -0.0000 p_z= -1.6584
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.312\pi$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.312\pi$
normalized amplitude magnitude $|\widetilde{A}|$



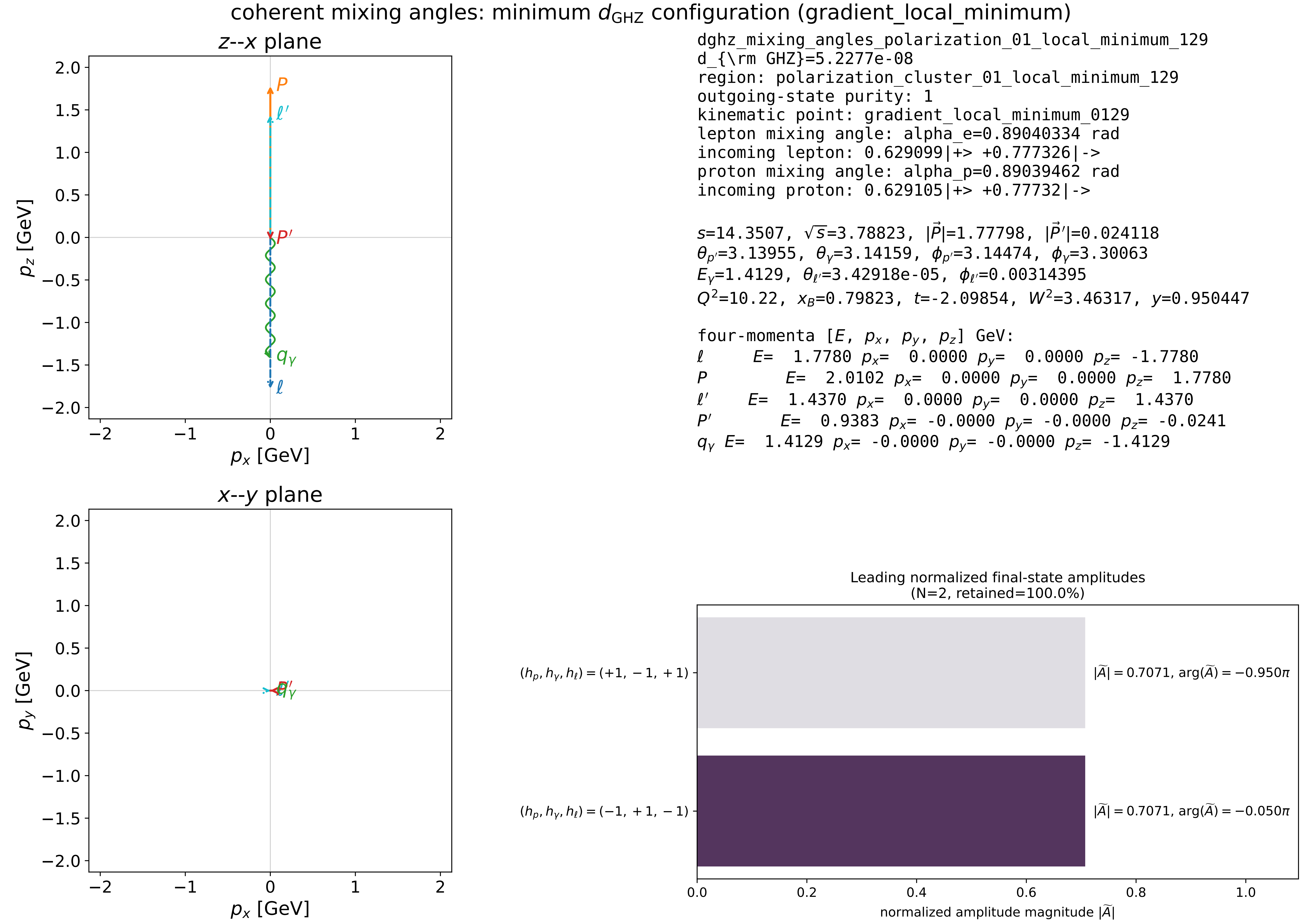
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
q_γ
P'
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_114
d_{\rm GHZ}=4.29606e-08
region: polarization_cluster_01_local_minimum_114
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0114
lepton mixing angle: alpha_e=0.94445649 rad
incoming lepton: 0.586183|+> +0.810178|->
proton mixing angle: alpha_p=0.62636419 rad
incoming proton: 0.810164|+> +0.586203|->
s=24.8065, $\sqrt{s}$=4.98061, $|\vec{P}|$=2.40198, $|\vec{P}'|$=2.40198
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.70525, ϕ_γ=5.75553
E_γ=1.06741, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.84684
Q^2=12.8224, x_B=0.546678, t=-23.078, W^2=11.5126, y=0.980293
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4020 p_x= 0.0000 p_y= 0.0000 p_z= -2.4020
P E= 2.5786 p_x= 0.0000 p_y= 0.0000 p_z= 2.4020
ℓ' E= 1.3346 p_x= 0.0000 p_y= -0.0000 p_z= 1.3346
P' E= 2.5786 p_x= -0.0000 p_y= 0.0000 p_z= -2.4020
q_γ E= 1.0674 p_x= 0.0000 p_y= -0.0000 p_z= 1.0674
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

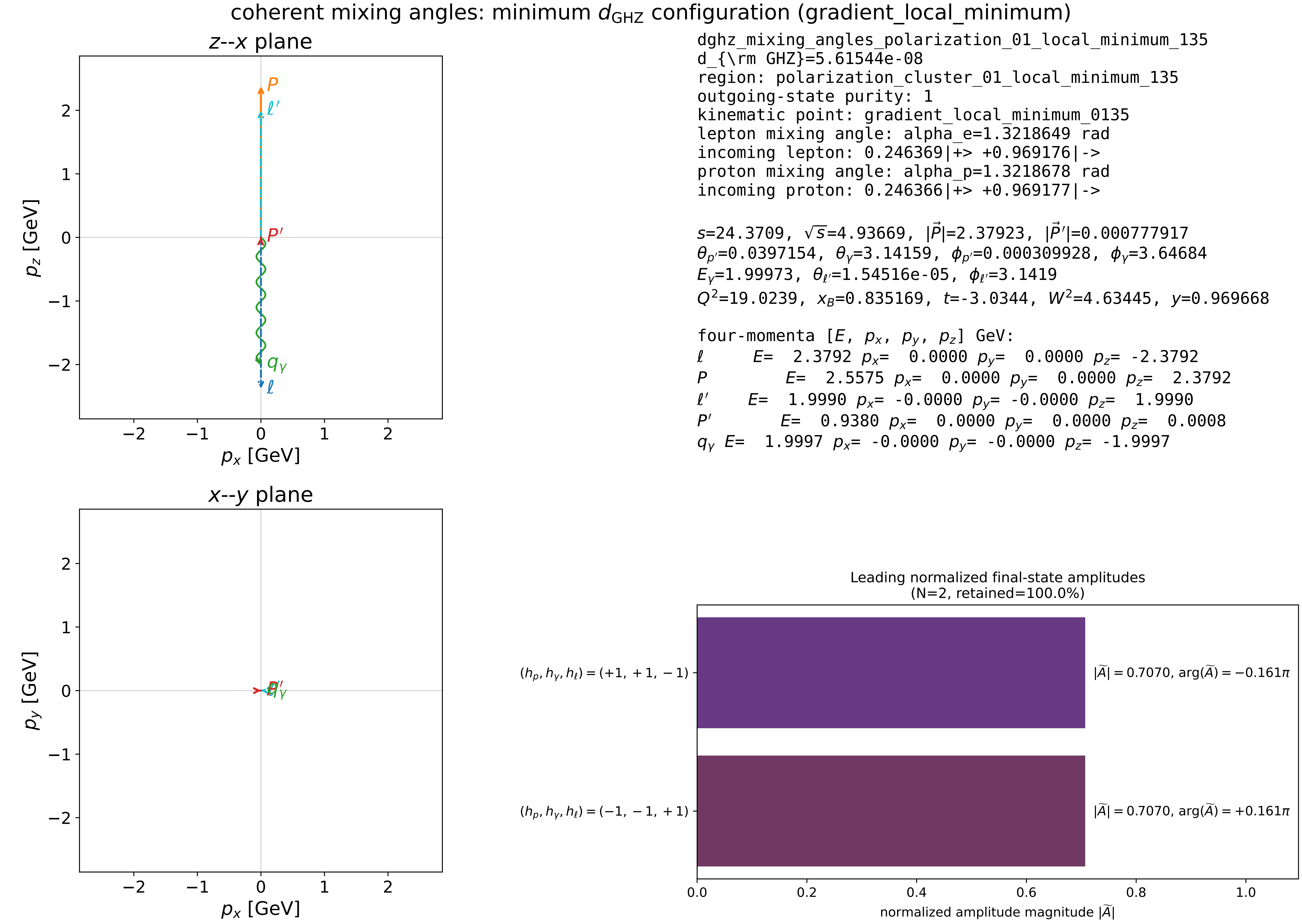

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_115
d_{\rm GHZ}=4.30756e-08
region: polarization_cluster_01_local_minimum_115
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0115
lepton mixing angle: alpha_e=0.76452518 rad
incoming lepton: 0.721711|+> +0.692194|->
proton mixing angle: alpha_p=0.76454044 rad
incoming proton: 0.721701|+> +0.692205|->
s=16.5413, $\sqrt{s}$=4.0671, $|\vec{P}|$=1.92538, $|\vec{P}'|$=0.000570618
$\theta_{p'}$=3.11934, θ_γ=3.14159, $\phi_{p'}$=3.14223, ϕ_γ=3.8405
E_γ=1.56427, $\theta_{\ell'}$=8.11432e-06, $\phi_{\ell'}$=0.000639486
Q^2=12.0516, x_B=0.804278, t=-2.26037, W^2=3.81262, y=0.95677
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9254 p_x= 0.0000 p_y= 0.0000 p_z= -1.9254
P E= 2.1417 p_x= 0.0000 p_y= 0.0000 p_z= 1.9254
ℓ' E= 1.5648 p_x= 0.0000 p_y= 0.0000 p_z= 1.5648
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0006
q_γ E= 1.5643 p_x= -0.0000 p_y= -0.0000 p_z= -1.5643
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.222\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.778\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_121
d_{\rm GHZ}=4.75198e-08
region: polarization_cluster_01_local_minimum_121
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0121
lepton mixing angle: alpha_e=0.80409735 rad
incoming lepton: 0.693762|+> +0.720205|->
proton mixing angle: alpha_p=0.80410559 rad
incoming proton: 0.693756|+> +0.72021|->
s=15.2928, $\sqrt{s}$=3.9106, $|\vec{P}|$=1.8428, $|\vec{P}'|$=0.0114504
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=3.39183, ϕ_γ=3.21079
E_γ=1.48054, $\theta_{\ell'}$=0, $\phi_{\ell'}$=0.0705831
Q^2=10.9978, x_B=0.800329, t=-2.16199, W^2=3.62364, y=0.953419
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.8428 p_x= 0.0000 p_y= 0.0000 p_z= -1.8428
P E= 2.0678 p_x= 0.0000 p_y= 0.0000 p_z= 1.8428
ℓ' E= 1.4920 p_x= 0.0000 p_y= 0.0000 p_z= 1.4920
P' E= 0.9381 p_x= -0.0000 p_y= -0.0000 p_z= -0.0115
q_γ E= 1.4805 p_x= -0.0000 p_y= -0.0000 p_z= -1.4805
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.978\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.022\pi$
normalized amplitude magnitude $|\widetilde{A}|$

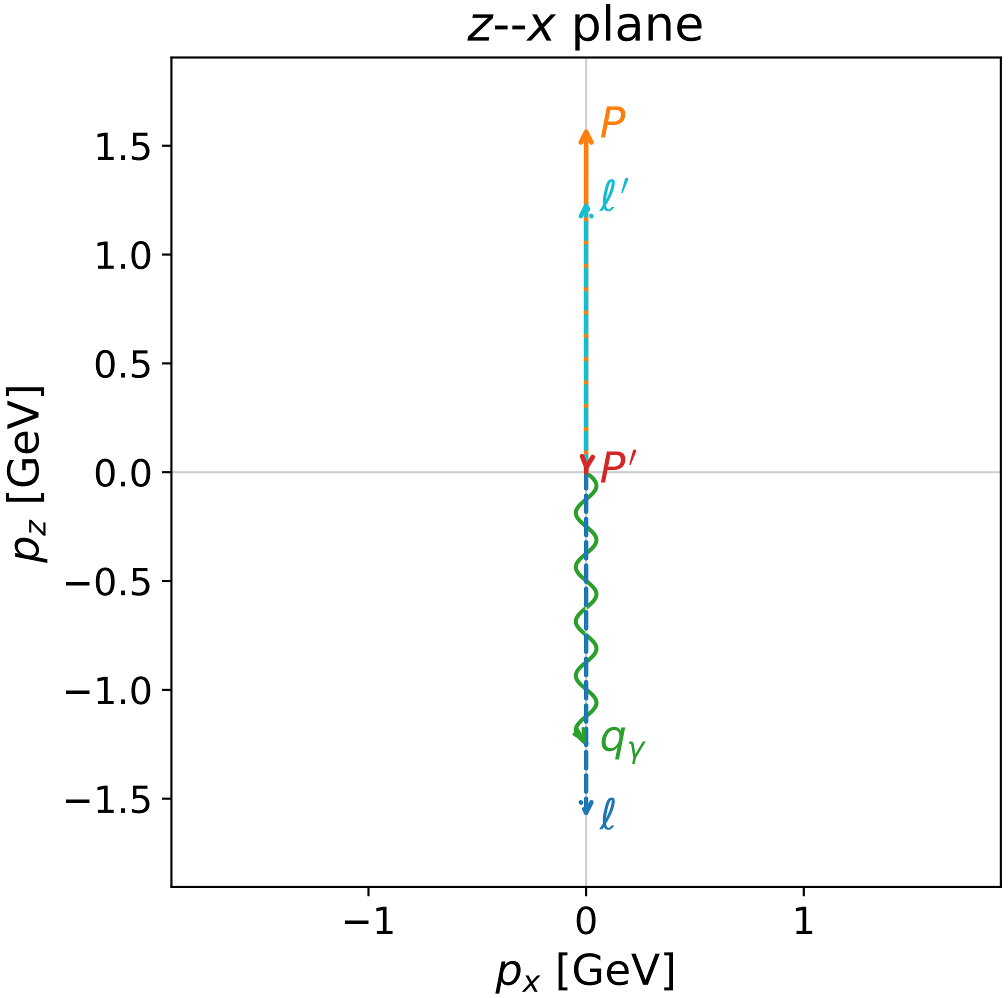


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_126
d_{\rm GHZ}=4.95818e-08
region: polarization_cluster_01_local_minimum_126
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0126
lepton mixing angle: alpha_e=0.79324484 rad
incoming lepton: 0.701537|+> +0.712633|->
proton mixing angle: alpha_p=0.79327627 rad
incoming proton: 0.701514|+> +0.712655|->
s=14.3543, $\sqrt{s}$=3.78871, $|\vec{P}|$=1.77824, $|\vec{P}'|$=0.000403116
$\theta_{p'}$=2.96794, θ_γ=3.14159, $\phi_{p'}$=3.16594, ϕ_γ=3.0289
E_γ=1.42516, $\theta_{\ell'}$=4.88583e-05, $\phi_{\ell'}$=0.0243458
Q^2=10.1399, x_B=0.791416, t=-2.01337, W^2=3.55231, y=0.950862
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7782 p_x= 0.0000 p_y= 0.0000 p_z= -1.7782
P E= 2.0105 p_x= 0.0000 p_y= 0.0000 p_z= 1.7782
ℓ' E= 1.4256 p_x= 0.0001 p_y= 0.0000 p_z= 1.4256
P' E= 0.9380 p_x= -0.0001 p_y= -0.0000 p_z= -0.0004
q_γ E= 1.4252 p_x= -0.0000 p_y= 0.0000 p_z= -1.4252
Leading normalized final-state amplitudes
(N=2, retained=99.2%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7045$, $\arg(\widetilde{A}) = +0.044\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7044$, $\arg(\widetilde{A}) = +0.956\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_129
d_{\rm GHZ}=5.2277e-08
region: polarization_cluster_01_local_minimum_129
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0129
lepton mixing angle: alpha_e=0.89040334 rad
incoming lepton: 0.629099|+> +0.777326|->
proton mixing angle: alpha_p=0.89039462 rad
incoming proton: 0.629105|+> +0.77732|->
s=14.3507, $\sqrt{s}$=3.78823, $|\vec{P}|$=1.77798, $|\vec{P}'|$=0.024118
$\theta_{p'}$=3.13955, θ_γ=3.14159, $\phi_{p'}$=3.14474, ϕ_γ=3.30063
E_γ=1.4129, $\theta_{\ell'}$=3.42918e-05, $\phi_{\ell'}$=0.00314395
Q^2=10.22, x_B=0.79823, t=-2.09854, W^2=3.46317, y=0.950447
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7780 p_x= 0.0000 p_y= 0.0000 p_z= -1.7780
P E= 2.0102 p_x= 0.0000 p_y= 0.0000 p_z= 1.7780
ℓ' E= 1.4370 p_x= 0.0000 p_y= 0.0000 p_z= 1.4370
P' E= 0.9383 p_x= -0.0000 p_y= -0.0000 p_z= -0.0241
q_γ E= 1.4129 p_x= -0.0000 p_y= -0.0000 p_z= -1.4129
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.950\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.050\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_135
d_{\rm GHZ}=5.61544e-08
region: polarization_cluster_01_local_minimum_135
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0135
lepton mixing angle: alpha_e=1.3218649 rad
incoming lepton: 0.246369|+> +0.969176|->
proton mixing angle: alpha_p=1.3218678 rad
incoming proton: 0.246366|+> +0.969177|->
s=24.3709, $\sqrt{s}$=4.93669, $|\vec{P}|$=2.37923, $|\vec{P}'|$=0.000777917
$\theta_{p'}$=0.0397154, θ_γ=3.14159, $\phi_{p'}$=0.000309928, ϕ_γ=3.64684
E_γ=1.99973, $\theta_{\ell'}$=1.54516e-05, $\phi_{\ell'}$=3.1419
Q^2=19.0239, x_B=0.835169, t=-3.0344, W^2=4.63445, y=0.969668
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3792 p_x= 0.0000 p_y= 0.0000 p_z= -2.3792
P E= 2.5575 p_x= 0.0000 p_y= 0.0000 p_z= 2.3792
ℓ' E= 1.9990 p_x= -0.0000 p_y= -0.0000 p_z= 1.9990
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0008
q_γ E= 1.9997 p_x= -0.0000 p_y= -0.0000 p_z= -1.9997
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = -0.161\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = +0.161\pi$
normalized amplitude magnitude $|\widetilde{A}|$

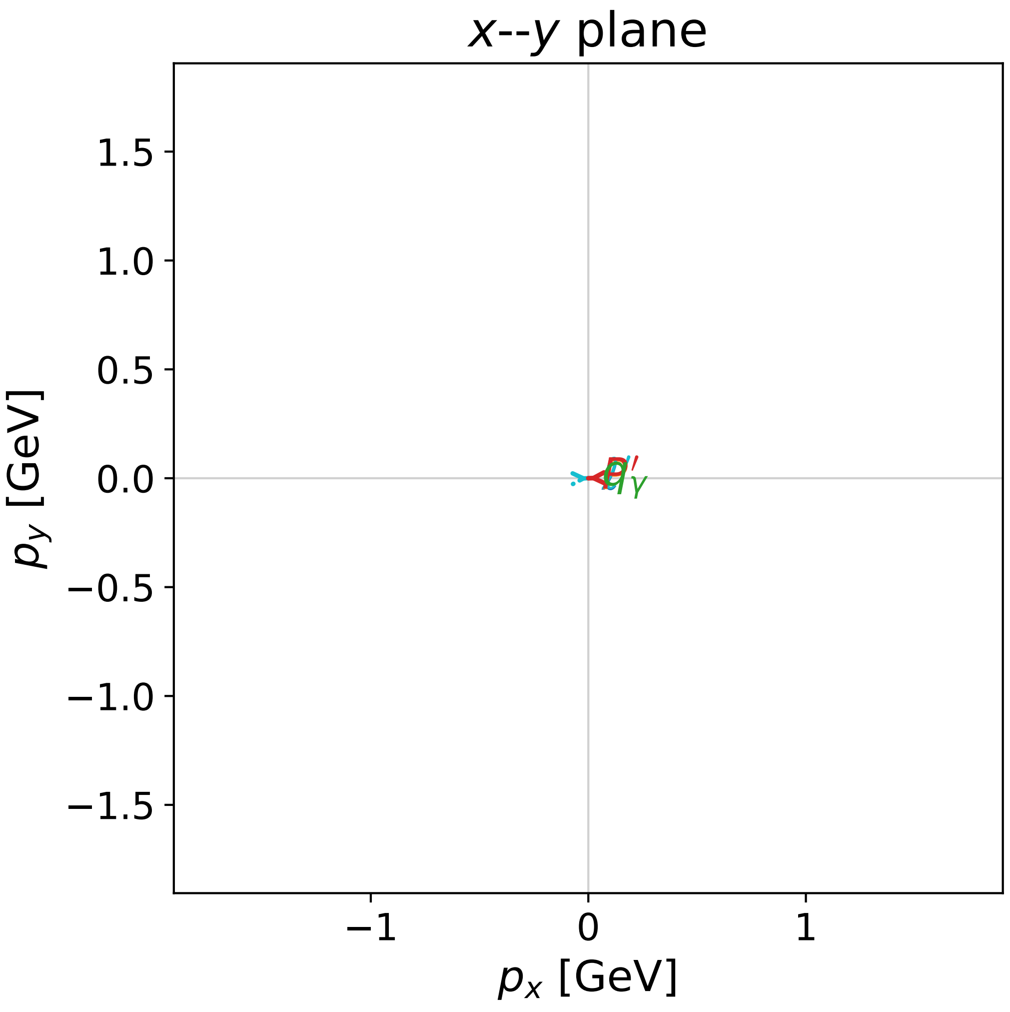
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

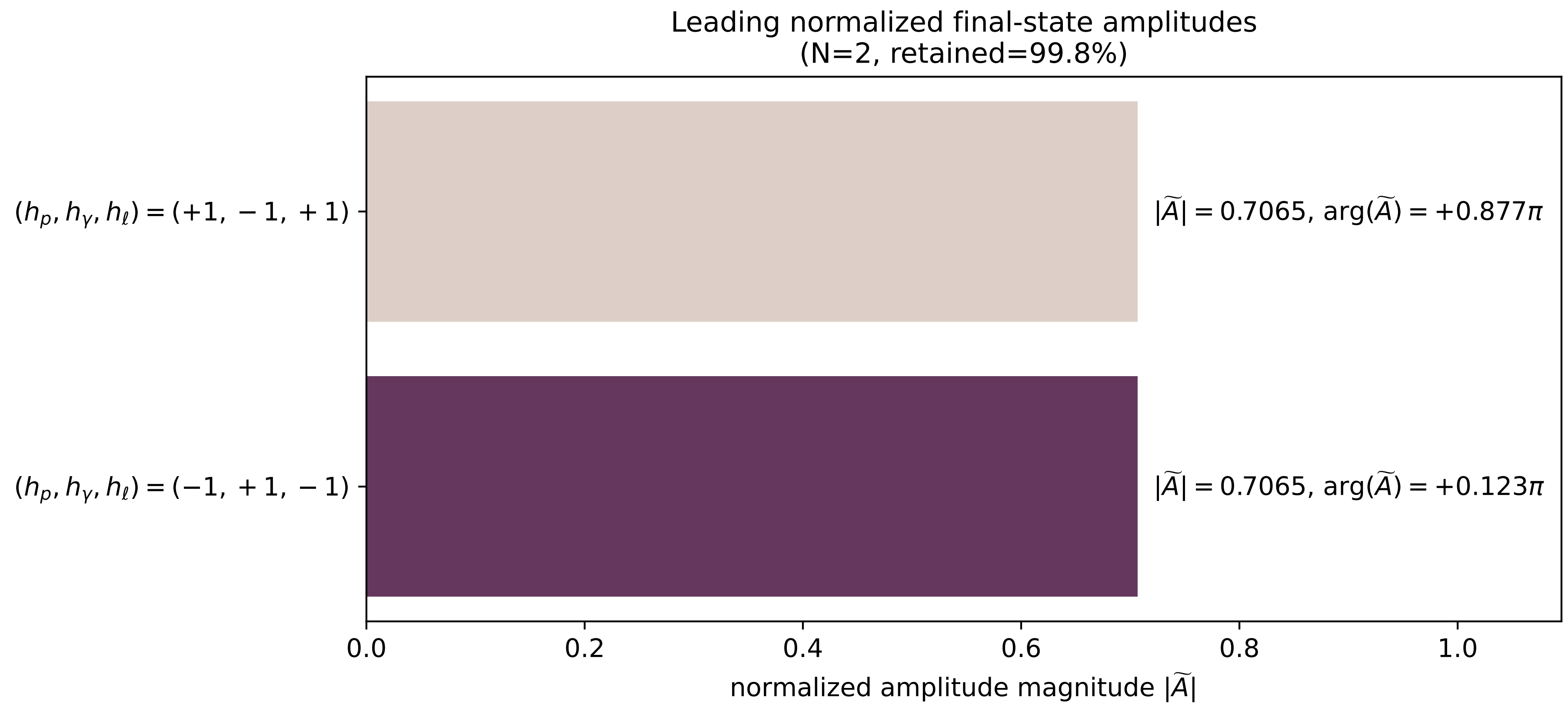


dghz_mixing_angles_polarization_01_local_minimum_138
d_{\rm GHZ}=6.56334e-08
region: polarization_cluster_01_local_minimum_138
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0138
lepton mixing angle: alpha_e=0.83474474 rad
incoming lepton: 0.671367|+> +0.741125|->
proton mixing angle: alpha_p=0.83473256 rad
incoming proton: 0.671376|+> +0.741117|->

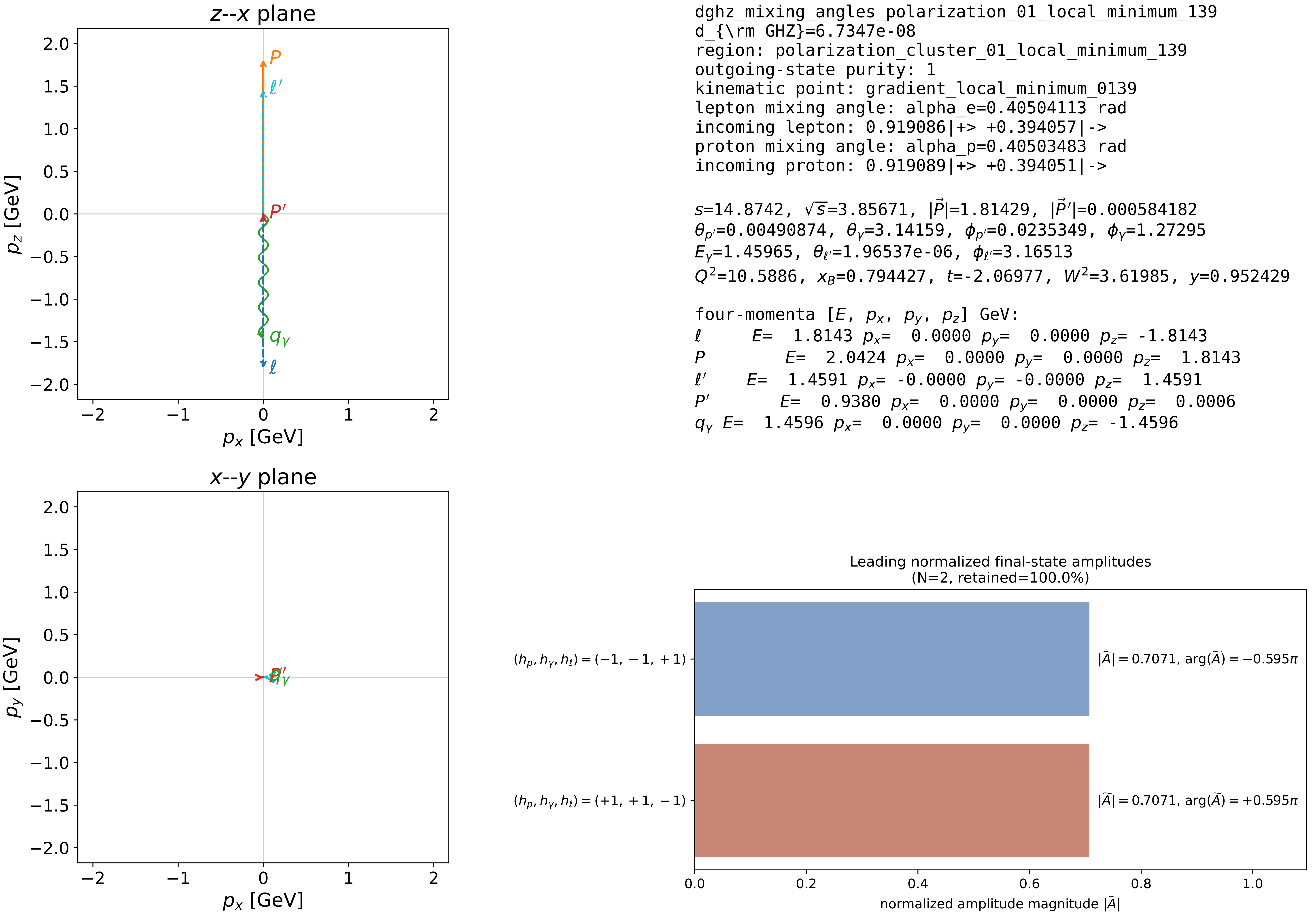
s=11.7789, $\sqrt{s}$=3.43203, $|\vec{P}|$=1.58784, $|\vec{P}'|$=0.00100941
$\theta_{p'}$=3.06084, θ_γ=3.14159, $\phi_{p'}$=3.17093, ϕ_γ=2.78545
E_γ=1.24651, $\theta_{\ell'}$=6.52663e-05, $\phi_{\ell'}$=0.0293406
Q^2=7.92343, x_B=0.772311, t=-1.70322, W^2=3.2158, y=0.941313

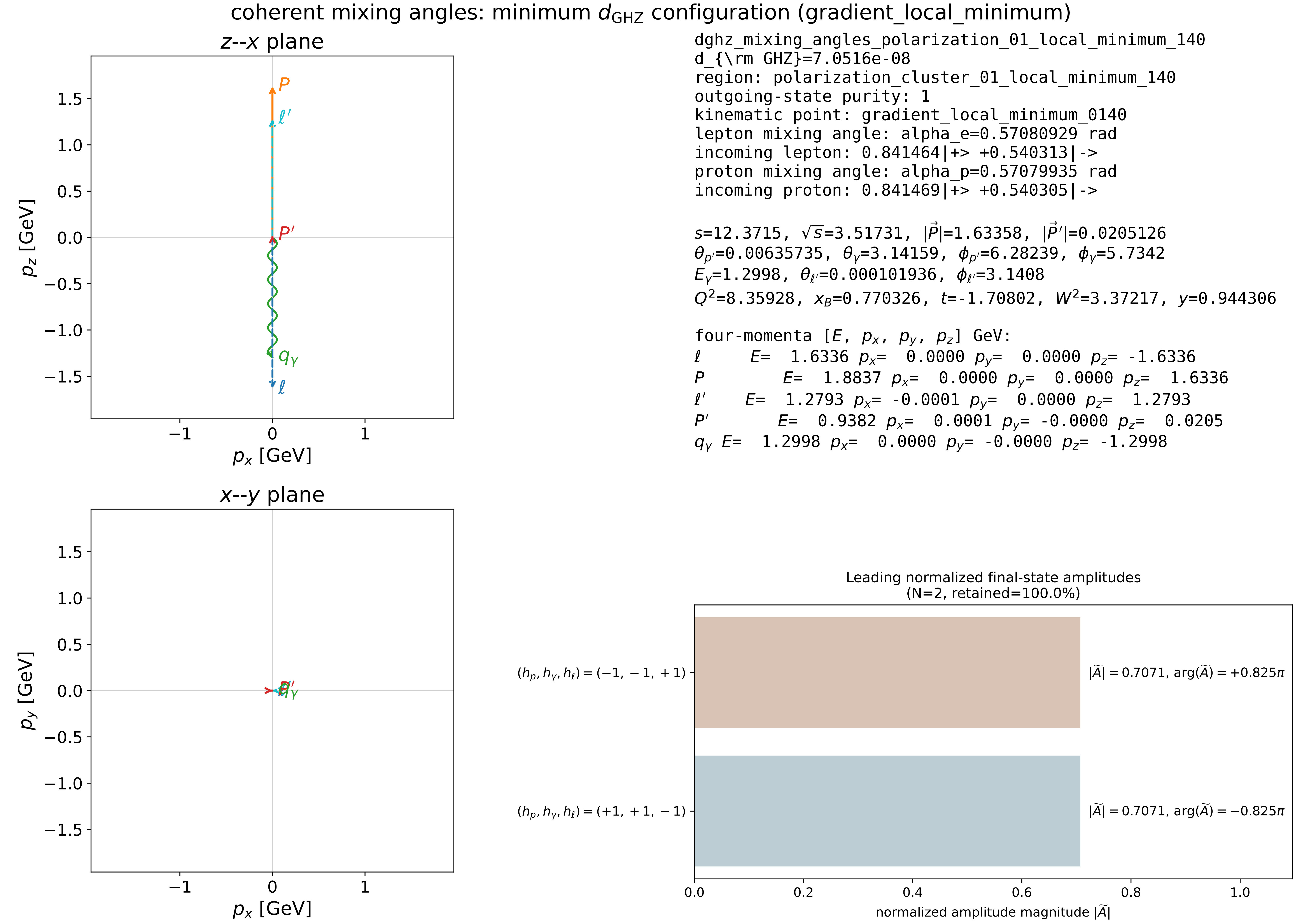
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5878 p_x= 0.0000 p_y= 0.0000 p_z= -1.5878
P E= 1.8442 p_x= 0.0000 p_y= 0.0000 p_z= 1.5878
ℓ' E= 1.2475 p_x= 0.0001 p_y= 0.0000 p_z= 1.2475
P' E= 0.9380 p_x= -0.0001 p_y= -0.0000 p_z= -0.0010
q_γ E= 1.2465 p_x= -0.0000 p_y= 0.0000 p_z= -1.2465





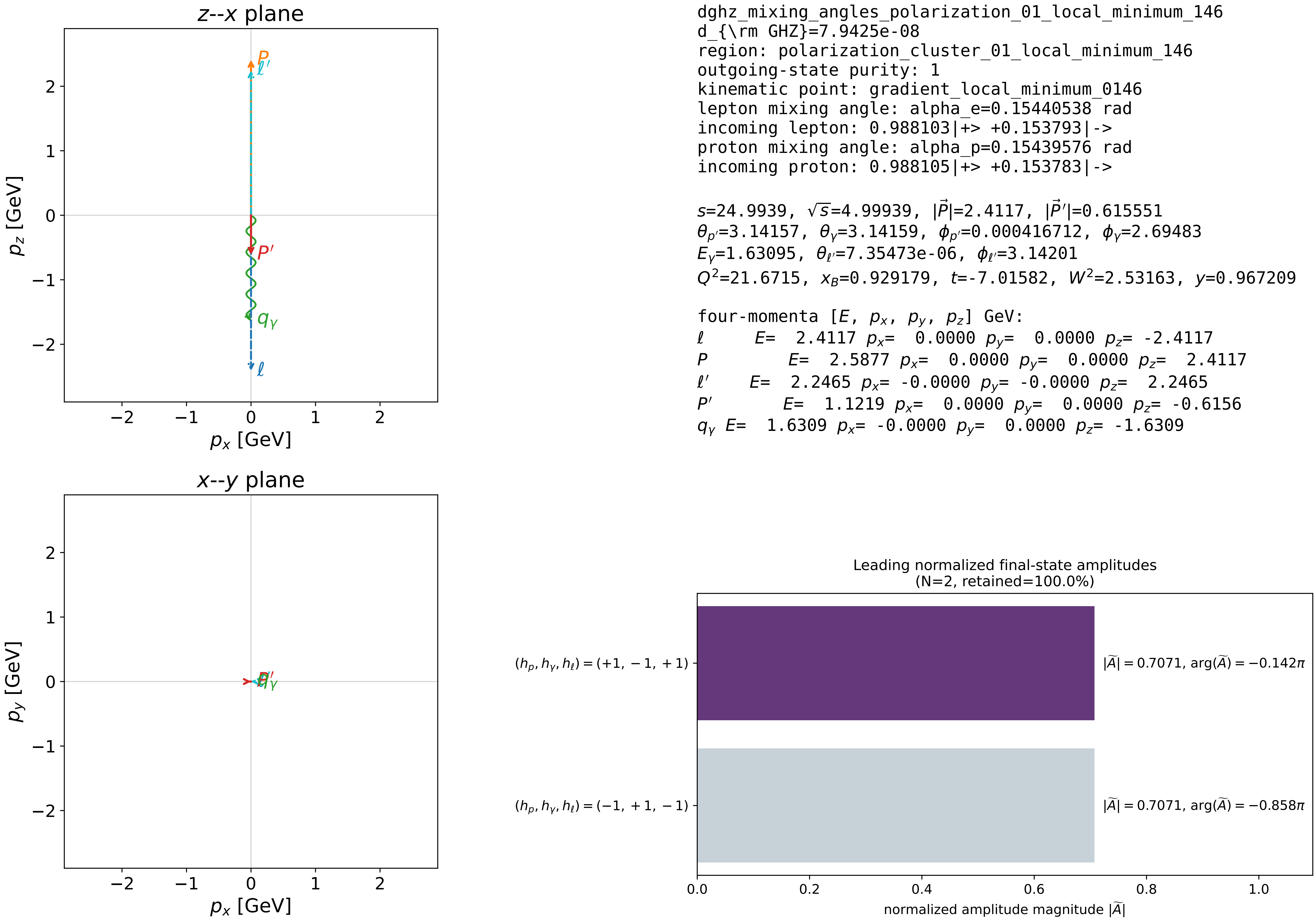
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

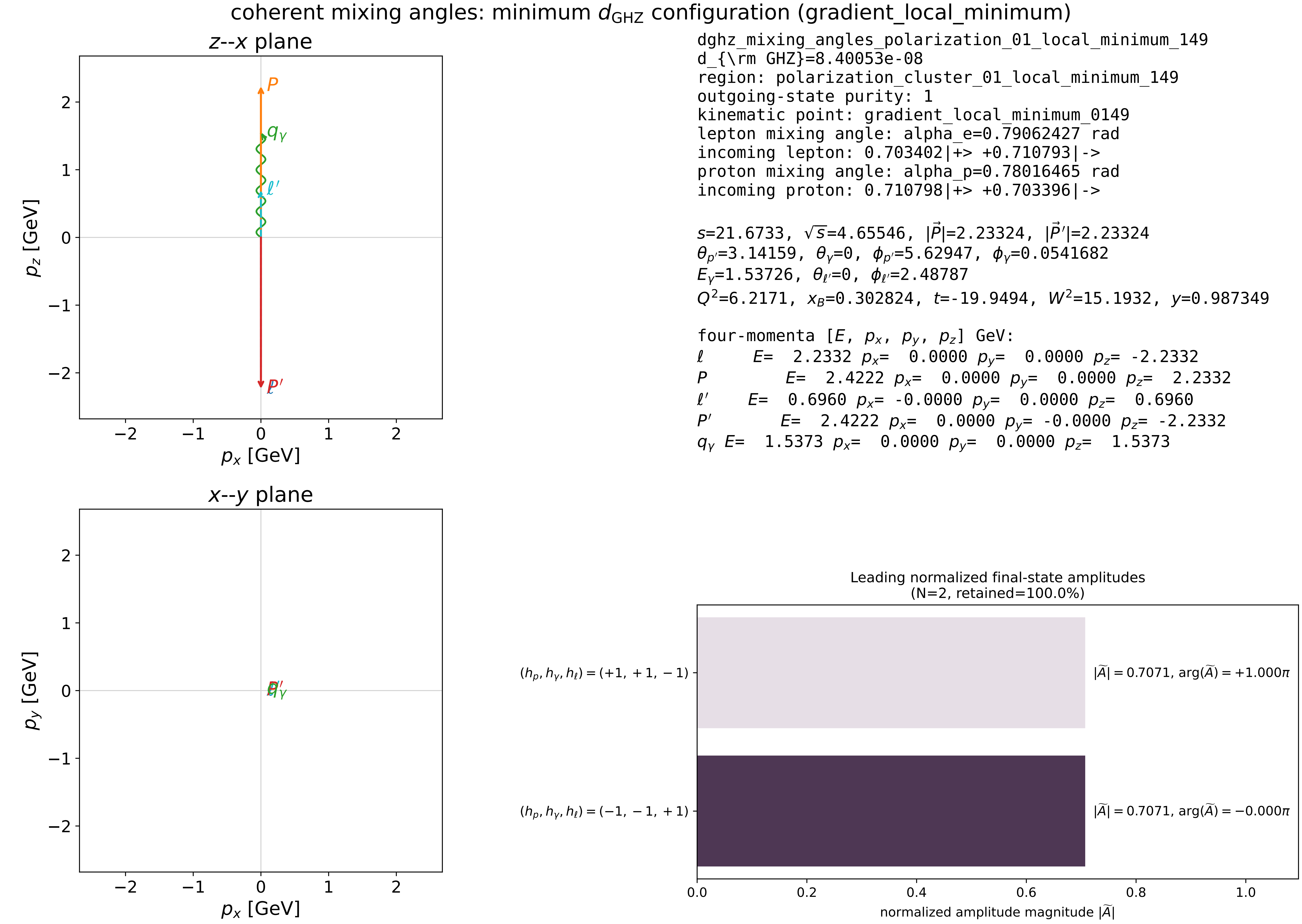




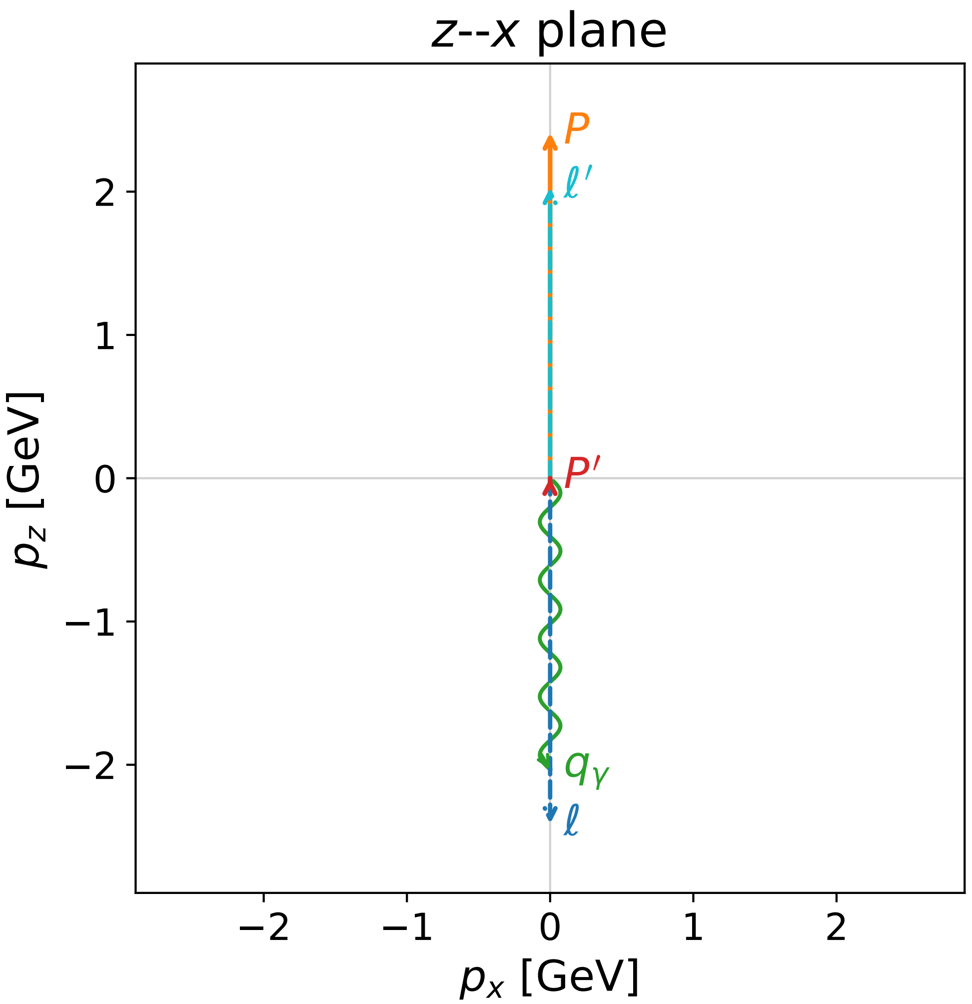
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_140
d_{\rm GHZ}=7.0516e-08
region: polarization_cluster_01_local_minimum_140
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0140
lepton mixing angle: alpha_e=0.57080929 rad
incoming lepton: 0.841464|+> +0.540313|->
proton mixing angle: alpha_p=0.57079935 rad
incoming proton: 0.841469|+> +0.540305|->
s=12.3715, $\sqrt{s}$=3.51731, $|\vec{P}|$=1.63358, $|\vec{P}'|$=0.0205126
$\theta_{p'}$=0.00635735, θ_γ=3.14159, $\phi_{p'}$=6.28239, ϕ_γ=5.7342
E_γ=1.2998, $\theta_{\ell'}$=0.000101936, $\phi_{\ell'}$=3.1408
Q^2=8.35928, x_B=0.770326, t=-1.70802, W^2=3.37217, y=0.944306
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.6336 p_x= 0.0000 p_y= 0.0000 p_z= -1.6336
P E= 1.8837 p_x= 0.0000 p_y= 0.0000 p_z= 1.6336
ℓ' E= 1.2793 p_x= -0.0001 p_y= 0.0000 p_z= 1.2793
P' E= 0.9382 p_x= 0.0001 p_y= -0.0000 p_z= 0.0205
q_γ E= 1.2998 p_x= 0.0000 p_y= -0.0000 p_z= -1.2998
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.825\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.825\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
q_γ
ℓ'
P'
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_149
d_{\rm GHZ}=8.40053e-08
region: polarization_cluster_01_local_minimum_149
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0149
lepton mixing angle: alpha_e=0.79062427 rad
incoming lepton: 0.703402|+> +0.710793|->
proton mixing angle: alpha_p=0.78016465 rad
incoming proton: 0.710798|+> +0.703396|->
s=21.6733, $\sqrt{s}$=4.65546, $|\vec{P}|$=2.23324, $|\vec{P}'|$=2.23324
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=5.62947, ϕ_γ=0.0541682
E_γ=1.53726, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.48787
Q^2=6.2171, x_B=0.302824, t=-19.9494, W^2=15.1932, y=0.987349
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2332 p_x= 0.0000 p_y= 0.0000 p_z= -2.2332
P E= 2.4222 p_x= 0.0000 p_y= 0.0000 p_z= 2.2332
ℓ' E= 0.6960 p_x= -0.0000 p_y= 0.0000 p_z= 0.6960
P' E= 2.4222 p_x= 0.0000 p_y= -0.0000 p_z= -2.2332
q_γ E= 1.5373 p_x= 0.0000 p_y= 0.0000 p_z= 1.5373
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

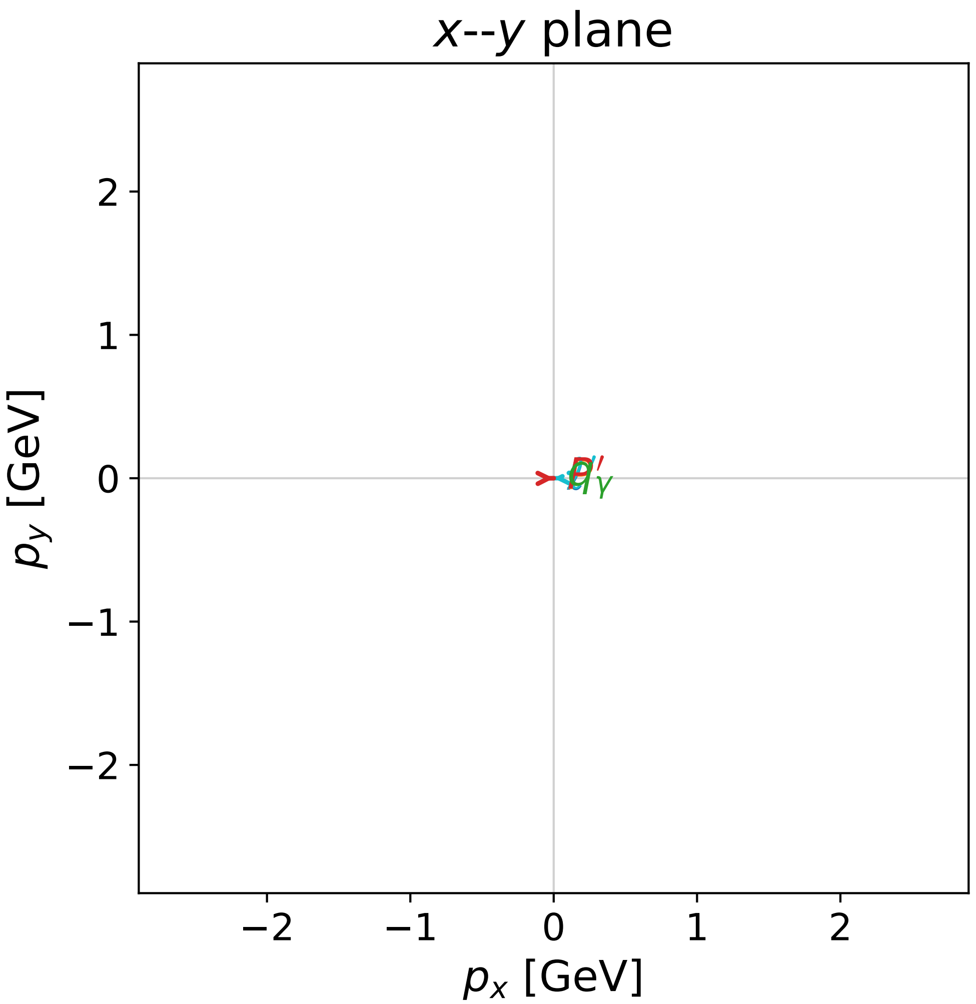
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

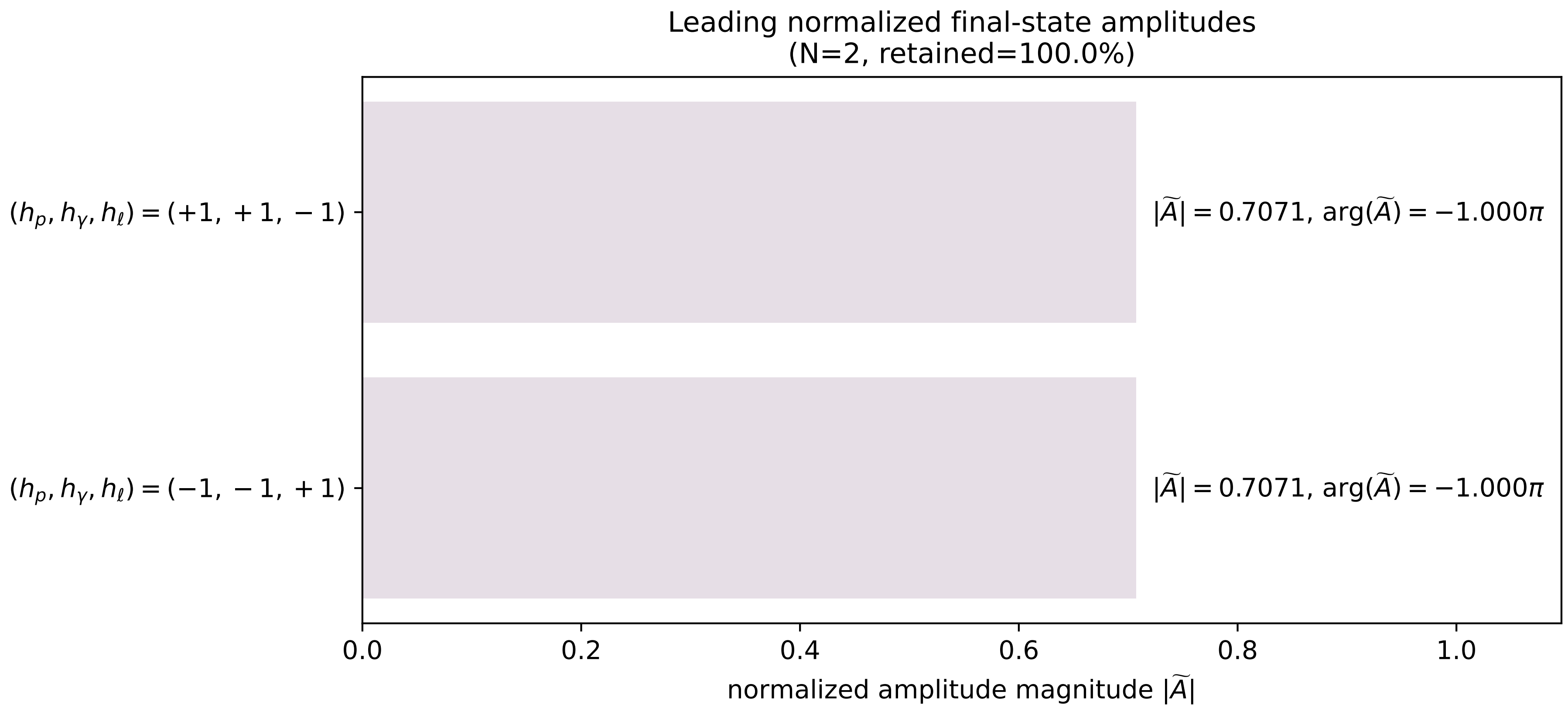


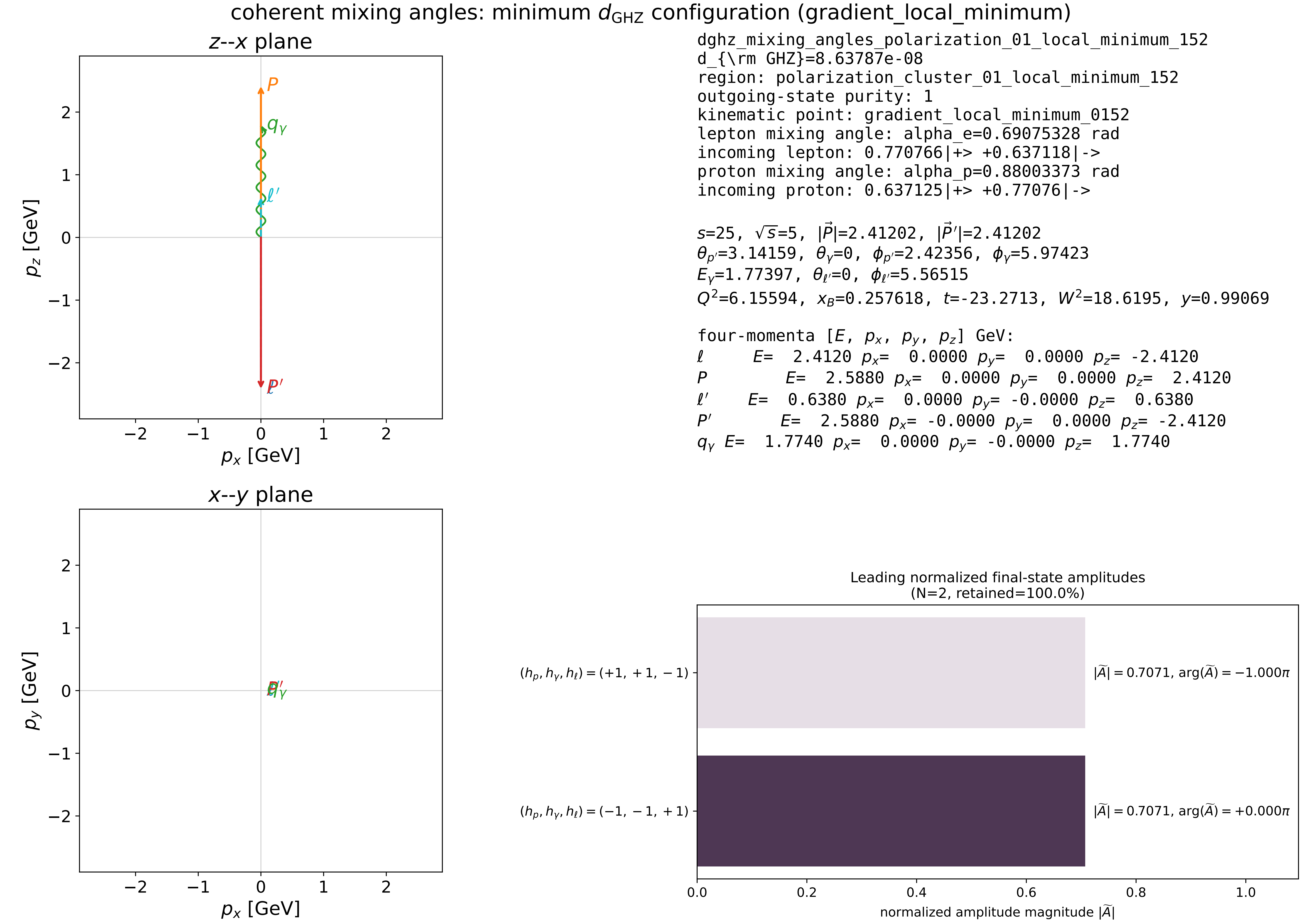
dghz_mixing_angles_polarization_01_local_minimum_151
d_{\rm GHZ}=8.54757e-08
region: polarization_cluster_01_local_minimum_151
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0151
lepton mixing angle: alpha_e=0.15707963 rad
incoming lepton: 0.987688|+> +0.156434|->
proton mixing angle: alpha_p=0.15707963 rad
incoming proton: 0.987688|+> +0.156434|->

s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=0.00283891
$\theta_{p'}$=0.00981748, θ_γ=3.14159, $\phi_{p'}$=0.00981748, ϕ_γ=0
E_γ=2.03242, $\theta_{\ell'}$=1.37321e-05, $\phi_{\ell'}$=3.15141
Q^2=19.5815, x_B=0.836606, t=-3.0817, W^2=4.70422, y=0.970386

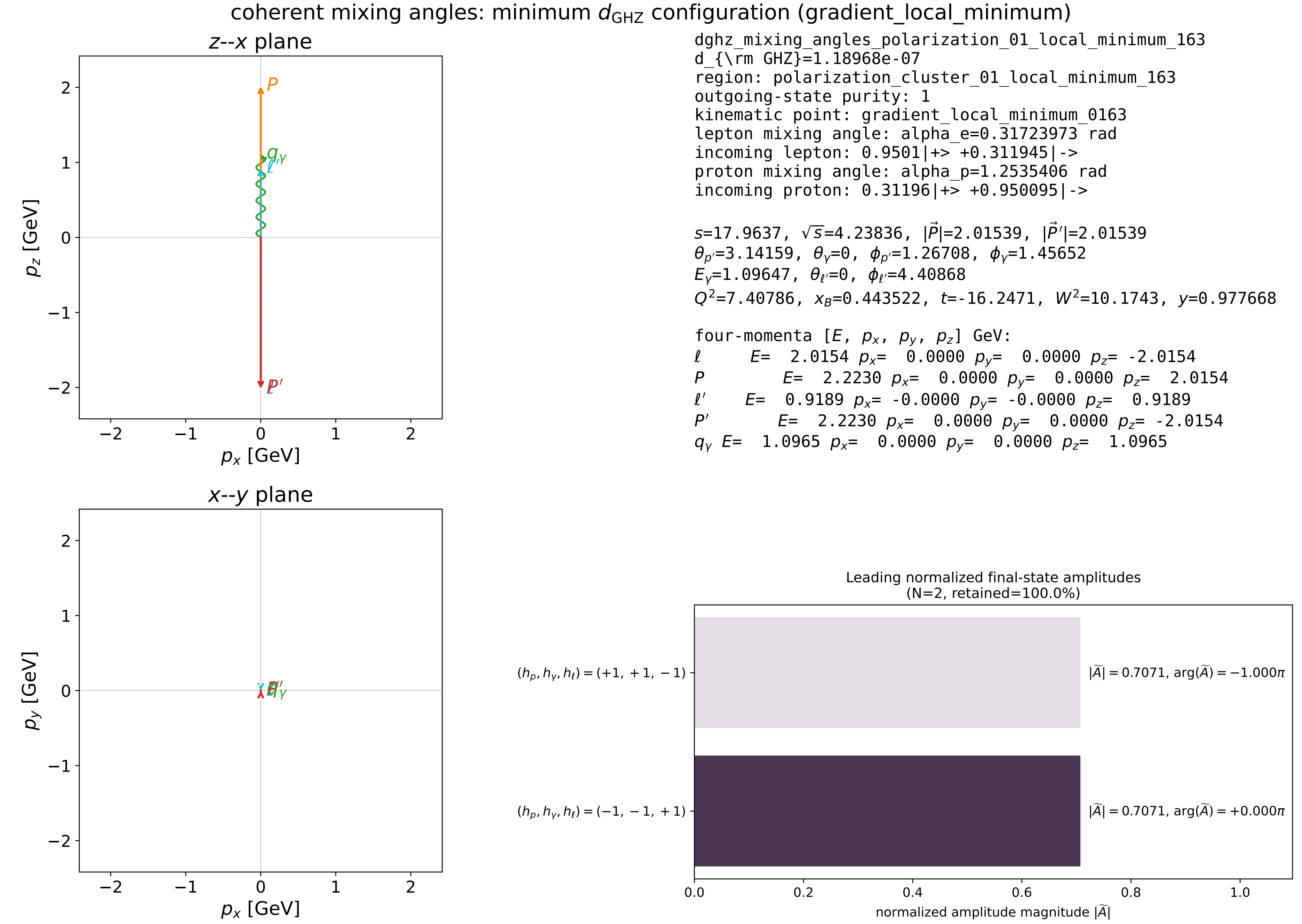
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 2.0296 p_x= -0.0000 p_y= -0.0000 p_z= 2.0296
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0028
q_γ E= 2.0324 p_x= 0.0000 p_y= 0.0000 p_z= -2.0324

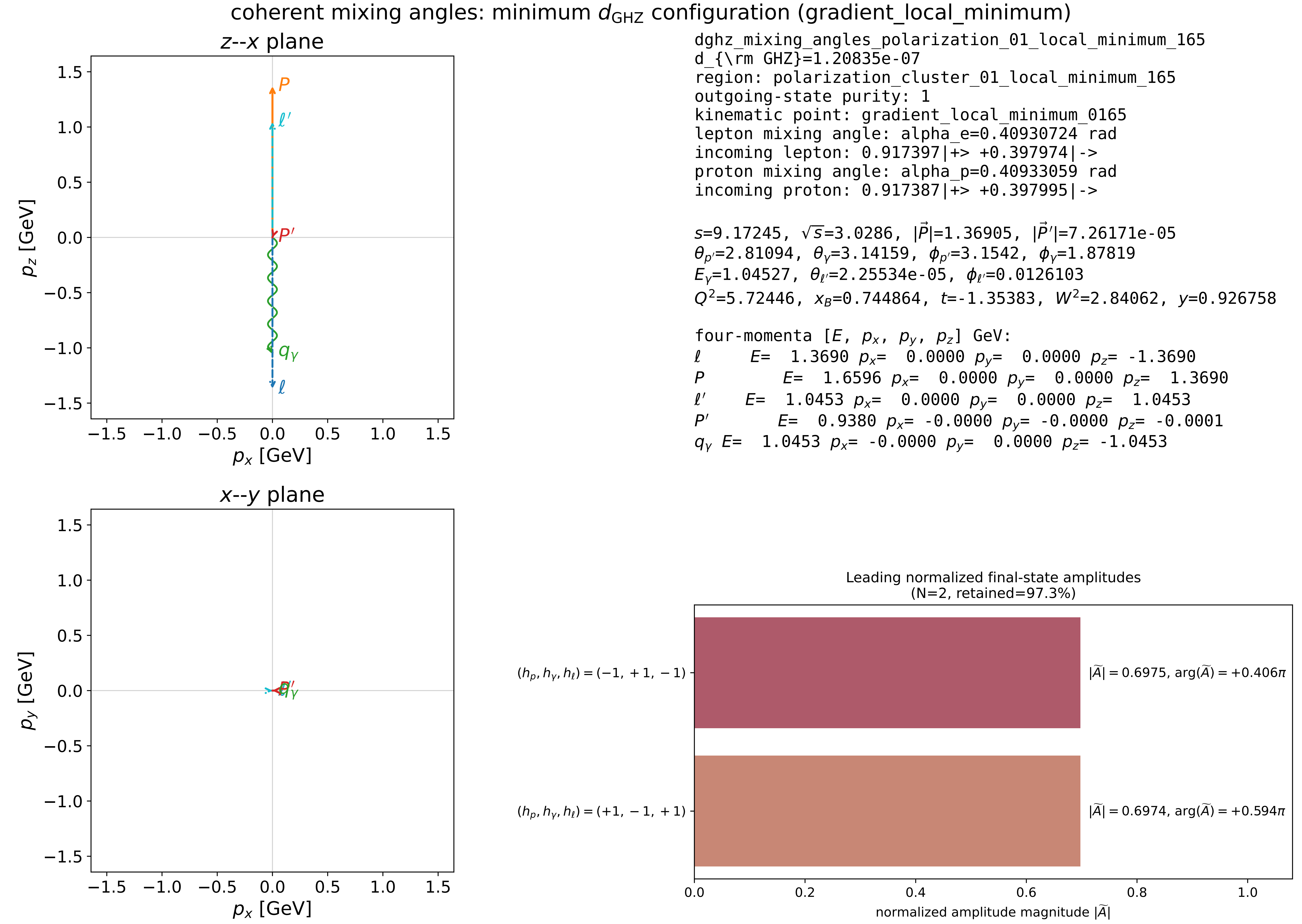


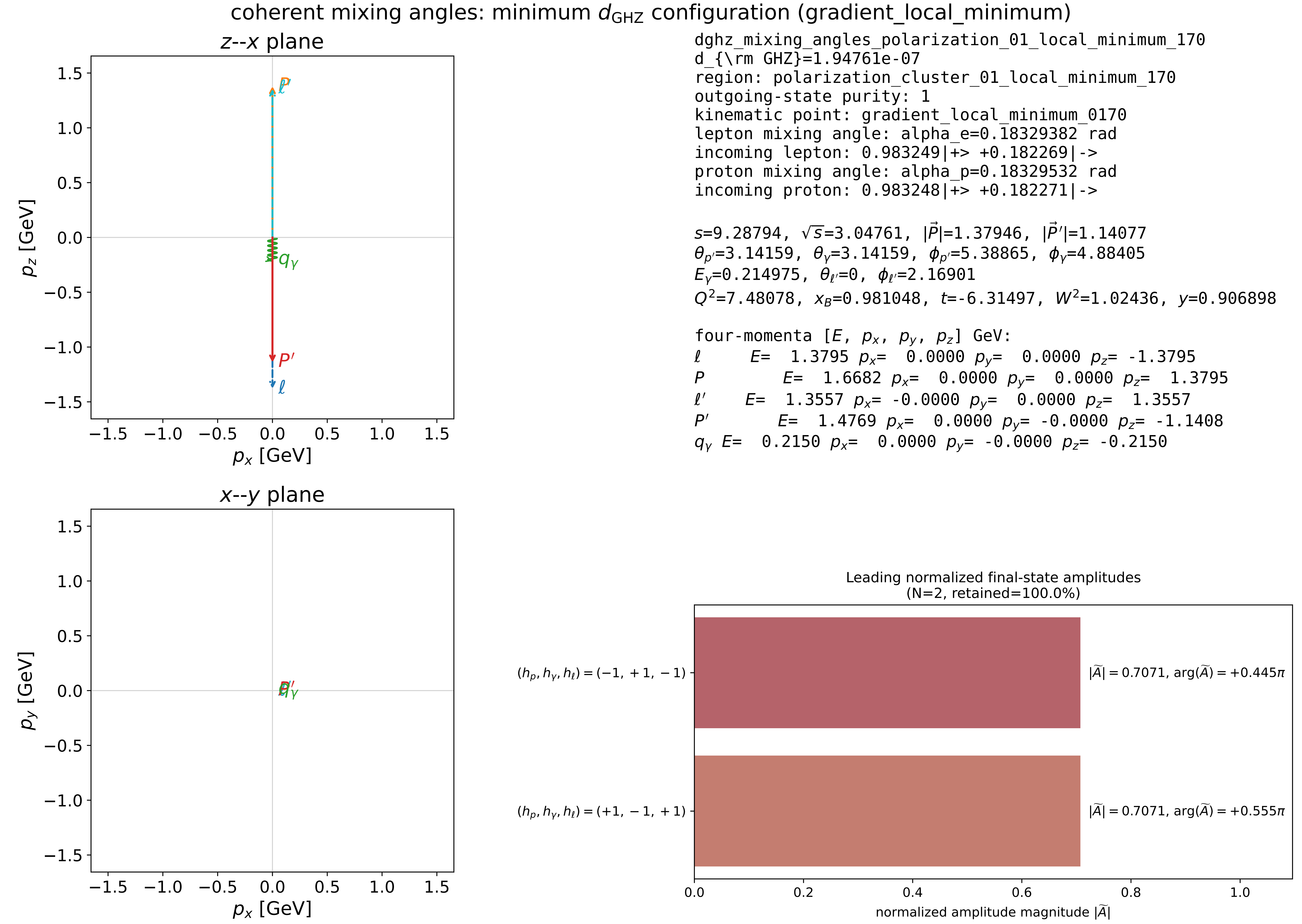


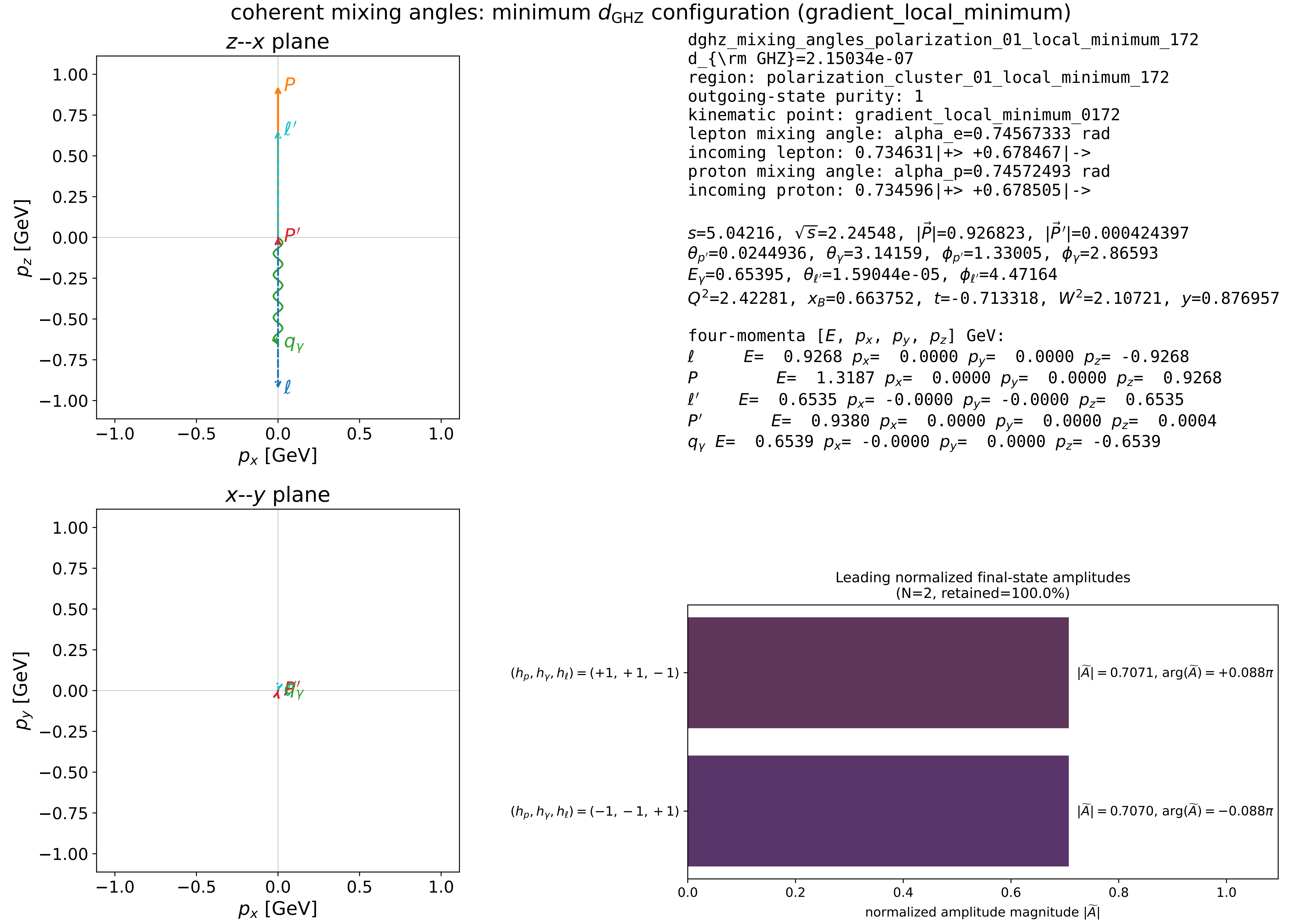


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
P
q_γ
ℓ'
P'
p_z [GeV]
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_152
d_{\rm GHZ}=8.63787e-08
region: polarization_cluster_01_local_minimum_152
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0152
lepton mixing angle: alpha_e=0.69075328 rad
incoming lepton: 0.770766|+> +0.637118|->
proton mixing angle: alpha_p=0.88003373 rad
incoming proton: 0.637125|+> +0.77076|->
s=25, $\sqrt{s}$=5, $|\vec{P}|$=2.41202, $|\vec{P}'|$=2.41202
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=2.42356, ϕ_γ=5.97423
E_γ=1.77397, $\theta_{\ell'}$=0, $\phi_{\ell'}$=5.56515
Q^2=6.15594, x_B=0.257618, t=-23.2713, W^2=18.6195, y=0.99069
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4120 p_x= 0.0000 p_y= 0.0000 p_z= -2.4120
P E= 2.5880 p_x= 0.0000 p_y= 0.0000 p_z= 2.4120
ℓ' E= 0.6380 p_x= 0.0000 p_y= -0.0000 p_z= 0.6380
P' E= 2.5880 p_x= -0.0000 p_y= 0.0000 p_z= -2.4120
q_γ E= 1.7740 p_x= 0.0000 p_y= -0.0000 p_z= 1.7740
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

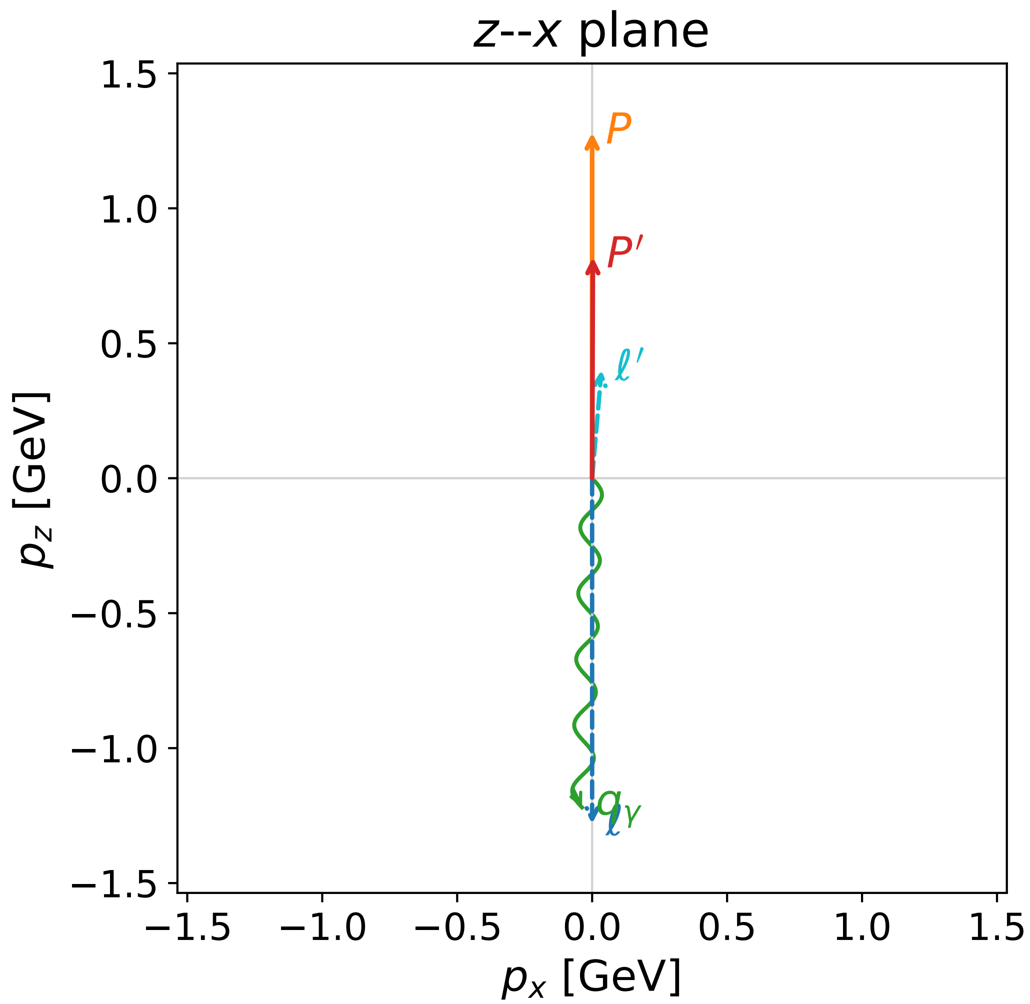

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
q_γ
ℓ'
P'
ℓ
dghz_mixing_angles_polarization_01_local_minimum_163
d_{\rm GHZ}=1.18968e-07
region: polarization_cluster_01_local_minimum_163
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0163
lepton mixing angle: alpha_e=0.31723973 rad
incoming lepton: 0.9501|+> +0.311945|->
proton mixing angle: alpha_p=1.2535406 rad
incoming proton: 0.31196|+> +0.950095|->
s=17.9637, $\sqrt{s}$=4.23836, $|\vec{P}|$=2.01539, $|\vec{P}'|$=2.01539
$\theta_{p'}$=3.14159, θ_γ=0, $\phi_{p'}$=1.26708, ϕ_γ=1.45652
E_γ=1.09647, $\theta_{\ell'}$=0, $\phi_{\ell'}$=4.40868
Q^2=7.40786, x_B=0.443522, t=-16.2471, W^2=10.1743, y=0.977668
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.0154 p_x= 0.0000 p_y= 0.0000 p_z= -2.0154
P E= 2.2230 p_x= 0.0000 p_y= 0.0000 p_z= 2.0154
ℓ' E= 0.9189 p_x= -0.0000 p_y= -0.0000 p_z= 0.9189
P' E= 2.2230 p_x= 0.0000 p_y= 0.0000 p_z= -2.0154
q_γ E= 1.0965 p_x= 0.0000 p_y= 0.0000 p_z= 1.0965
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_165
d_{\rm GHZ}=1.20835e-07
region: polarization_cluster_01_local_minimum_165
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0165
lepton mixing angle: alpha_e=0.40930724 rad
incoming lepton: 0.917397|+> +0.397974|->
proton mixing angle: alpha_p=0.40933059 rad
incoming proton: 0.917387|+> +0.397995|->
s=9.17245, $\sqrt{s}$=3.0286, $|\vec{P}|$=1.36905, $|\vec{P}'|$=7.26171e-05
$\theta_{p'}$=2.81094, θ_γ=3.14159, $\phi_{p'}$=3.1542, ϕ_γ=1.87819
E_γ=1.04527, $\theta_{\ell'}$=2.25534e-05, $\phi_{\ell'}$=0.0126103
Q^2=5.72446, x_B=0.744864, t=-1.35383, W^2=2.84062, y=0.926758
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.3690 p_x= 0.0000 p_y= 0.0000 p_z= -1.3690
P E= 1.6596 p_x= 0.0000 p_y= 0.0000 p_z= 1.3690
ℓ' E= 1.0453 p_x= 0.0000 p_y= 0.0000 p_z= 1.0453
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0001
q_γ E= 1.0453 p_x= -0.0000 p_y= 0.0000 p_z= -1.0453
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=97.3%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.6975$, $\arg(\widetilde{A}) = +0.406\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.6974$, $\arg(\widetilde{A}) = +0.594\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
q_γ
P'
ℓ
dghz_mixing_angles_polarization_01_local_minimum_170
d_{\rm GHZ}=1.94761e-07
region: polarization_cluster_01_local_minimum_170
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0170
lepton mixing angle: alpha_e=0.18329382 rad
incoming lepton: 0.983249|+> +0.182269|->
proton mixing angle: alpha_p=0.18329532 rad
incoming proton: 0.983248|+> +0.182271|->
s=9.28794, $\sqrt{s}$=3.04761, $|\vec{P}|$=1.37946, $|\vec{P}'|$=1.14077
$\theta_{p'}$=3.14159, θ_γ=3.14159, $\phi_{p'}$=5.38865, ϕ_γ=4.88405
E_γ=0.214975, $\theta_{\ell'}$=0, $\phi_{\ell'}$=2.16901
Q^2=7.48078, x_B=0.981048, t=-6.31497, W^2=1.02436, y=0.906898
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.3795 p_x= 0.0000 p_y= 0.0000 p_z= -1.3795
P E= 1.6682 p_x= 0.0000 p_y= 0.0000 p_z= 1.3795
ℓ' E= 1.3557 p_x= -0.0000 p_y= 0.0000 p_z= 1.3557
P' E= 1.4769 p_x= 0.0000 p_y= -0.0000 p_z= -1.1408
q_γ E= 0.2150 p_x= 0.0000 p_y= -0.0000 p_z= -0.2150
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.445\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.555\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_172
d_{\rm GHZ}=2.15034e-07
region: polarization_cluster_01_local_minimum_172
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0172
lepton mixing angle: alpha_e=0.74567333 rad
incoming lepton: 0.734631|+> +0.678467|->
proton mixing angle: alpha_p=0.74572493 rad
incoming proton: 0.734596|+> +0.678505|->
s=5.04216, $\sqrt{s}$=2.24548, $|\vec{P}|$=0.926823, $|\vec{P}'|$=0.000424397
$\theta_{p'}$=0.0244936, θ_γ=3.14159, $\phi_{p'}$=1.33005, ϕ_γ=2.86593
E_γ=0.65395, $\theta_{\ell'}$=1.59044e-05, $\phi_{\ell'}$=4.47164
Q^2=2.42281, x_B=0.663752, t=-0.713318, W^2=2.10721, y=0.876957
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.9268 p_x= 0.0000 p_y= 0.0000 p_z= -0.9268
P E= 1.3187 p_x= 0.0000 p_y= 0.0000 p_z= 0.9268
ℓ' E= 0.6535 p_x= -0.0000 p_y= -0.0000 p_z= 0.6535
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= 0.0004
q_γ E= 0.6539 p_x= -0.0000 p_y= 0.0000 p_z= -0.6539
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.7071$, $\arg(\widetilde{A}) = +0.088\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7070$, $\arg(\widetilde{A}) = -0.088\pi$
normalized amplitude magnitude $|\widetilde{A}|$

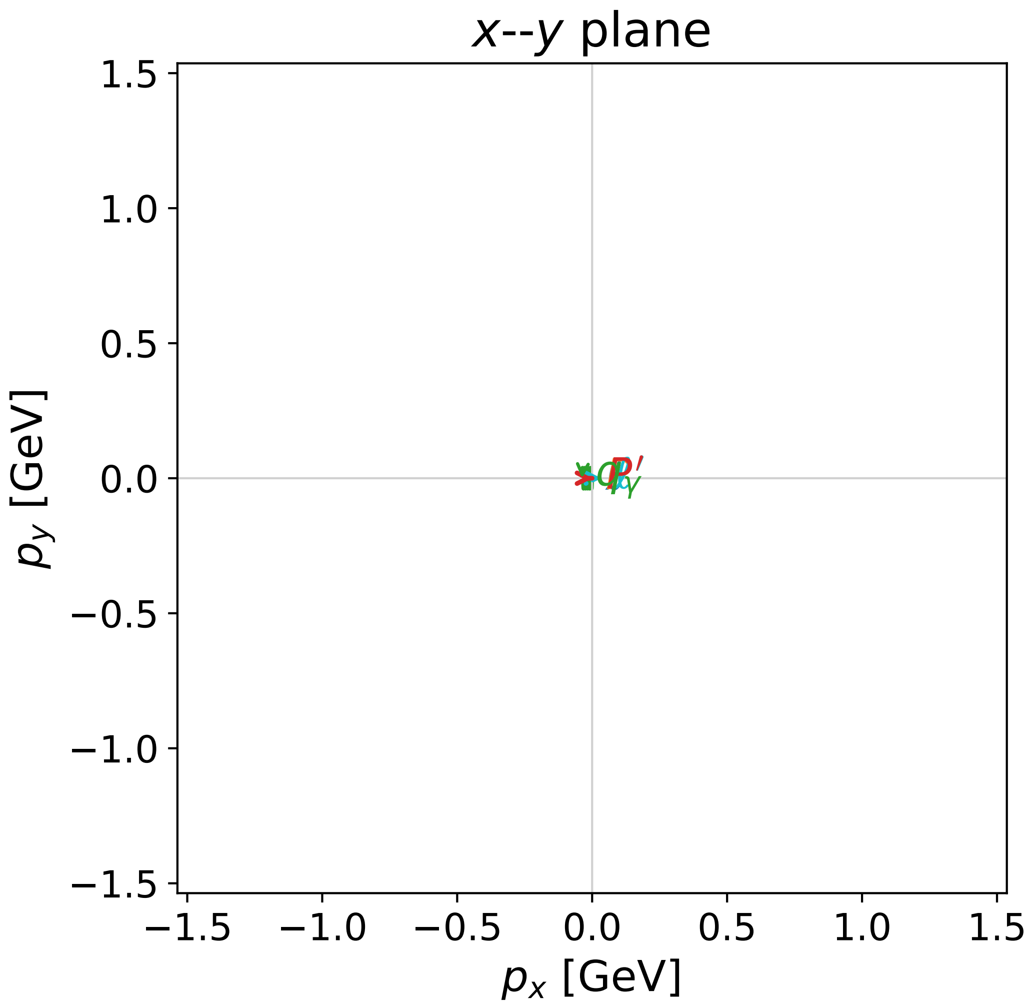
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

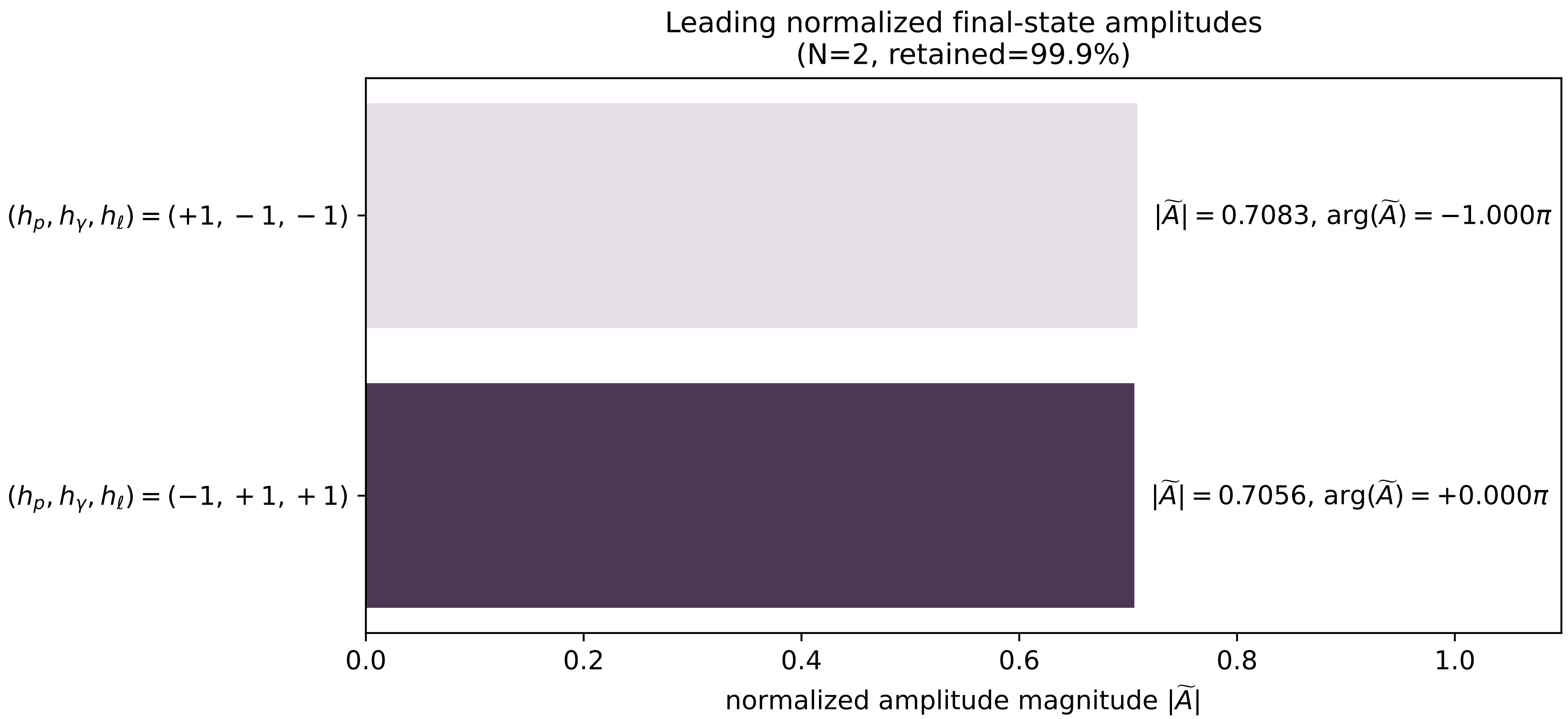


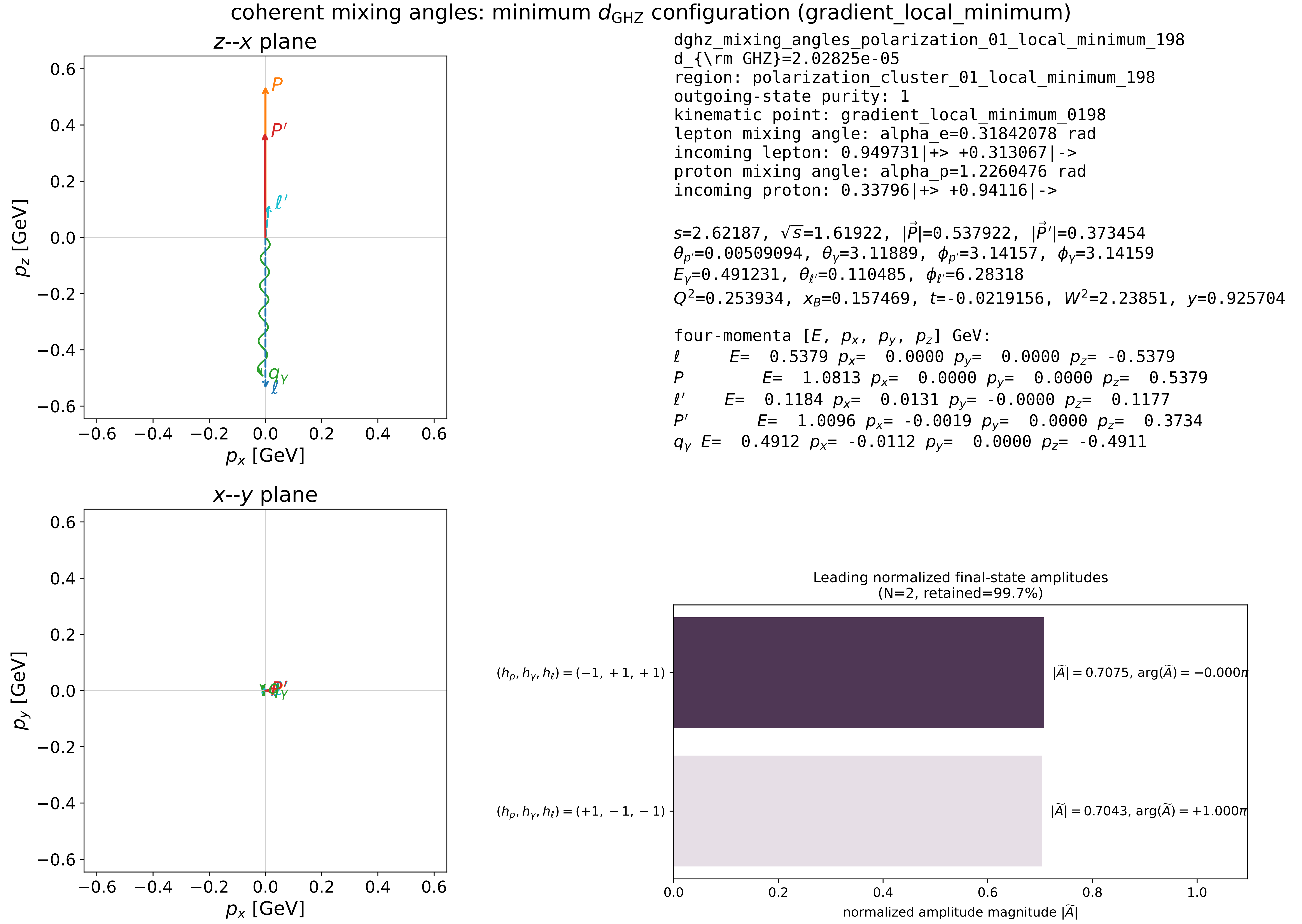
dghz_mixing_angles_polarization_01_local_minimum_193
d_{\rm GHZ}=1.77496e-05
region: polarization_cluster_01_local_minimum_193
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0193
lepton mixing angle: alpha_e=0.24273564 rad
incoming lepton: 0.970684|+> +0.240359|->
proton mixing angle: alpha_p=1.3360225 rad
incoming proton: 0.232623|+> +0.972567|->

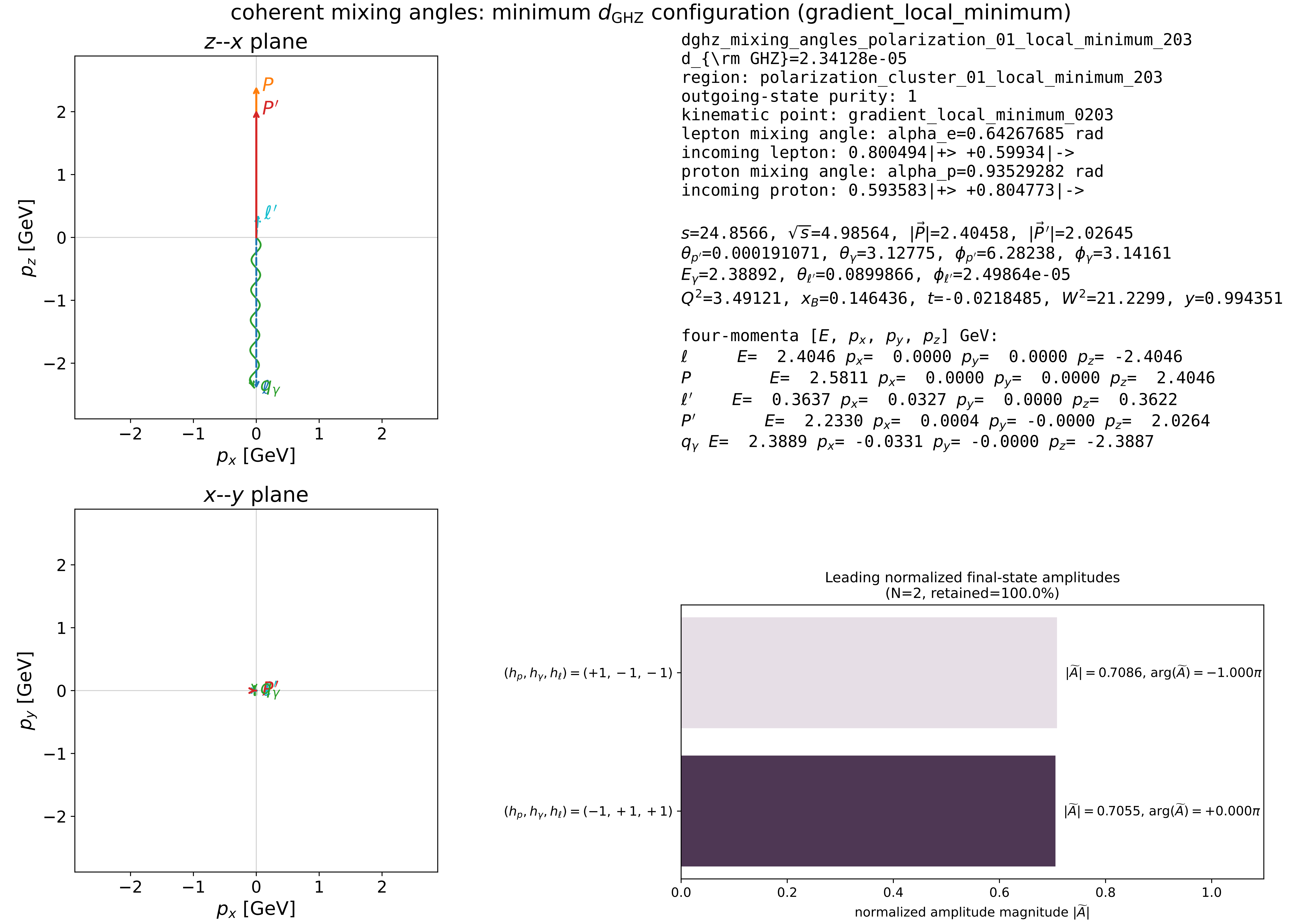
s=8.22294, $\sqrt{s}$=2.86757, $|\vec{P}|$=1.28037, $|\vec{P}'|$=0.818827
$\theta_{p'}$=0.00352053, θ_γ=3.11084, $\phi_{p'}$=6.28317, ϕ_γ=3.14158
E_γ=1.22018, $\theta_{\ell'}$=0.086202, $\phi_{\ell'}$=6.28317
Q^2=2.0564, x_B=0.289943, t=-0.0960184, W^2=5.91586, y=0.965862

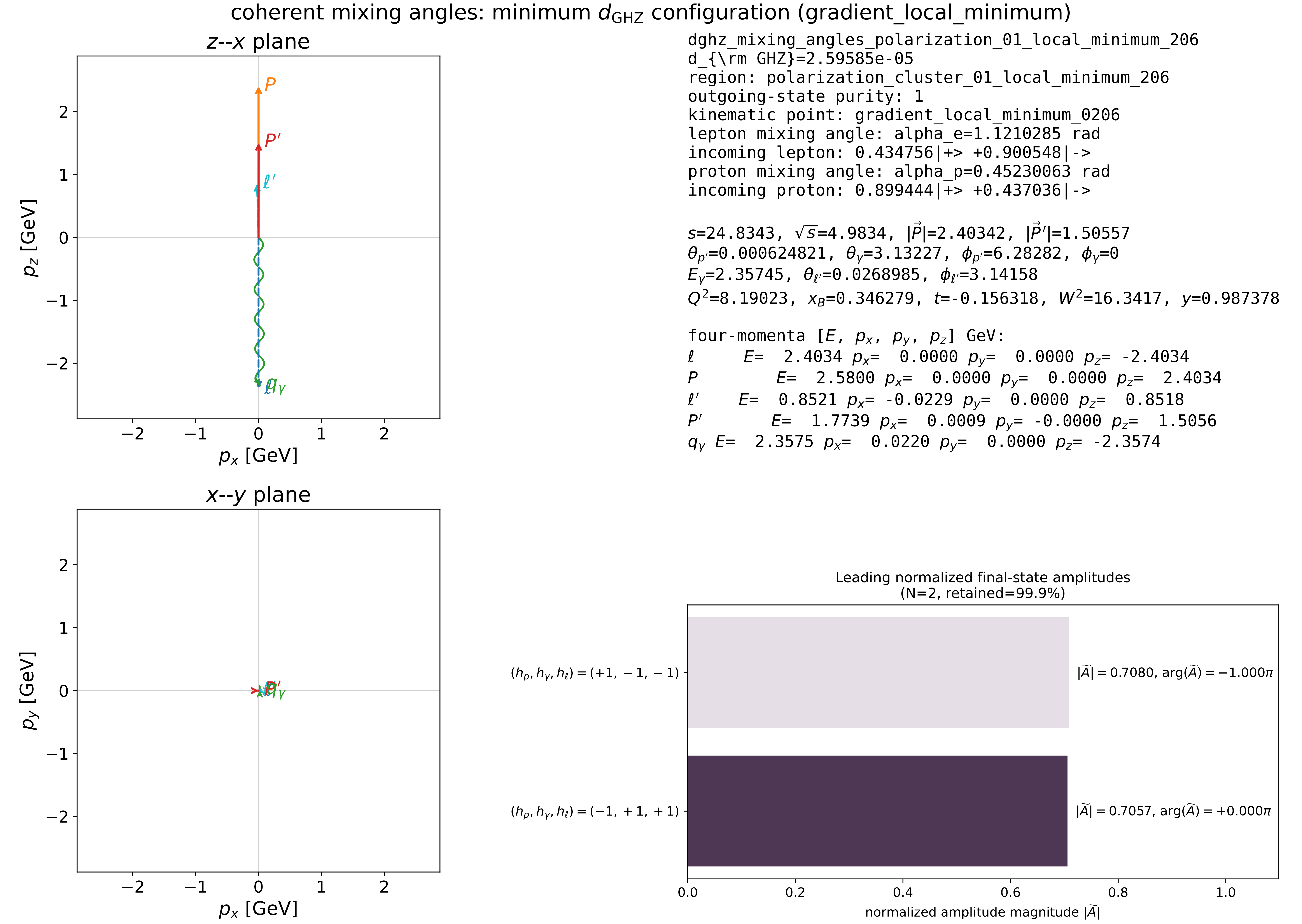
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2804 p_x= 0.0000 p_y= 0.0000 p_z= -1.2804
P E= 1.5872 p_x= 0.0000 p_y= 0.0000 p_z= 1.2804
ℓ' E= 0.4023 p_x= 0.0346 p_y= -0.0000 p_z= 0.4008
P' E= 1.2451 p_x= 0.0029 p_y= -0.0000 p_z= 0.8188
q_γ E= 1.2202 p_x= -0.0375 p_y= 0.0000 p_z= -1.2196

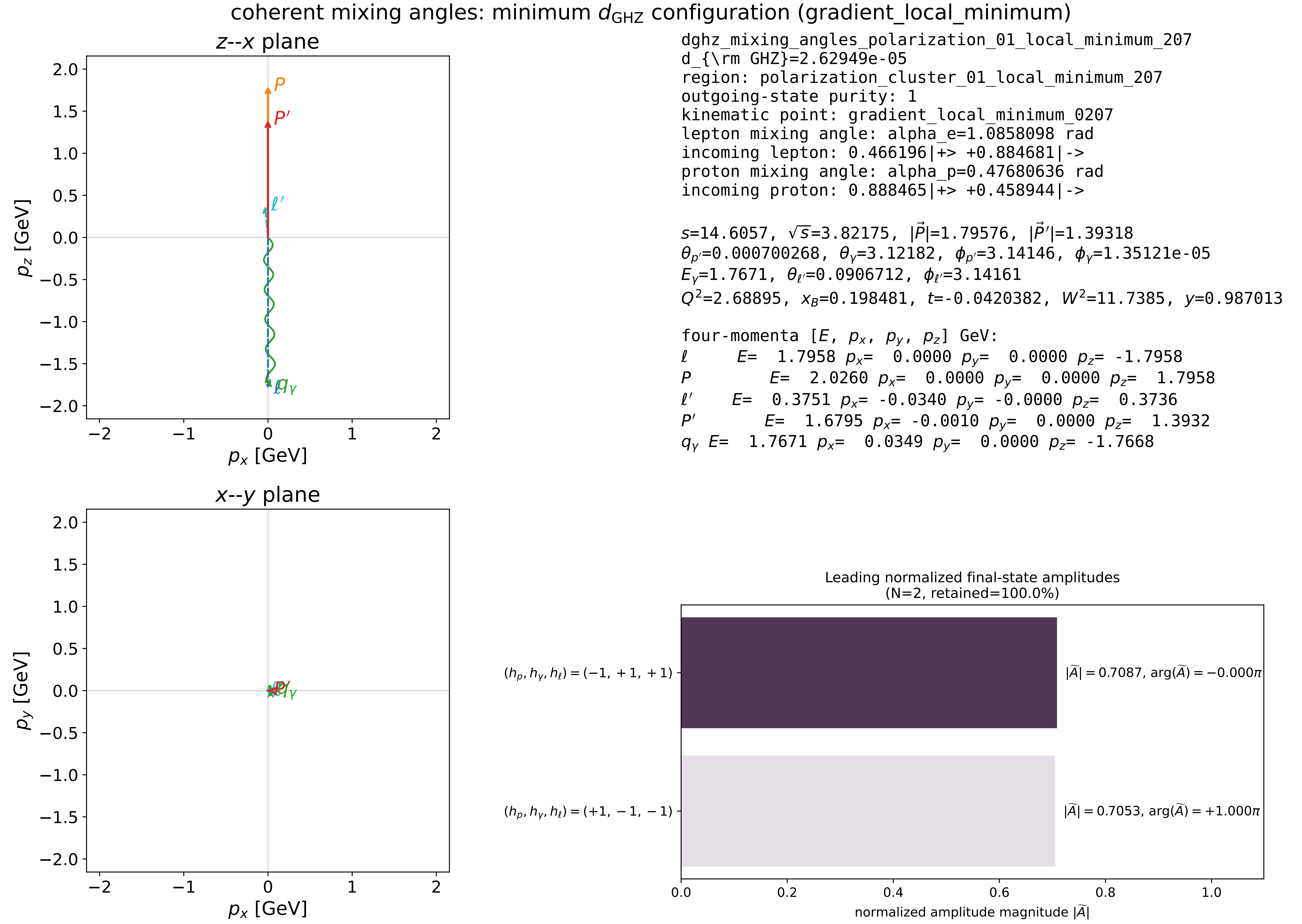


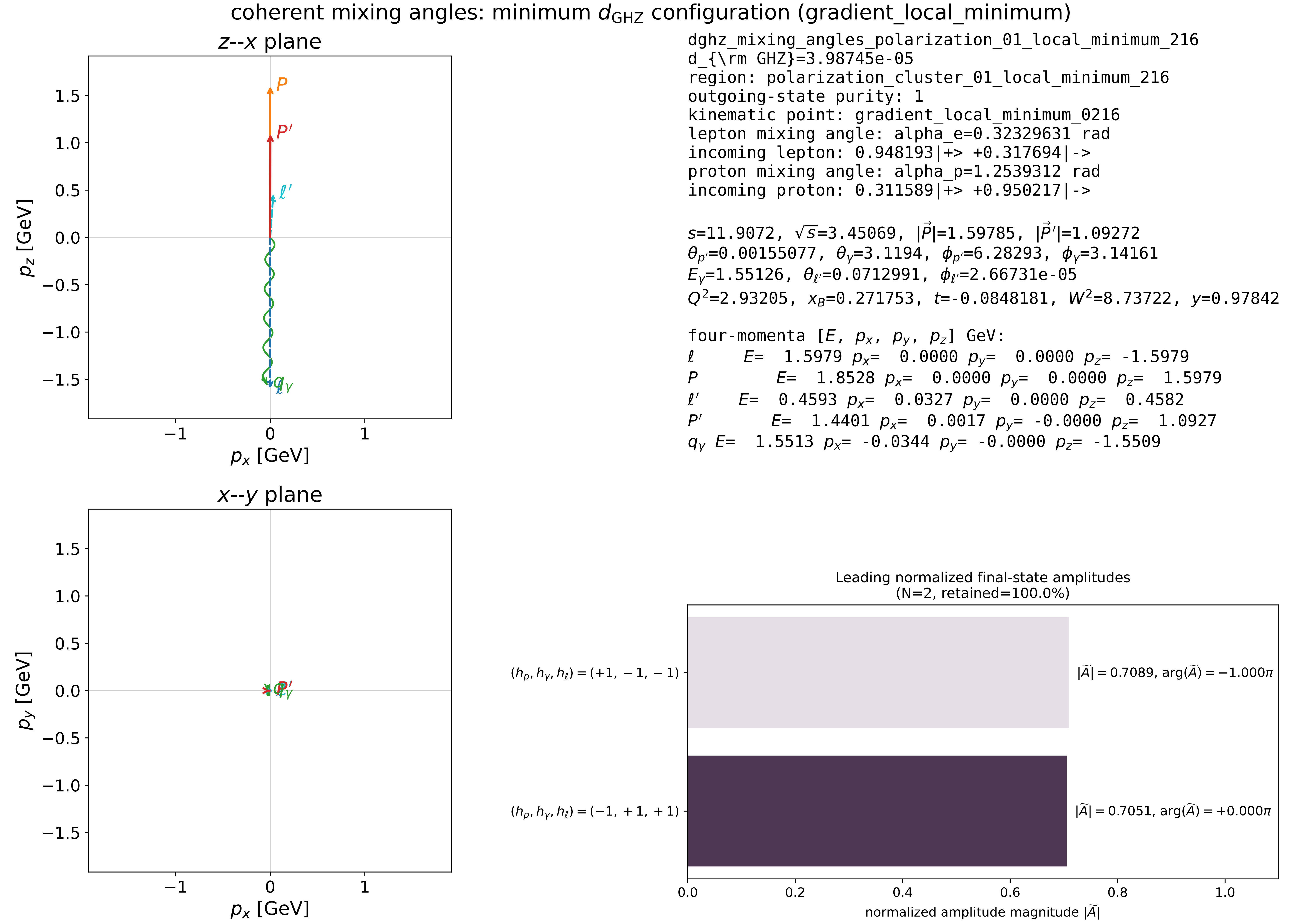


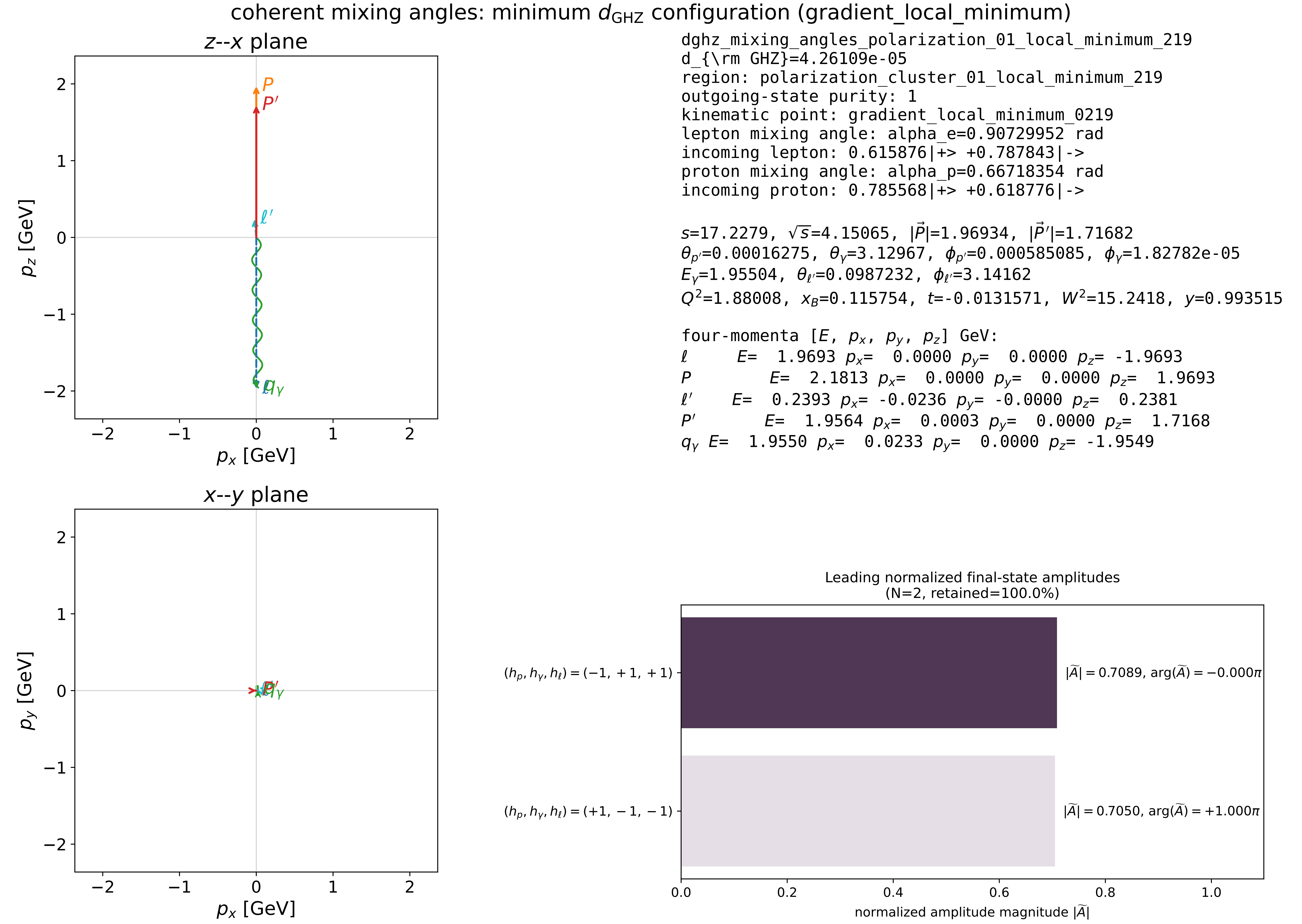

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
P'
ℓ'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_198
d_{\rm GHZ}=2.02825e-05
region: polarization_cluster_01_local_minimum_198
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0198
lepton mixing angle: alpha_e=0.31842078 rad
incoming lepton: 0.949731|+> +0.313067|->
proton mixing angle: alpha_p=1.2260476 rad
incoming proton: 0.33796|+> +0.94116|->
s=2.62187, $\sqrt{s}$=1.61922, $|\vec{P}|$=0.537922, $|\vec{P}'|$=0.373454
$\theta_{p'}$=0.00509094, θ_γ=3.11889, $\phi_{p'}$=3.14157, ϕ_γ=3.14159
E_γ=0.491231, $\theta_{\ell'}$=0.110485, $\phi_{\ell'}$=6.28318
Q^2=0.253934, x_B=0.157469, t=-0.0219156, W^2=2.23851, y=0.925704
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5379 p_x= 0.0000 p_y= 0.0000 p_z= -0.5379
P E= 1.0813 p_x= 0.0000 p_y= 0.0000 p_z= 0.5379
ℓ' E= 0.1184 p_x= 0.0131 p_y= -0.0000 p_z= 0.1177
P' E= 1.0096 p_x= -0.0019 p_y= 0.0000 p_z= 0.3734
q_γ E= 0.4912 p_x= -0.0112 p_y= 0.0000 p_z= -0.4911
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=99.7%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7075$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7043$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P'
ℓ'
ℓ
q_γ
x--y plane
p_y [GeV]
dghz_mixing_angles_polarization_01_local_minimum_203
d_{\rm GHZ}=2.34128e-05
region: polarization_cluster_01_local_minimum_203
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0203
lepton mixing angle: alpha_e=0.64267685 rad
incoming lepton: 0.800494|+> +0.59934|->
proton mixing angle: alpha_p=0.93529282 rad
incoming proton: 0.593583|+> +0.804773|->
s=24.8566, $\sqrt{s}$=4.98564, $|\vec{P}|$=2.40458, $|\vec{P}'|$=2.02645
$\theta_{p'}$=0.000191071, θ_γ=3.12775, $\phi_{p'}$=6.28238, ϕ_γ=3.14161
E_γ=2.38892, $\theta_{\ell'}$=0.0899866, $\phi_{\ell'}$=2.49864e-05
Q^2=3.49121, x_B=0.146436, t=-0.0218485, W^2=21.2299, y=0.994351
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4046 p_x= 0.0000 p_y= 0.0000 p_z= -2.4046
P E= 2.5811 p_x= 0.0000 p_y= 0.0000 p_z= 2.4046
ℓ' E= 0.3637 p_x= 0.0327 p_y= 0.0000 p_z= 0.3622
P' E= 2.2330 p_x= 0.0004 p_y= -0.0000 p_z= 2.0264
q_γ E= 2.3889 p_x= -0.0331 p_y= -0.0000 p_z= -2.3887
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7086$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7055$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

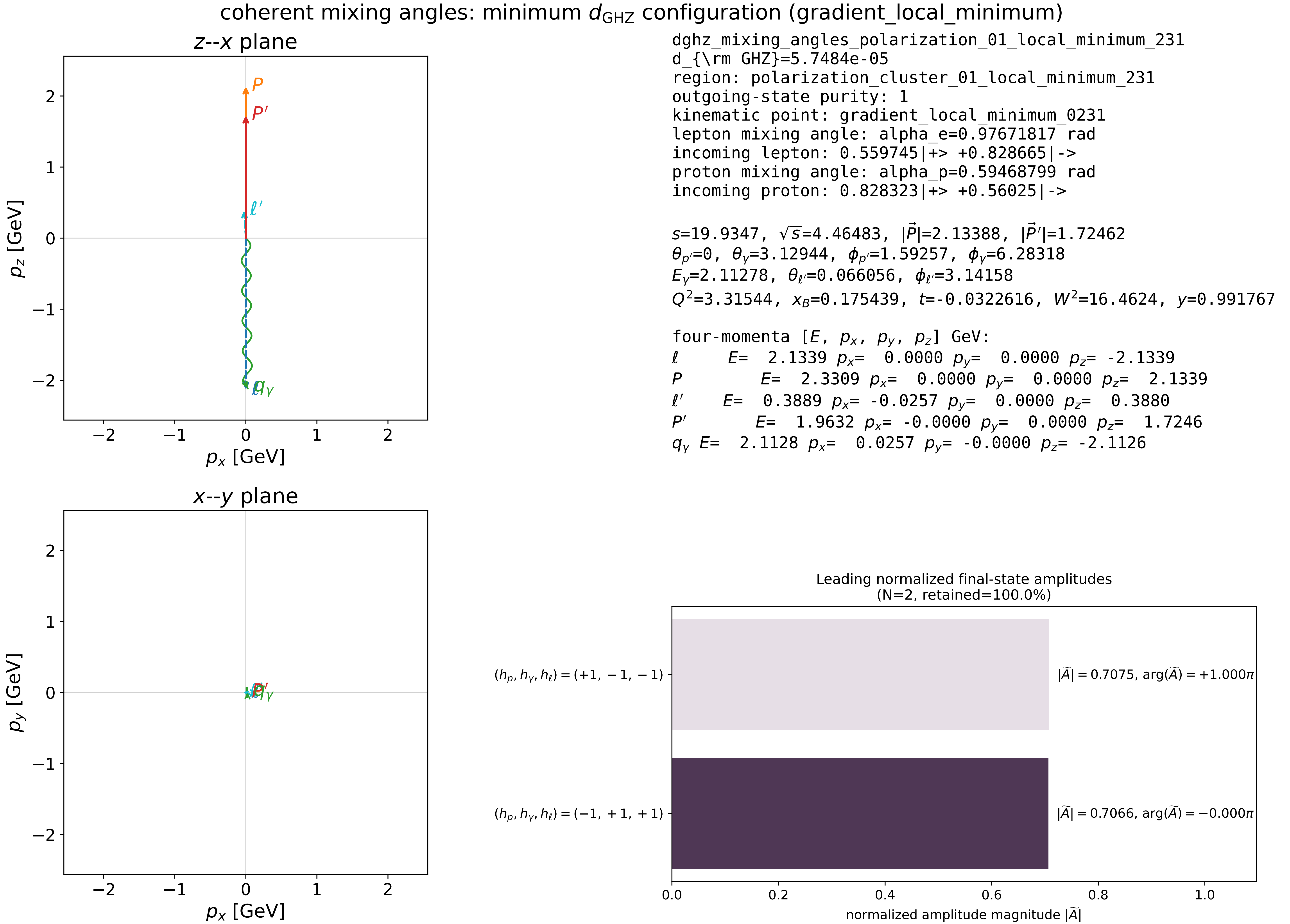

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
P'
ℓ'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_206
d_{\rm GHZ}=2.59585e-05
region: polarization_cluster_01_local_minimum_206
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0206
lepton mixing angle: alpha_e=1.1210285 rad
incoming lepton: 0.434756|+> +0.900548|->
proton mixing angle: alpha_p=0.45230063 rad
incoming proton: 0.899444|+> +0.437036|->
s=24.8343, $\sqrt{s}$=4.9834, $|\vec{P}|$=2.40342, $|\vec{P}'|$=1.50557
$\theta_{p'}$=0.000624821, θ_γ=3.13227, $\phi_{p'}$=6.28282, ϕ_γ=0
E_γ=2.35745, $\theta_{\ell'}$=0.0268985, $\phi_{\ell'}$=3.14158
Q^2=8.19023, x_B=0.346279, t=-0.156318, W^2=16.3417, y=0.987378
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.4034 p_x= 0.0000 p_y= 0.0000 p_z= -2.4034
P E= 2.5800 p_x= 0.0000 p_y= 0.0000 p_z= 2.4034
ℓ' E= 0.8521 p_x= -0.0229 p_y= 0.0000 p_z= 0.8518
P' E= 1.7739 p_x= 0.0009 p_y= -0.0000 p_z= 1.5056
q_γ E= 2.3575 p_x= 0.0220 p_y= 0.0000 p_z= -2.3574
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7080$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7057$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
P'
ℓ'
ℓ
q_γ
dghz_mixing_angles_polarization_01_local_minimum_207
d_{\rm GHZ}=2.62949e-05
region: polarization_cluster_01_local_minimum_207
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0207
lepton mixing angle: alpha_e=1.0858098 rad
incoming lepton: 0.466196|+> +0.884681|->
proton mixing angle: alpha_p=0.47680636 rad
incoming proton: 0.888465|+> +0.458944|->
s=14.6057, $\sqrt{s}$=3.82175, $|\vec{P}|$=1.79576, $|\vec{P}'|$=1.39318
$\theta_{p'}$=0.000700268, θ_γ=3.12182, $\phi_{p'}$=3.14146, ϕ_γ=1.35121e-05
E_γ=1.7671, $\theta_{\ell'}$=0.0906712, $\phi_{\ell'}$=3.14161
Q^2=2.68895, x_B=0.198481, t=-0.0420382, W^2=11.7385, y=0.987013
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7958 p_x= 0.0000 p_y= 0.0000 p_z= -1.7958
P E= 2.0260 p_x= 0.0000 p_y= 0.0000 p_z= 1.7958
ℓ' E= 0.3751 p_x= -0.0340 p_y= -0.0000 p_z= 0.3736
P' E= 1.6795 p_x= -0.0010 p_y= 0.0000 p_z= 1.3932
q_γ E= 1.7671 p_x= 0.0349 p_y= 0.0000 p_z= -1.7668
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7087$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7053$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

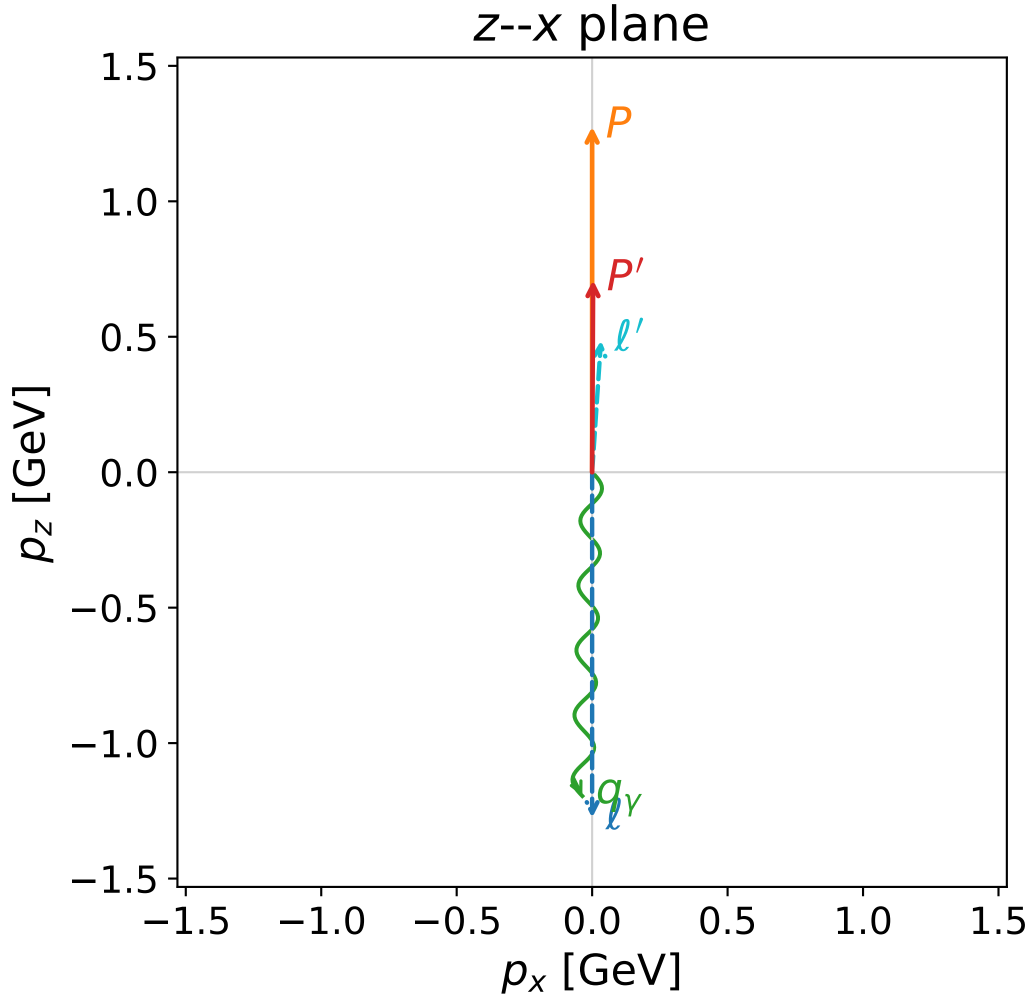

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_216
d_{\rm GHZ}=3.98745e-05
region: polarization_cluster_01_local_minimum_216
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0216
lepton mixing angle: alpha_e=0.32329631 rad
incoming lepton: 0.948193|+> +0.317694|->
proton mixing angle: alpha_p=1.2539312 rad
incoming proton: 0.311589|+> +0.950217|->
s=11.9072, $\sqrt{s}$=3.45069, $|\vec{P}|$=1.59785, $|\vec{P}'|$=1.09272
$\theta_{p'}$=0.00155077, θ_γ=3.1194, $\phi_{p'}$=6.28293, ϕ_γ=3.14161
E_γ=1.55126, $\theta_{\ell'}$=0.0712991, $\phi_{\ell'}$=2.66731e-05
Q^2=2.93205, x_B=0.271753, t=-0.0848181, W^2=8.73722, y=0.97842
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5979 p_x= 0.0000 p_y= 0.0000 p_z= -1.5979
P E= 1.8528 p_x= 0.0000 p_y= 0.0000 p_z= 1.5979
ℓ' E= 0.4593 p_x= 0.0327 p_y= 0.0000 p_z= 0.4582
P' E= 1.4401 p_x= 0.0017 p_y= -0.0000 p_z= 1.0927
q_γ E= 1.5513 p_x= -0.0344 p_y= -0.0000 p_z= -1.5509
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7089$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7051$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
p_x [GeV]
p_z [GeV]
p_y [GeV]
P
P'
ℓ'
q_γ
dghz_mixing_angles_polarization_01_local_minimum_219
d_{\rm GHZ}=4.26109e-05
region: polarization_cluster_01_local_minimum_219
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0219
lepton mixing angle: alpha_e=0.90729952 rad
incoming lepton: 0.615876|+> +0.787843|->
proton mixing angle: alpha_p=0.66718354 rad
incoming proton: 0.785568|+> +0.618776|->
s=17.2279, $\sqrt{s}$=4.15065, $|\vec{P}|$=1.96934, $|\vec{P}'|$=1.71682
$\theta_{p'}$=0.00016275, θ_γ=3.12967, $\phi_{p'}$=0.000585085, ϕ_γ=1.82782e-05
E_γ=1.95504, $\theta_{\ell'}$=0.0987232, $\phi_{\ell'}$=3.14162
Q^2=1.88008, x_B=0.115754, t=-0.0131571, W^2=15.2418, y=0.993515
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.9693 p_x= 0.0000 p_y= 0.0000 p_z= -1.9693
P E= 2.1813 p_x= 0.0000 p_y= 0.0000 p_z= 1.9693
ℓ' E= 0.2393 p_x= -0.0236 p_y= -0.0000 p_z= 0.2381
P' E= 1.9564 p_x= 0.0003 p_y= 0.0000 p_z= 1.7168
q_γ E= 1.9550 p_x= 0.0233 p_y= 0.0000 p_z= -1.9549
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7089$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7050$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
P'
ℓ'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_231
d_{\rm GHZ}=5.7484e-05
region: polarization_cluster_01_local_minimum_231
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0231
lepton mixing angle: alpha_e=0.97671817 rad
incoming lepton: 0.559745|+> +0.828665|->
proton mixing angle: alpha_p=0.59468799 rad
incoming proton: 0.828323|+> +0.56025|->
s=19.9347, $\sqrt{s}$=4.46483, $|\vec{P}|$=2.13388, $|\vec{P}'|$=1.72462
$\theta_{p'}$=0, θ_γ=3.12944, $\phi_{p'}$=1.59257, ϕ_γ=6.28318
E_γ=2.11278, $\theta_{\ell'}$=0.066056, $\phi_{\ell'}$=3.14158
Q^2=3.31544, x_B=0.175439, t=-0.0322616, W^2=16.4624, y=0.991767
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.1339 p_x= 0.0000 p_y= 0.0000 p_z= -2.1339
P E= 2.3309 p_x= 0.0000 p_y= 0.0000 p_z= 2.1339
ℓ' E= 0.3889 p_x= -0.0257 p_y= 0.0000 p_z= 0.3880
P' E= 1.9632 p_x= -0.0000 p_y= 0.0000 p_z= 1.7246
q_γ E= 2.1128 p_x= 0.0257 p_y= -0.0000 p_z= -2.1126
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7075$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7066$, $\arg(\widetilde{A}) = -0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

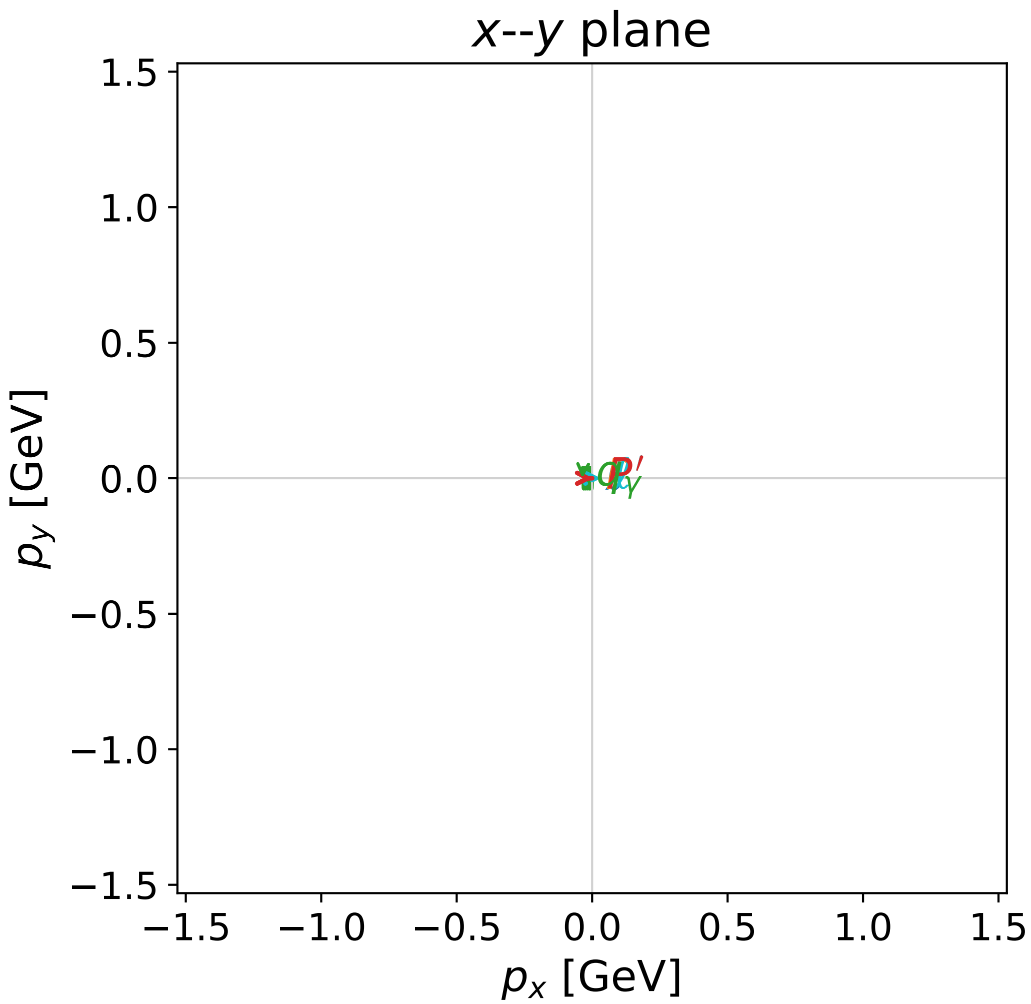
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

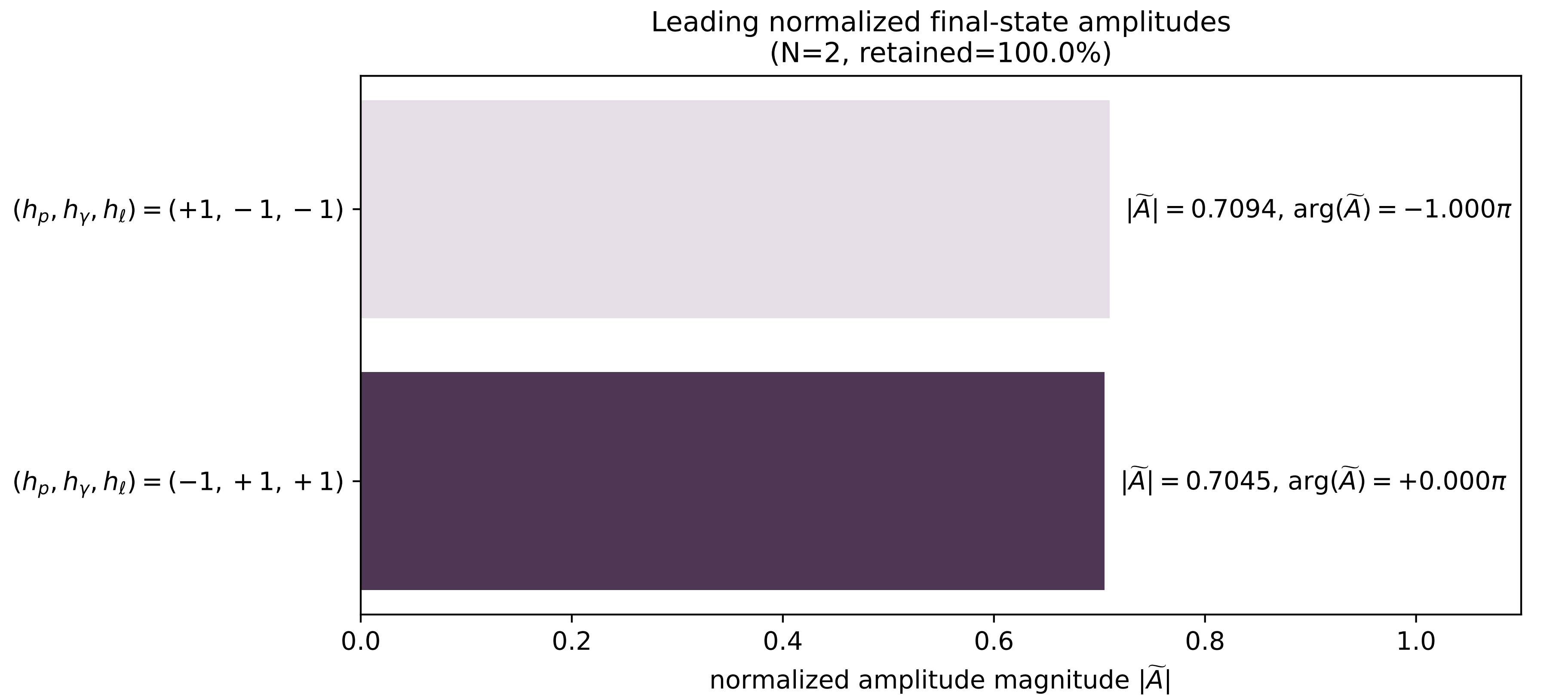


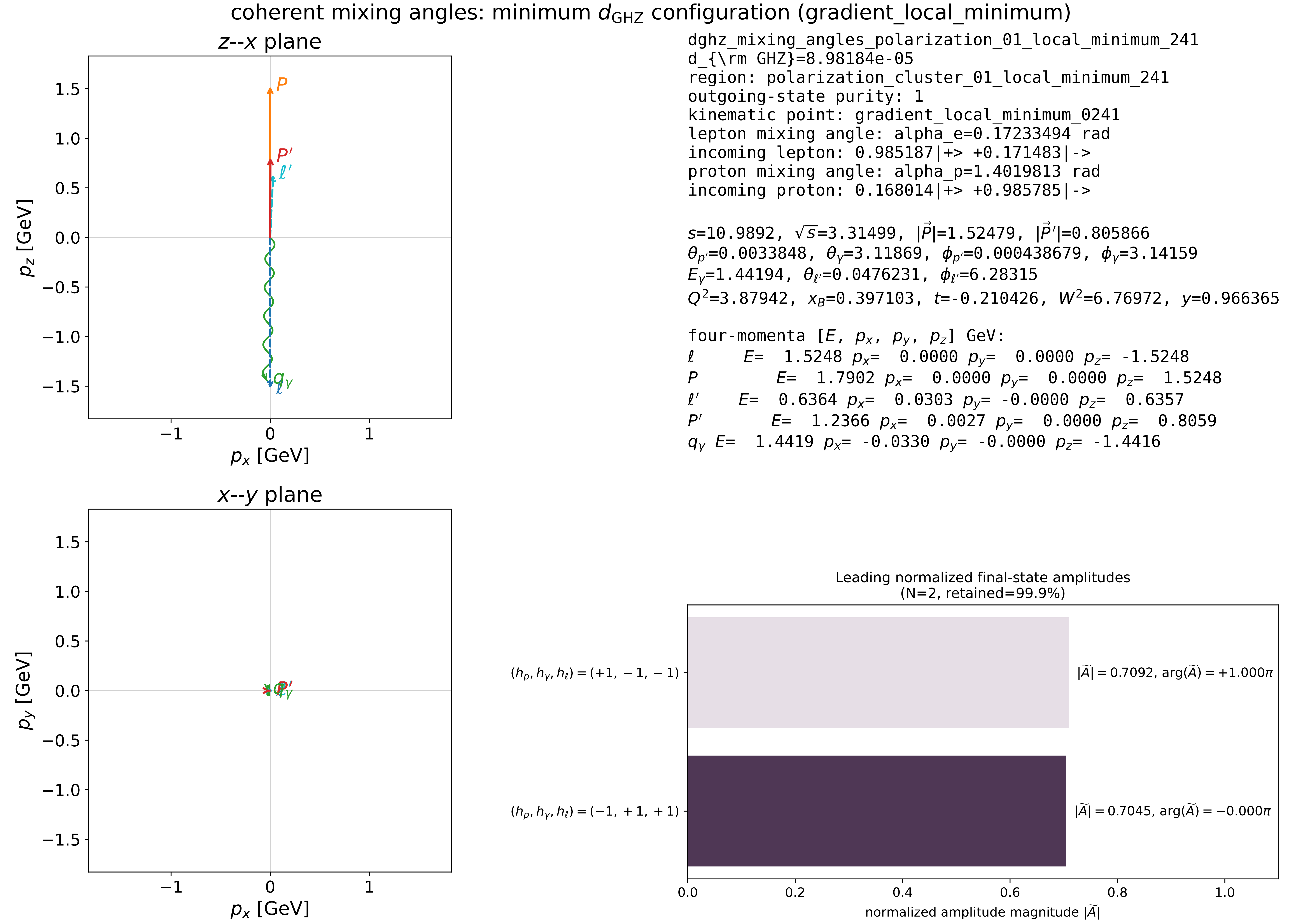

```
dghz_mixing_angles_polarization_01_local_minimum_236
d_{\rm GHZ}=6.73836e-05
region: polarization_cluster_01_local_minimum_236
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0236
lepton mixing angle: alpha_e=0.17076605 rad
incoming lepton: 0.985455|+> +0.169937|->
proton mixing angle: alpha_p=1.4059166 rad
incoming proton: 0.164134|+> +0.986438|->
```

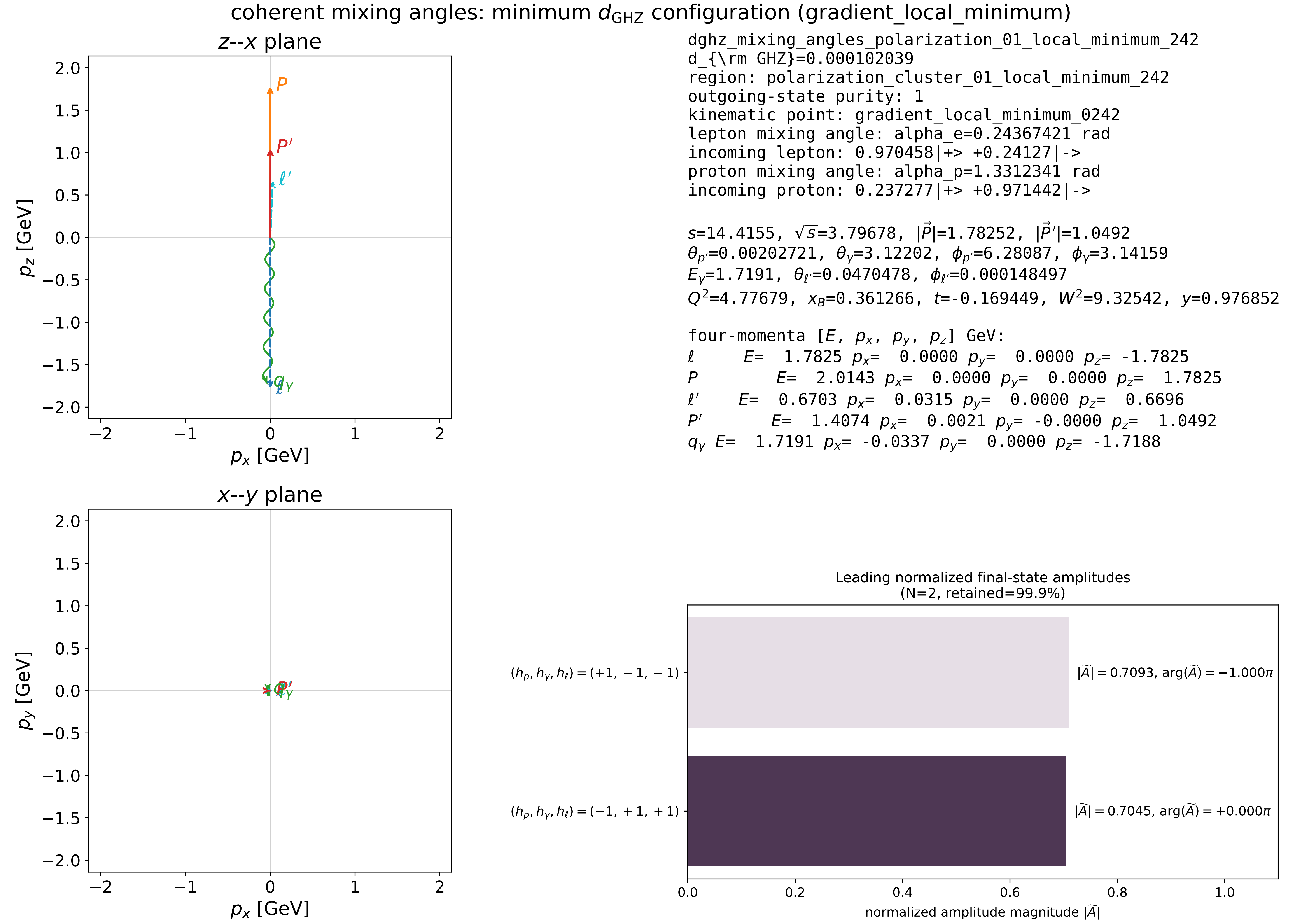
s=8.17465, $\sqrt{s}$=2.85913, $|\vec{P}|$=1.2757, $|\vec{P}'|$=0.709145
$\theta_{p'}$=0.00496652, θ_γ=3.11136, $\phi_{p'}$=6.28315, ϕ_γ=3.14159
E_γ=1.19591, $\theta_{\ell'}$=0.0670059, $\phi_{\ell'}$=4.64873e-06
Q^2=2.48393, x_B=0.355249, t=-0.154922, W^2=5.38799, y=0.958502

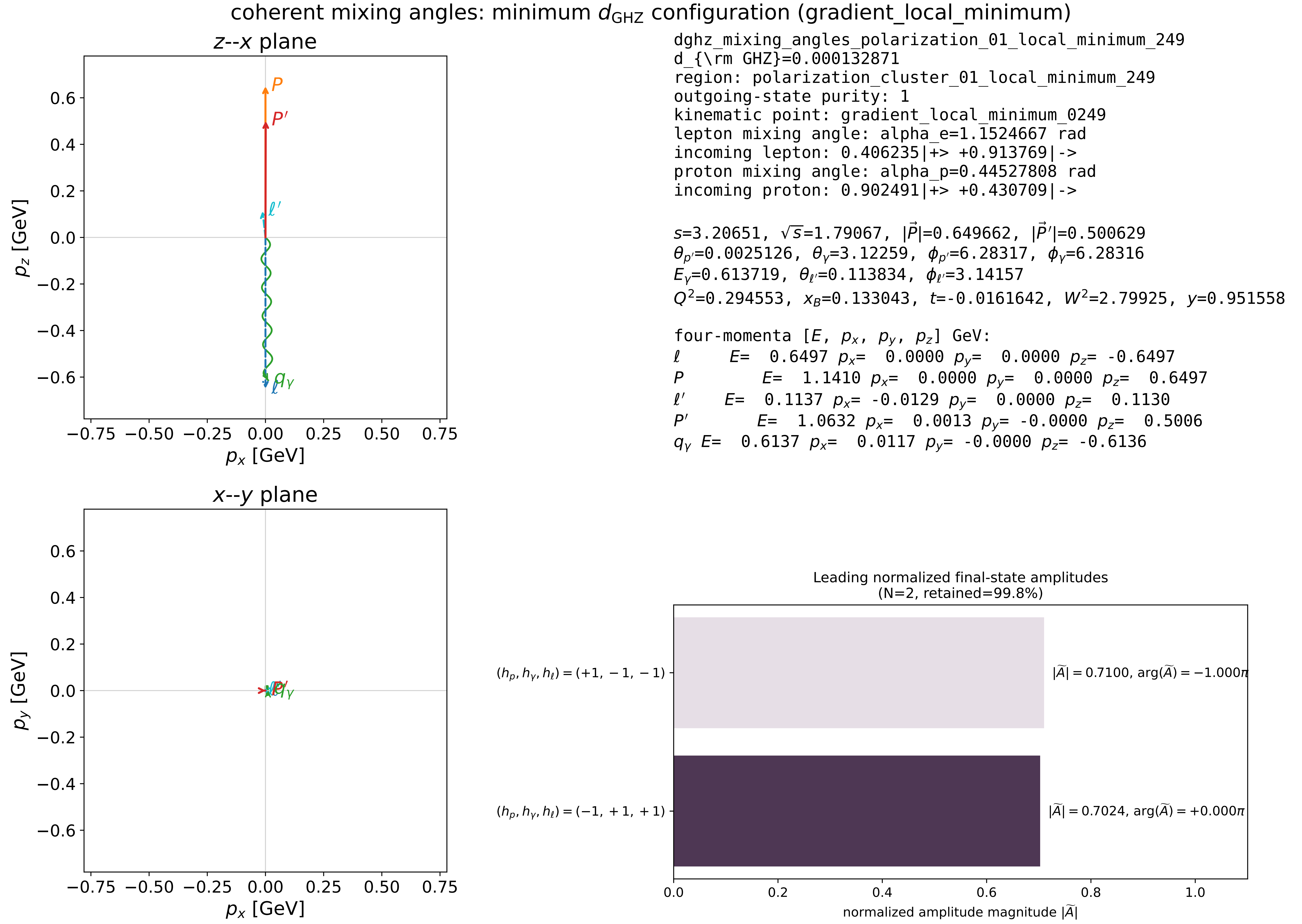
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2757 p_x= 0.0000 p_y= 0.0000 p_z= -1.2757
P E= 1.5834 p_x= 0.0000 p_y= 0.0000 p_z= 1.2757
ℓ' E= 0.4873 p_x= 0.0326 p_y= 0.0000 p_z= 0.4862
P' E= 1.1759 p_x= 0.0035 p_y= -0.0000 p_z= 0.7091
q_γ E= 1.1959 p_x= -0.0362 p_y= -0.0000 p_z= -1.1954

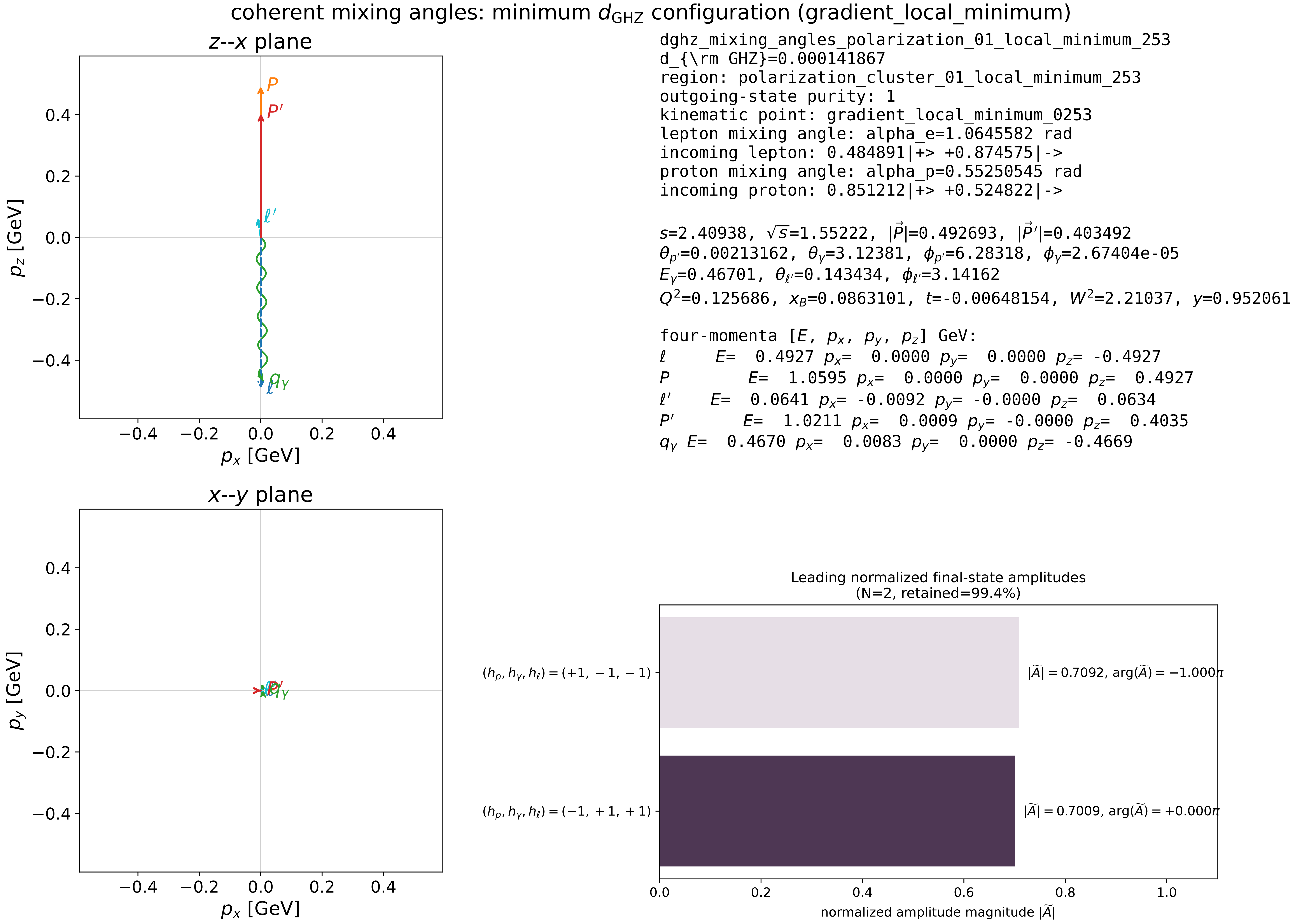


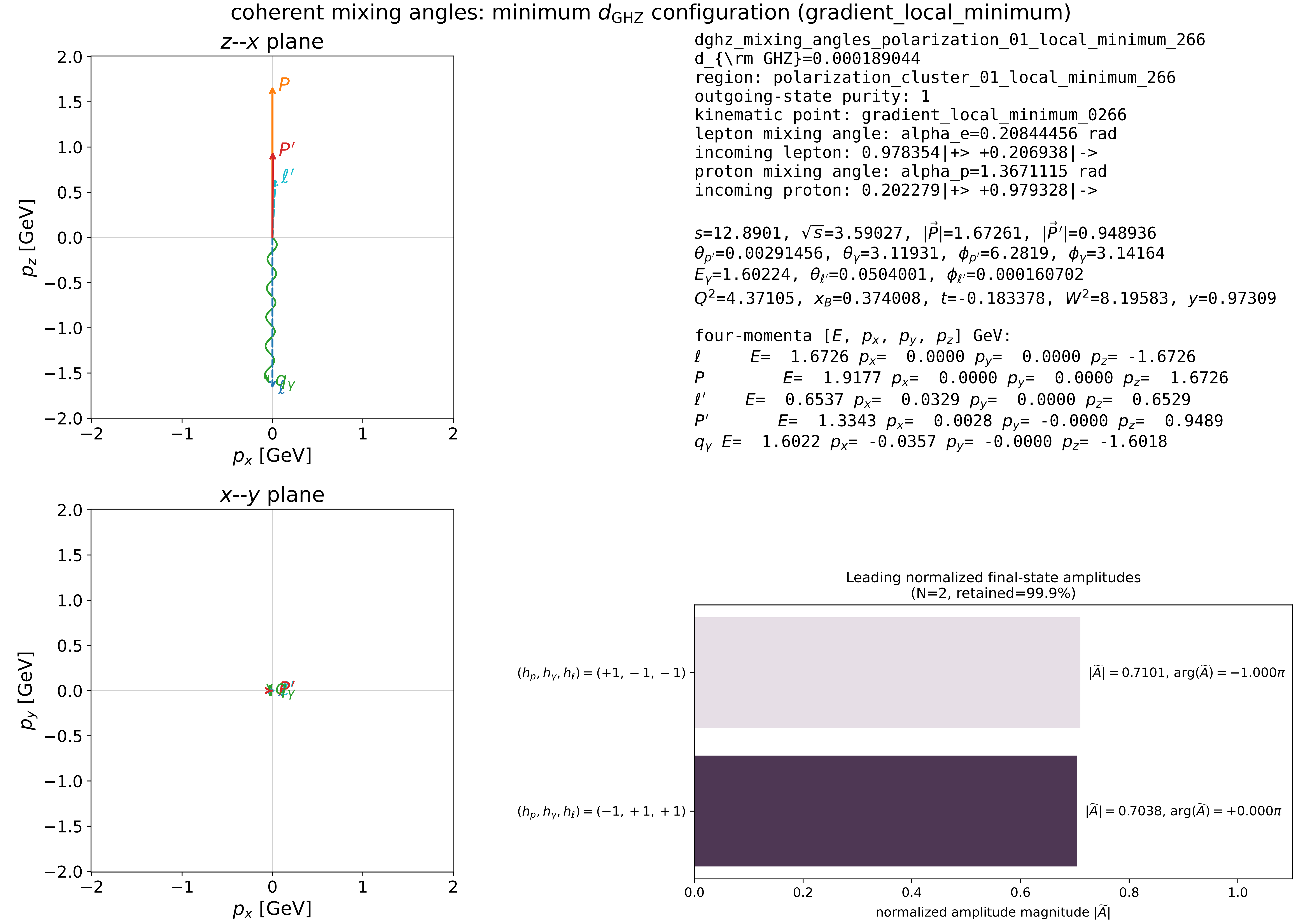



coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P′
ℓ′
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_241
d_{\rm GHZ}=8.98184e-05
region: polarization_cluster_01_local_minimum_241
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0241
lepton mixing angle: alpha_e=0.17233494 rad
incoming lepton: 0.985187|+> +0.171483|->
proton mixing angle: alpha_p=1.4019813 rad
incoming proton: 0.168014|+> +0.985785|->
s=10.9892, $\sqrt{s}$=3.31499, $|\vec{P}|$=1.52479, $|\vec{P}'|$=0.805866
$\theta_{p'}$=0.0033848, θ_γ=3.11869, $\phi_{p'}$=0.000438679, ϕ_γ=3.14159
E_γ=1.44194, $\theta_{\ell'}$=0.0476231, $\phi_{\ell'}$=6.28315
Q^2=3.87942, x_B=0.397103, t=-0.210426, W^2=6.76972, y=0.966365
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5248 p_x= 0.0000 p_y= 0.0000 p_z= -1.5248
P E= 1.7902 p_x= 0.0000 p_y= 0.0000 p_z= 1.5248
ℓ' E= 0.6364 p_x= 0.0303 p_y= -0.0000 p_z= 0.6357
P' E= 1.2366 p_x= 0.0027 p_y= 0.0000 p_z= 0.8059
q_γ E= 1.4419 p_x= -0.0330 p_y= -0.0000 p_z= -1.4416
Leading normalized final-state amplitudes
(N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7092$, $\arg(\widetilde{A}) = +1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7045$, $\arg(\widetilde{A}) = -0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

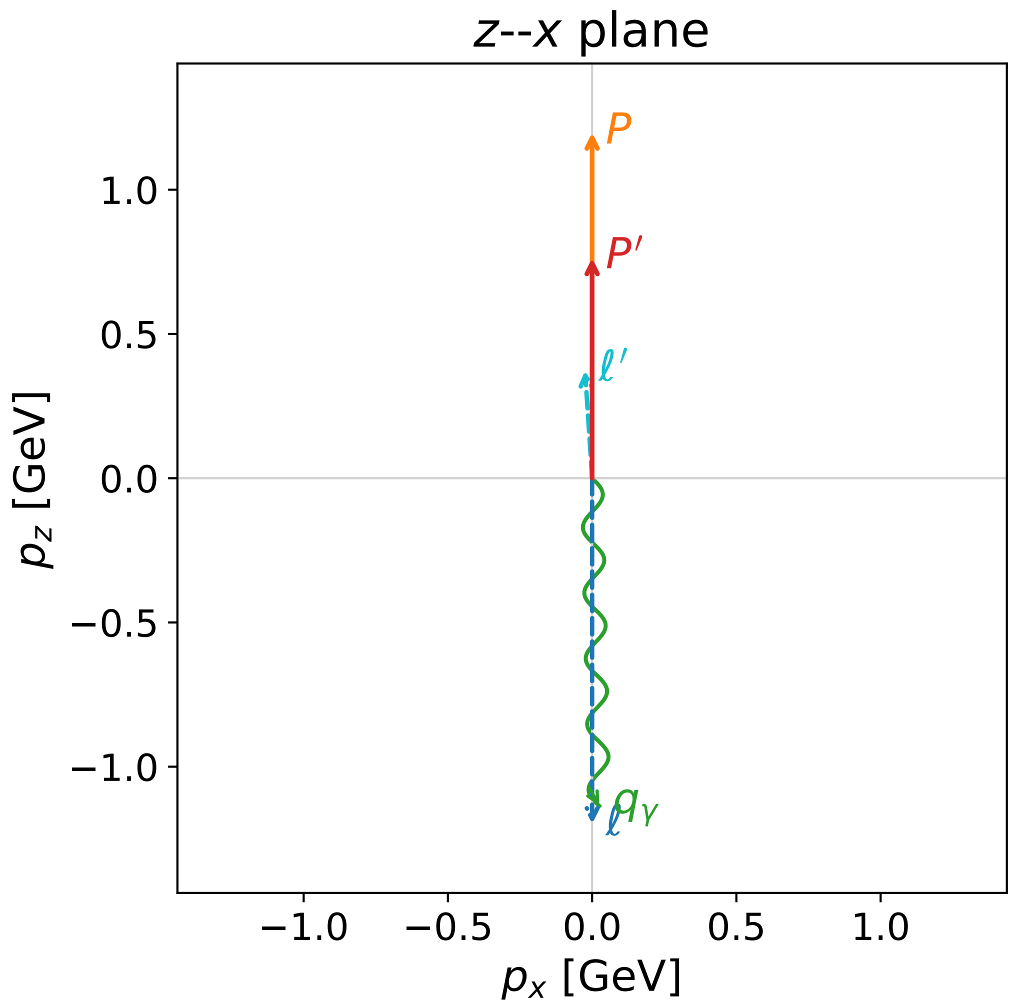

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P'
ℓ'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_242
d_{\rm GHZ}=0.000102039
region: polarization_cluster_01_local_minimum_242
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0242
lepton mixing angle: alpha_e=0.24367421 rad
incoming lepton: 0.970458|+> +0.24127|->
proton mixing angle: alpha_p=1.3312341 rad
incoming proton: 0.237277|+> +0.971442|->
s=14.4155, $\sqrt{s}$=3.79678, $|\vec{P}|$=1.78252, $|\vec{P}'|$=1.0492
$\theta_{p'}$=0.00202721, θ_γ=3.12202, $\phi_{p'}$=6.28087, ϕ_γ=3.14159
E_γ=1.7191, $\theta_{\ell'}$=0.0470478, $\phi_{\ell'}$=0.000148497
Q^2=4.77679, x_B=0.361266, t=-0.169449, W^2=9.32542, y=0.976852
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.7825 p_x= 0.0000 p_y= 0.0000 p_z= -1.7825
P E= 2.0143 p_x= 0.0000 p_y= 0.0000 p_z= 1.7825
ℓ' E= 0.6703 p_x= 0.0315 p_y= 0.0000 p_z= 0.6696
P' E= 1.4074 p_x= 0.0021 p_y= -0.0000 p_z= 1.0492
q_γ E= 1.7191 p_x= -0.0337 p_y= 0.0000 p_z= -1.7188
Leading normalized final-state amplitudes
(N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7093$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7045$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P'
ℓ'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_249
d_{\rm GHZ}=0.000132871
region: polarization_cluster_01_local_minimum_249
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0249
lepton mixing angle: alpha_e=1.1524667 rad
incoming lepton: 0.406235|+> +0.913769|->
proton mixing angle: alpha_p=0.44527808 rad
incoming proton: 0.902491|+> +0.430709|->
s=3.20651, $\sqrt{s}$=1.79067, $|\vec{P}|$=0.649662, $|\vec{P}'|$=0.500629
$\theta_{p'}$=0.0025126, θ_γ=3.12259, $\phi_{p'}$=6.28317, ϕ_γ=6.28316
E_γ=0.613719, $\theta_{\ell'}$=0.113834, $\phi_{\ell'}$=3.14157
Q^2=0.294553, x_B=0.133043, t=-0.0161642, W^2=2.79925, y=0.951558
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6497 p_x= 0.0000 p_y= 0.0000 p_z= -0.6497
P E= 1.1410 p_x= 0.0000 p_y= 0.0000 p_z= 0.6497
ℓ' E= 0.1137 p_x= -0.0129 p_y= 0.0000 p_z= 0.1130
P' E= 1.0632 p_x= 0.0013 p_y= -0.0000 p_z= 0.5006
q_γ E= 0.6137 p_x= 0.0117 p_y= -0.0000 p_z= -0.6136
Leading normalized final-state amplitudes
(N=2, retained=99.8%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7100$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7024$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_253
d_{\rm GHZ}=0.000141867
region: polarization_cluster_01_local_minimum_253
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0253
lepton mixing angle: alpha_e=1.0645582 rad
incoming lepton: 0.484891|+> +0.874575|->
proton mixing angle: alpha_p=0.55250545 rad
incoming proton: 0.851212|+> +0.524822|->
s=2.40938, $\sqrt{s}$=1.55222, $|\vec{P}|$=0.492693, $|\vec{P}'|$=0.403492
$\theta_{p'}$=0.00213162, θ_γ=3.12381, $\phi_{p'}$=6.28318, ϕ_γ=2.67404e-05
E_γ=0.46701, $\theta_{\ell'}$=0.143434, $\phi_{\ell'}$=3.14162
Q^2=0.125686, x_B=0.0863101, t=-0.00648154, W^2=2.21037, y=0.952061
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.4927 p_x= 0.0000 p_y= 0.0000 p_z= -0.4927
P E= 1.0595 p_x= 0.0000 p_y= 0.0000 p_z= 0.4927
ℓ' E= 0.0641 p_x= -0.0092 p_y= -0.0000 p_z= 0.0634
P' E= 1.0211 p_x= 0.0009 p_y= -0.0000 p_z= 0.4035
q_γ E= 0.4670 p_x= 0.0083 p_y= 0.0000 p_z= -0.4669
Leading normalized final-state amplitudes
(N=2, retained=99.4%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7092$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7009$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
P'
ℓ'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_266
d_{\rm GHZ}=0.000189044
region: polarization_cluster_01_local_minimum_266
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0266
lepton mixing angle: alpha_e=0.20844456 rad
incoming lepton: 0.978354|+> +0.206938|->
proton mixing angle: alpha_p=1.3671115 rad
incoming proton: 0.202279|+> +0.979328|->
s=12.8901, $\sqrt{s}$=3.59027, $|\vec{P}|$=1.67261, $|\vec{P}'|$=0.948936
$\theta_{p'}$=0.00291456, θ_γ=3.11931, $\phi_{p'}$=6.2819, ϕ_γ=3.14164
E_γ=1.60224, $\theta_{\ell'}$=0.0504001, $\phi_{\ell'}$=0.000160702
Q^2=4.37105, x_B=0.374008, t=-0.183378, W^2=8.19583, y=0.97309
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.6726 p_x= 0.0000 p_y= 0.0000 p_z= -1.6726
P E= 1.9177 p_x= 0.0000 p_y= 0.0000 p_z= 1.6726
ℓ' E= 0.6537 p_x= 0.0329 p_y= 0.0000 p_z= 0.6529
P' E= 1.3343 p_x= 0.0028 p_y= -0.0000 p_z= 0.9489
q_γ E= 1.6022 p_x= -0.0357 p_y= -0.0000 p_z= -1.6018
Leading normalized final-state amplitudes
(N=2, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.7101$, $\arg(\widetilde{A}) = -1.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7038$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$

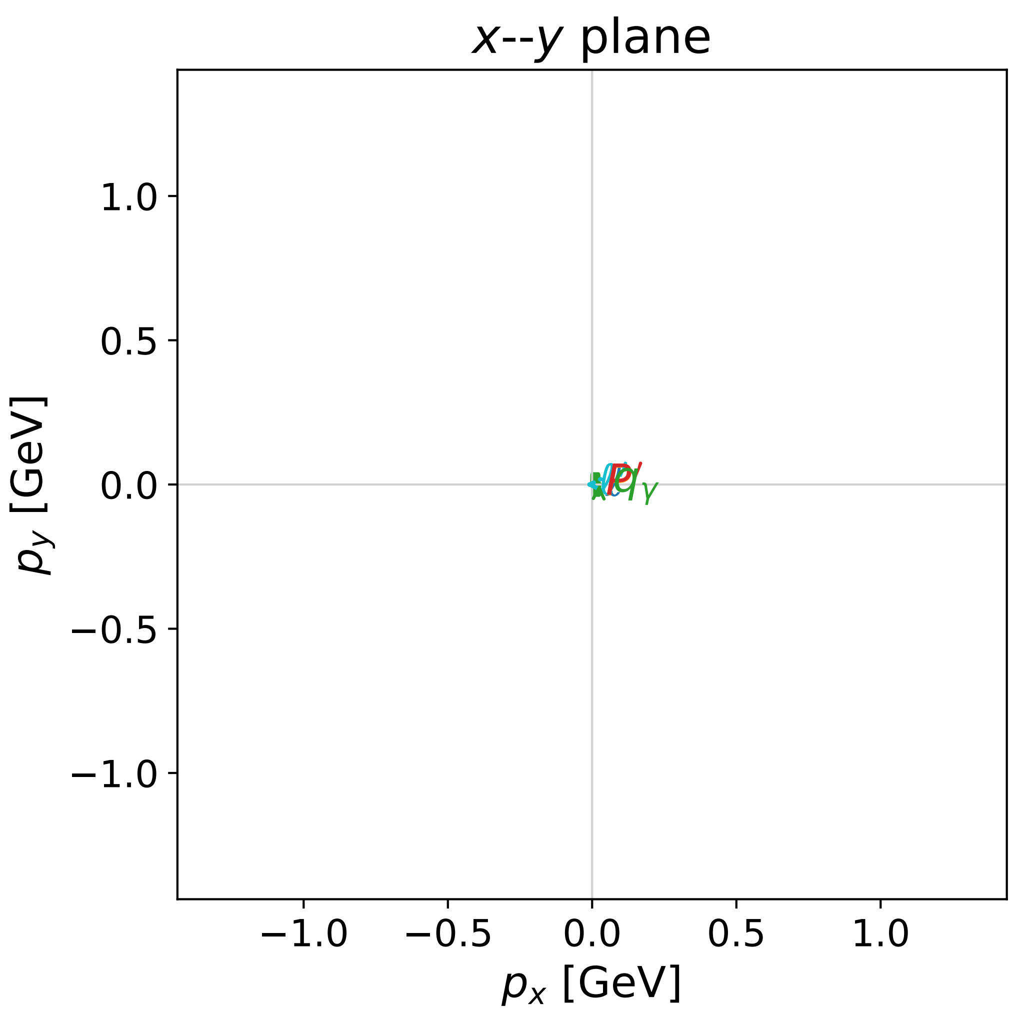
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

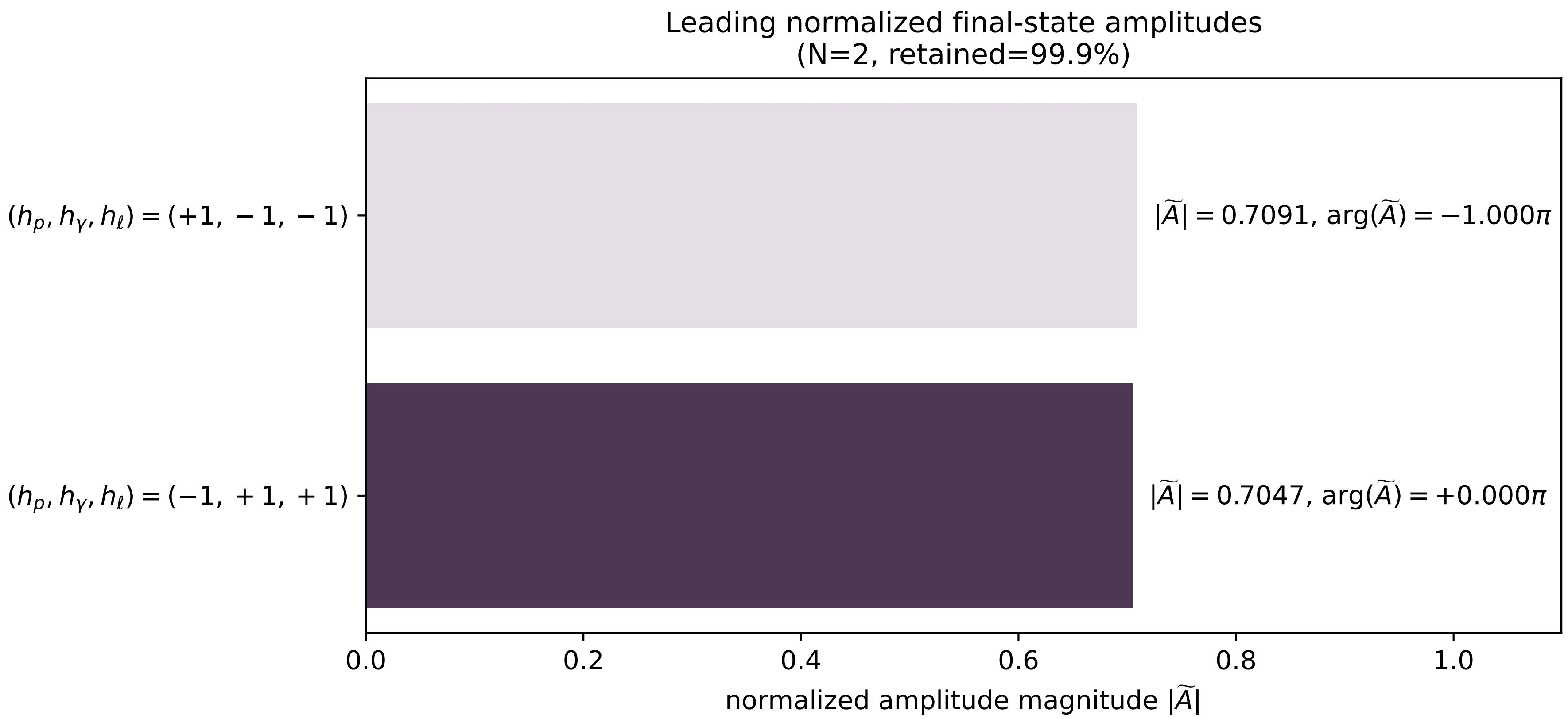


dghz_mixing_angles_polarization_01_local_minimum_314
d_{\rm GHZ}=0.000387584
region: polarization_cluster_01_local_minimum_314
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0314
lepton mixing angle: alpha_e=1.3236924 rad
incoming lepton: 0.244597|+> +0.969625|->
proton mixing angle: alpha_p=0.24858017 rad
incoming proton: 0.969263|+> +0.246028|->

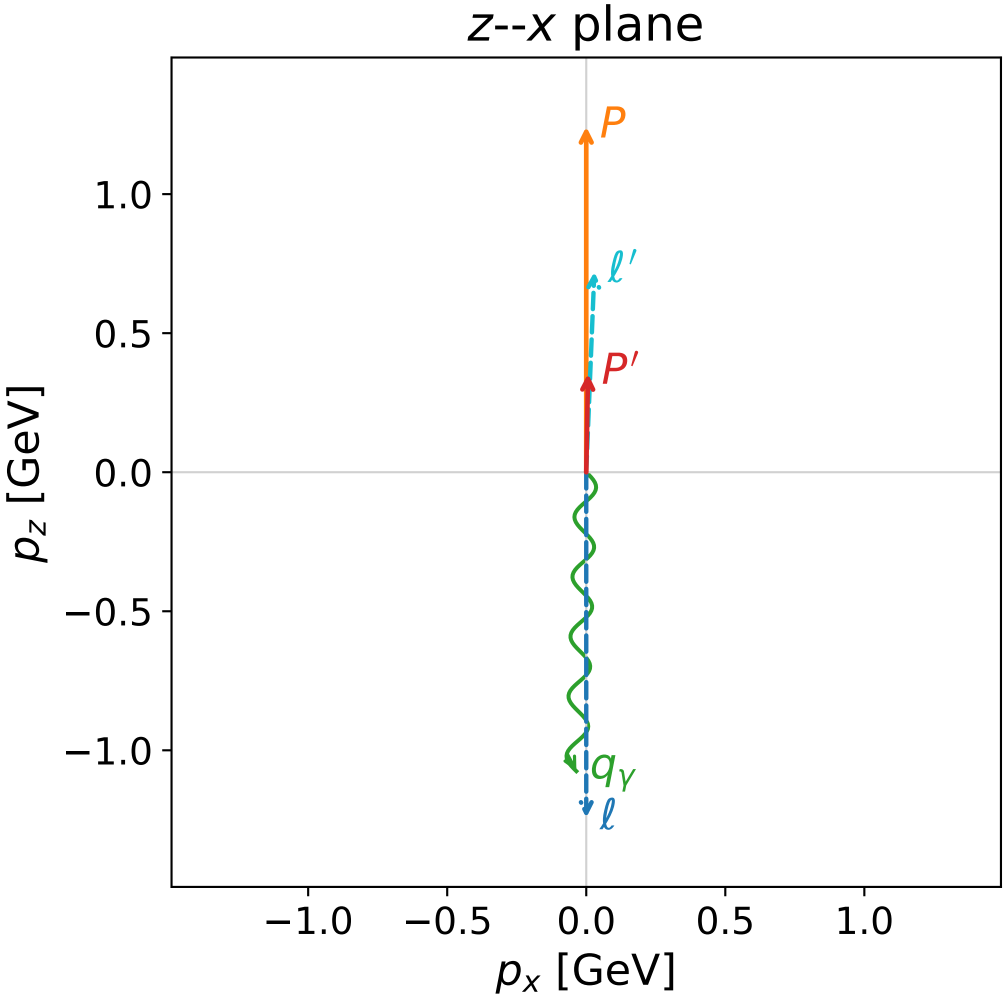
s=7.3952, $\sqrt{s}$=2.71941, $|\vec{P}|$=1.19794, $|\vec{P}'|$=0.763049
$\theta_{p'}$=0, θ_γ=3.11984, $\phi_{p'}$=3.82165, ϕ_γ=2.0169e-05
E_γ=1.13637, $\theta_{\ell'}$=0.0661538, $\phi_{\ell'}$=3.14161
Q^2=1.78954, x_B=0.285346, t=-0.091589, W^2=5.36178, y=0.962567

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.1979 p_x= 0.0000 p_y= 0.0000 p_z= -1.1979
P E= 1.5215 p_x= 0.0000 p_y= 0.0000 p_z= 1.1979
ℓ' E= 0.3739 p_x= -0.0247 p_y= -0.0000 p_z= 0.3731
P' E= 1.2092 p_x= -0.0000 p_y= -0.0000 p_z= 0.7630
q_γ E= 1.1364 p_x= 0.0247 p_y= 0.0000 p_z= -1.1361





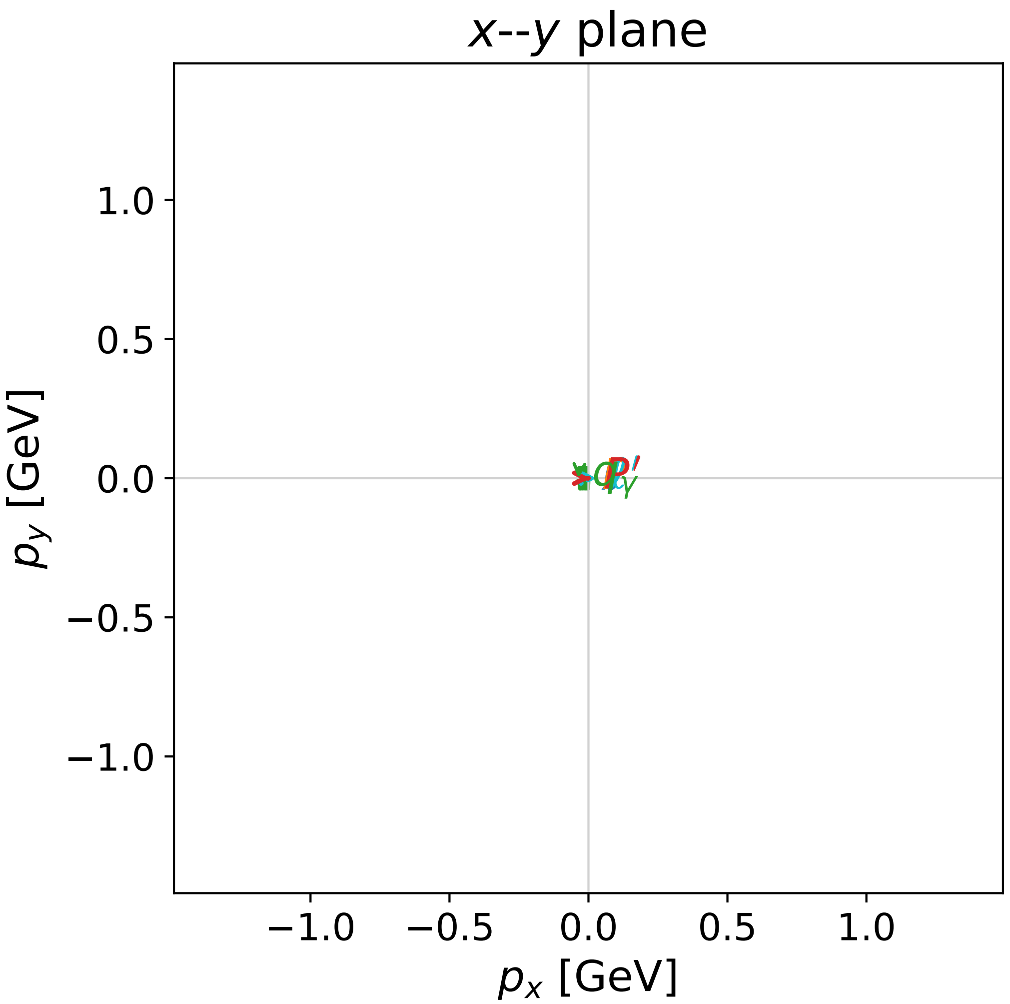
coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)

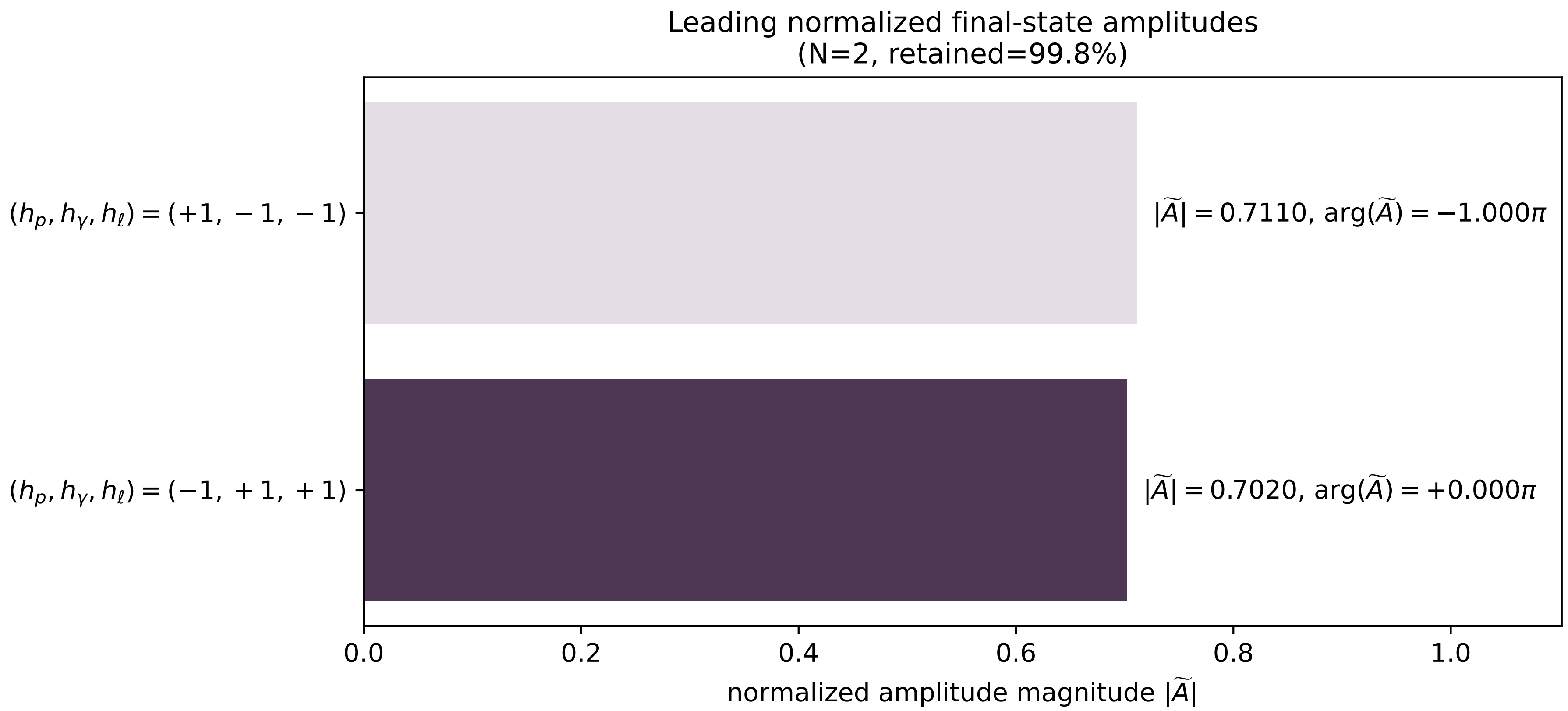


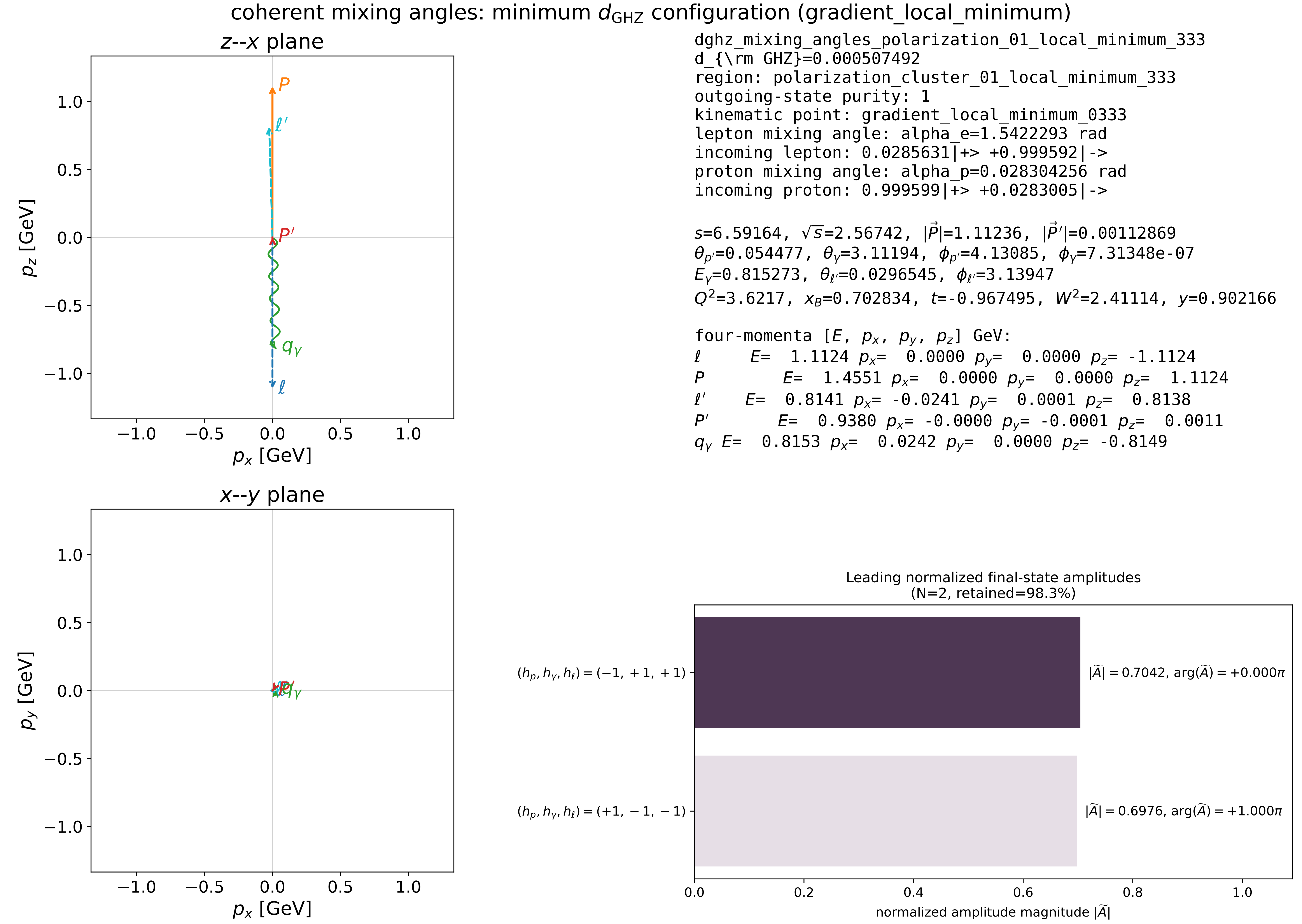

```
dghz_mixing_angles_polarization_01_local_minimum_331
d_{\rm GHZ}=0.000498586
region: polarization_cluster_01_local_minimum_331
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0331
lepton mixing angle: alpha_e=0.063309233 rad
incoming lepton: 0.997997|+> +0.063267|->
proton mixing angle: alpha_p=1.5095946 rad
incoming proton: 0.0611635|+> +0.998128|->
```

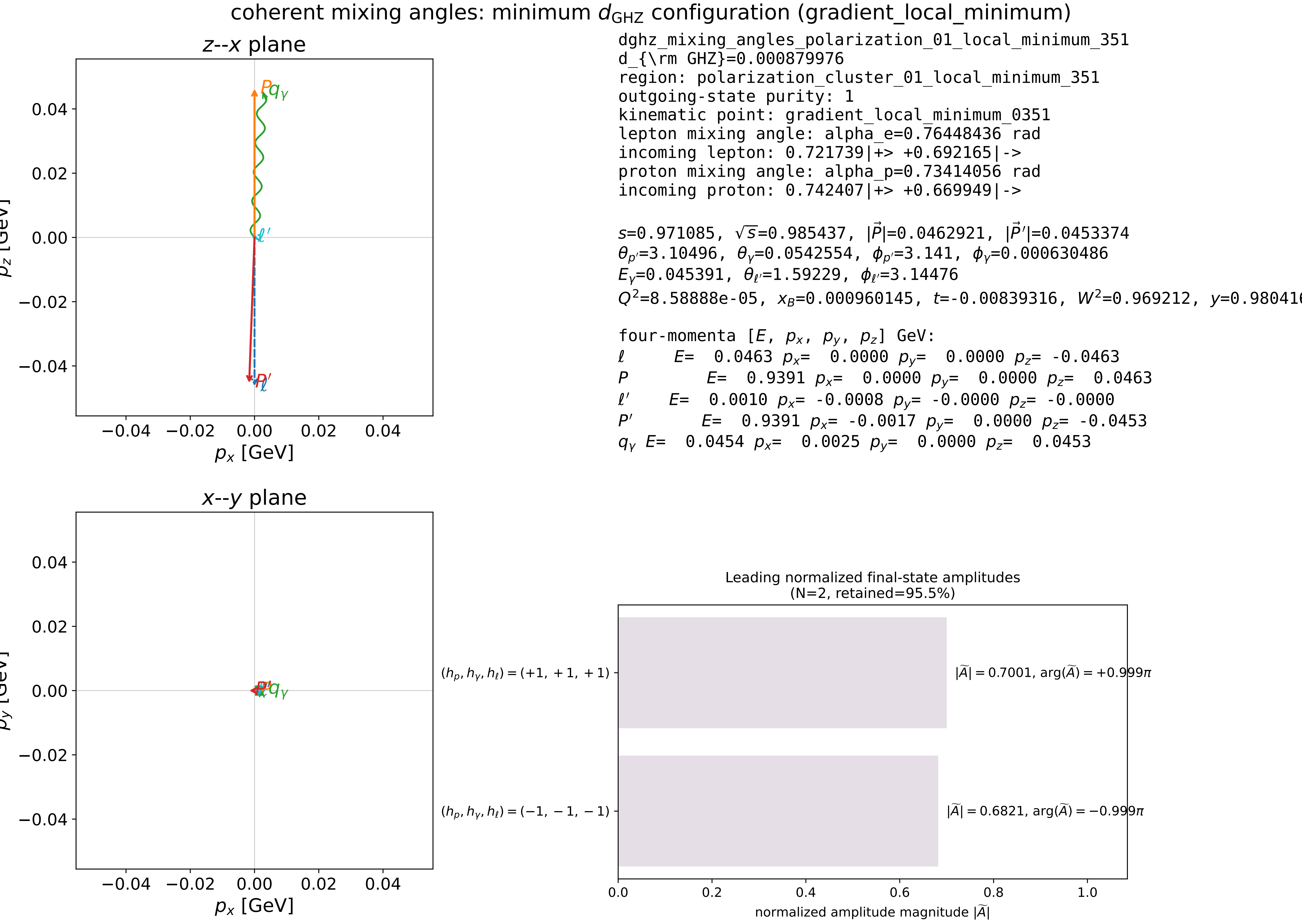
s=7.83845, $\sqrt{s}$=2.79972, $|\vec{P}|$=1.24273, $|\vec{P}'|$=0.354558
$\theta_{p'}$=0.0179191, θ_γ=3.1086, $\phi_{p'}$=6.28227, ϕ_γ=3.14163
E_γ=1.07572, $\theta_{\ell'}$=0.0404006, $\phi_{\ell'}$=0.000247116
Q^2=3.5837, x_B=0.551013, t=-0.481835, W^2=3.79999, y=0.934646

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.2427 p_x= 0.0000 p_y= 0.0000 p_z= -1.2427
P E= 1.5570 p_x= 0.0000 p_y= 0.0000 p_z= 1.2427
ℓ' E= 0.7212 p_x= 0.0291 p_y= 0.0000 p_z= 0.7206
P' E= 1.0028 p_x= 0.0064 p_y= -0.0000 p_z= 0.3545
q_γ E= 1.0757 p_x= -0.0355 p_y= -0.0000 p_z= -1.0751

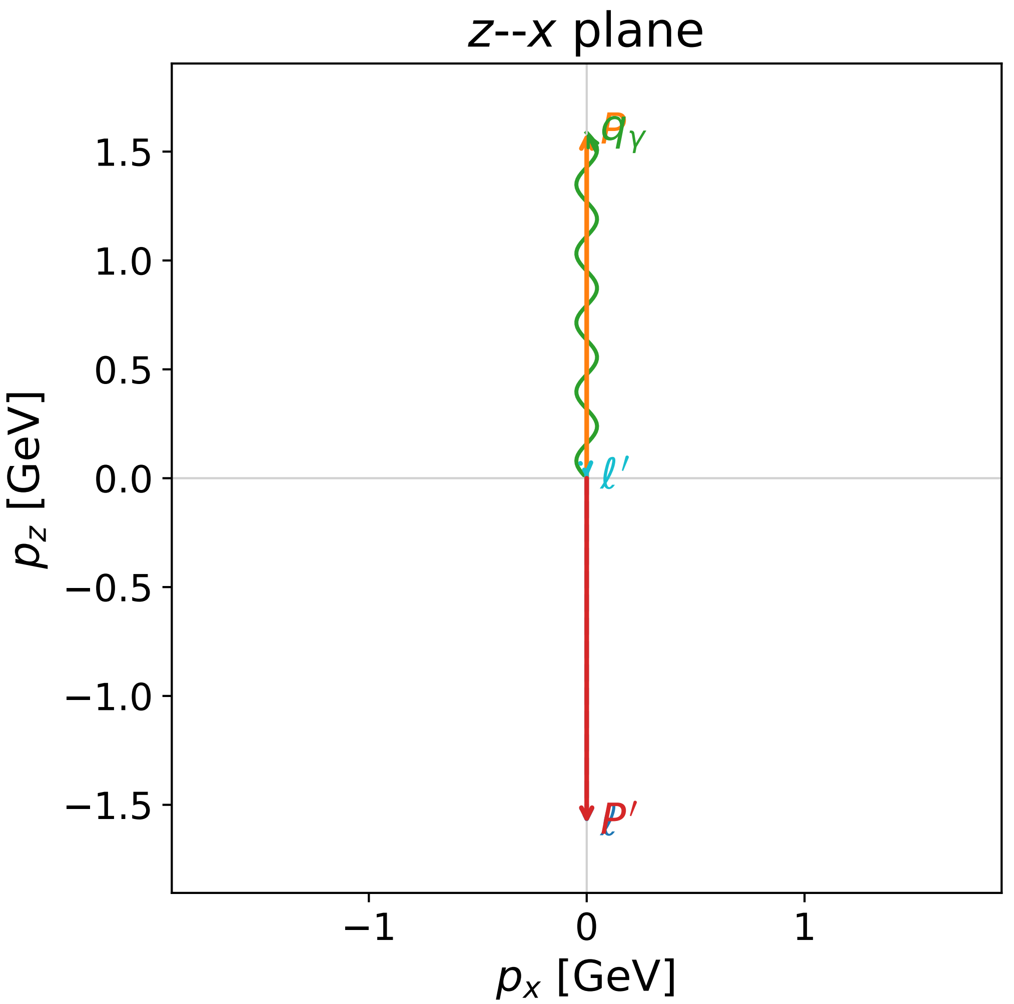





coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
x--y plane
p_z [GeV]
p_x [GeV]
p_y [GeV]
P
ℓ'
P'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_333
d_{\rm GHZ}=0.000507492
region: polarization_cluster_01_local_minimum_333
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0333
lepton mixing angle: alpha_e=1.5422293 rad
incoming lepton: 0.0285631|+> +0.999592|->
proton mixing angle: alpha_p=0.028304256 rad
incoming proton: 0.999599|+> +0.0283005|->
s=6.59164, $\sqrt{s}$=2.56742, $|\vec{P}|$=1.11236, $|\vec{P}'|$=0.00112869
$\theta_{p'}$=0.054477, θ_γ=3.11194, $\phi_{p'}$=4.13085, ϕ_γ=7.31348e-07
E_γ=0.815273, $\theta_{\ell'}$=0.0296545, $\phi_{\ell'}$=3.13947
Q^2=3.6217, x_B=0.702834, t=-0.967495, W^2=2.41114, y=0.902166
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.1124 p_x= 0.0000 p_y= 0.0000 p_z= -1.1124
P E= 1.4551 p_x= 0.0000 p_y= 0.0000 p_z= 1.1124
ℓ' E= 0.8141 p_x= -0.0241 p_y= 0.0001 p_z= 0.8138
P' E= 0.9380 p_x= -0.0000 p_y= -0.0001 p_z= 0.0011
q_γ E= 0.8153 p_x= 0.0242 p_y= 0.0000 p_z= -0.8149
Leading normalized final-state amplitudes
(N=2, retained=98.3%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
$|\widetilde{A}| = 0.7042$, $\arg(\widetilde{A}) = +0.000\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$
$|\widetilde{A}| = 0.6976$, $\arg(\widetilde{A}) = +1.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
q_γ
ℓ'
P'
ℓ
0.04
0.02
0.00
−0.02
−0.04
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_351
d_{\rm GHZ}=0.000879976
region: polarization_cluster_01_local_minimum_351
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0351
lepton mixing angle: alpha_e=0.76448436 rad
incoming lepton: 0.721739|+> +0.692165|->
proton mixing angle: alpha_p=0.73414056 rad
incoming proton: 0.742407|+> +0.669949|->
s=0.971085, $\sqrt{s}$=0.985437, $|\vec{P}|$=0.0462921, $|\vec{P}'|$=0.0453374
$\theta_{p'}$=3.10496, θ_γ=0.0542554, $\phi_{p'}$=3.141, ϕ_γ=0.000630486
E_γ=0.045391, $\theta_{\ell'}$=1.59229, $\phi_{\ell'}$=3.14476
Q^2=8.58888e-05, x_B=0.000960145, t=-0.00839316, W^2=0.969212, y=0.98041
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0463 p_x= 0.0000 p_y= 0.0000 p_z= -0.0463
P E= 0.9391 p_x= 0.0000 p_y= 0.0000 p_z= 0.0463
ℓ' E= 0.0010 p_x= -0.0008 p_y= -0.0000 p_z= -0.0000
P' E= 0.9391 p_x= -0.0017 p_y= 0.0000 p_z= -0.0453
q_γ E= 0.0454 p_x= 0.0025 p_y= 0.0000 p_z= 0.0453
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=95.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7001$, $\arg(\widetilde{A}) = +0.999\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.6821$, $\arg(\widetilde{A}) = -0.999\pi$
0.0
0.2
0.4
0.6
0.8
1.0
normalized amplitude magnitude $|\widetilde{A}|$

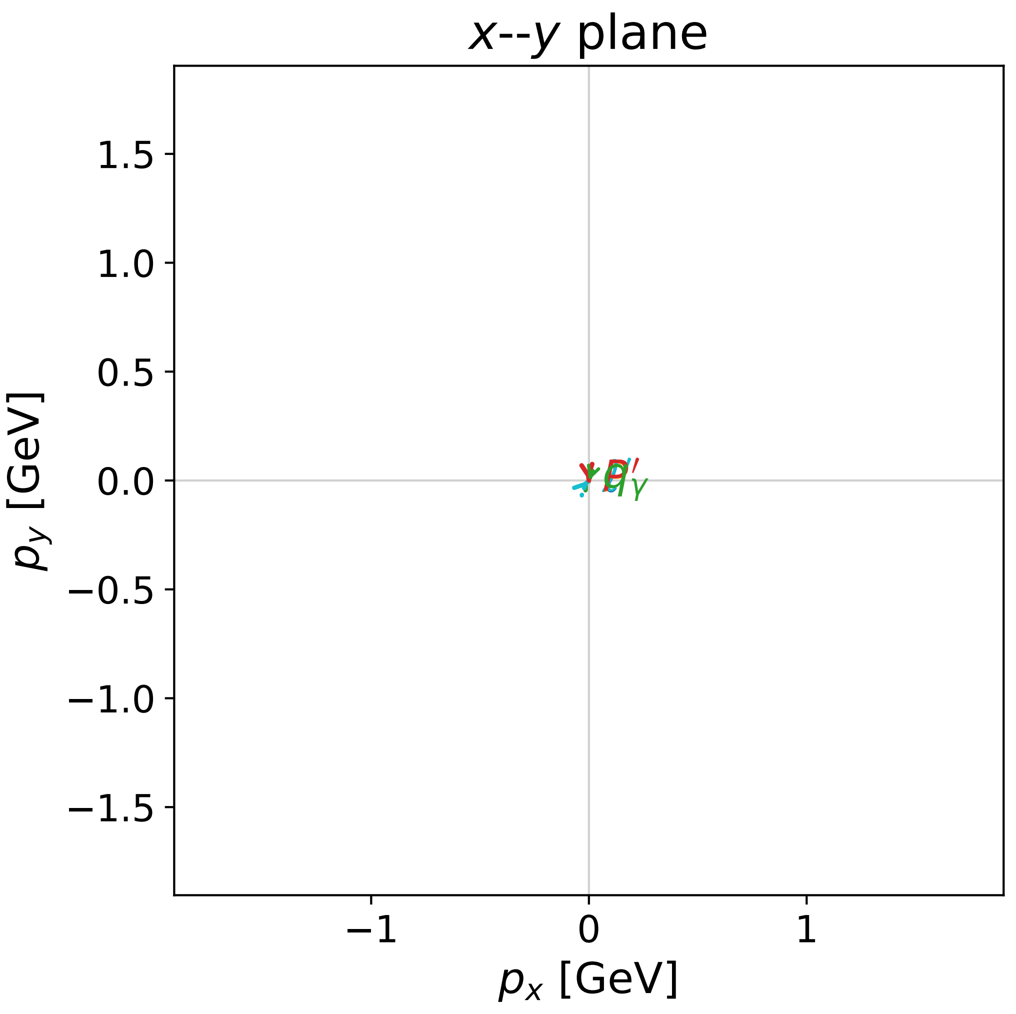
coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)

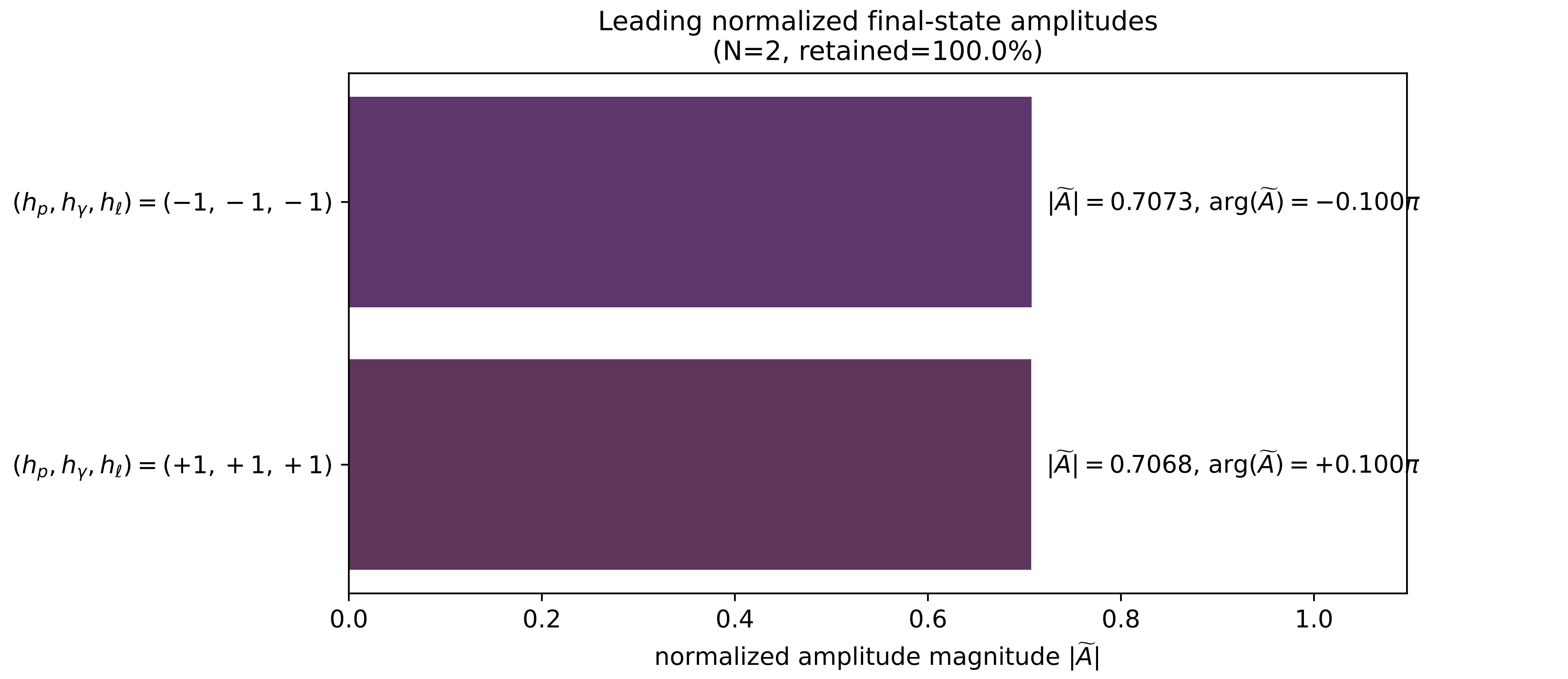


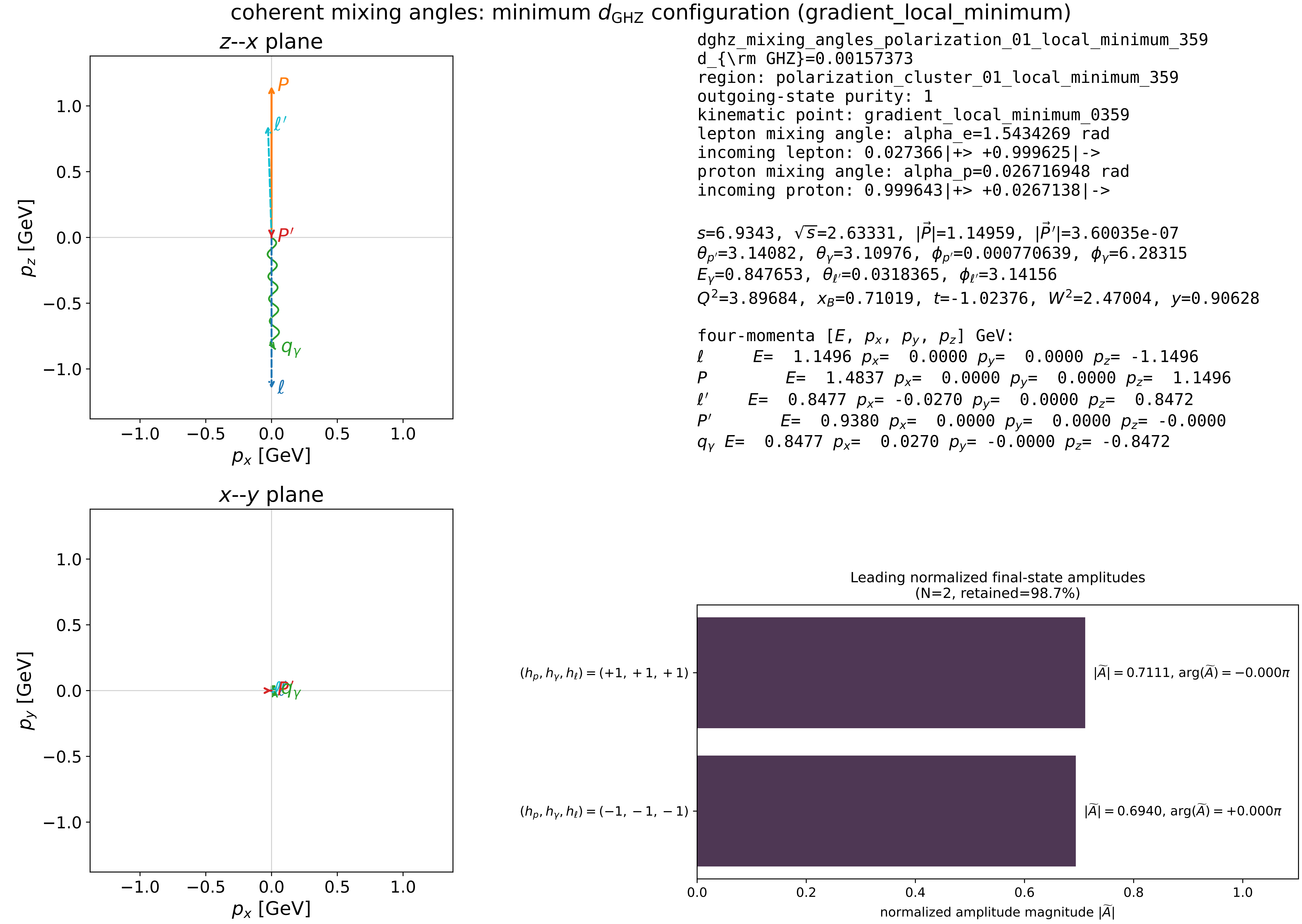

```
dghz_mixing_angles_polarization_01_local_minimum_356
d_{\rm GHZ}=0.0011903
region: polarization_cluster_01_local_minimum_356
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0356
lepton mixing angle: alpha_e=0.78547747 rad
incoming lepton: 0.707051|+> +0.707163|->
proton mixing angle: alpha_p=0.78569818 rad
incoming proton: 0.706895|+> +0.707319|->
```

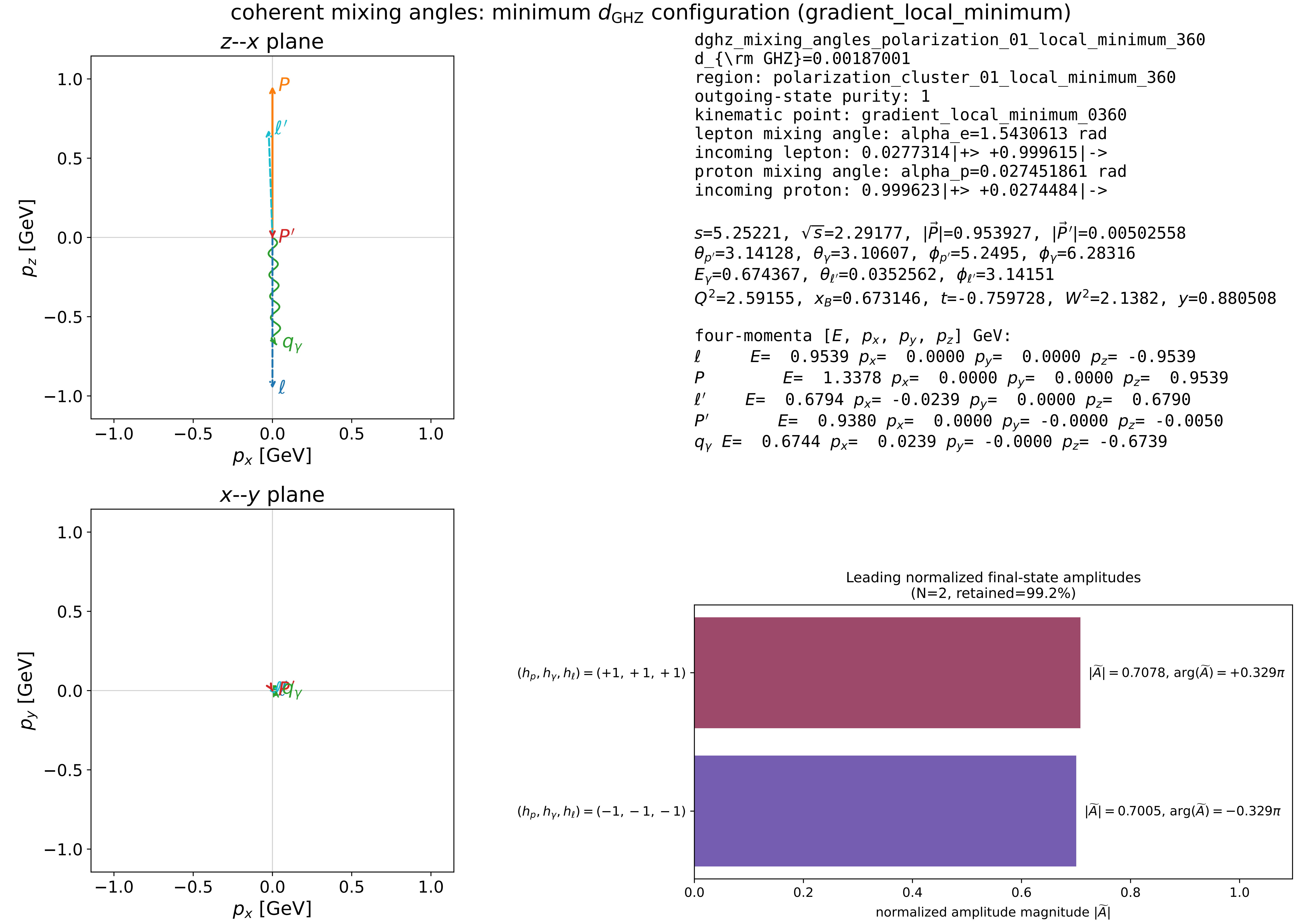
s=11.7704, $\sqrt{s}$=3.4308, $|\vec{P}|$=1.58717, $|\vec{P}'|$=1.58616
$\theta_{p'}$=3.14154, θ_γ=4.27948e-05, $\phi_{p'}$=4.8388, ϕ_γ=2.80317
E_γ=1.58703, $\theta_{\ell'}$=3.05347, $\phi_{\ell'}$=0.792465
Q^2=0.000448243, x_B=4.11835e-05, t=-10.07, W^2=11.7634, y=0.999402

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.5872 p_x= 0.0000 p_y= 0.0000 p_z= -1.5872
P E= 1.8436 p_x= 0.0000 p_y= 0.0000 p_z= 1.5872
ℓ' E= 0.0010 p_x= 0.0001 p_y= 0.0001 p_z= -0.0009
P' E= 1.8428 p_x= 0.0000 p_y= -0.0001 p_z= -1.5862
q_γ E= 1.5870 p_x= -0.0001 p_y= 0.0000 p_z= 1.5870

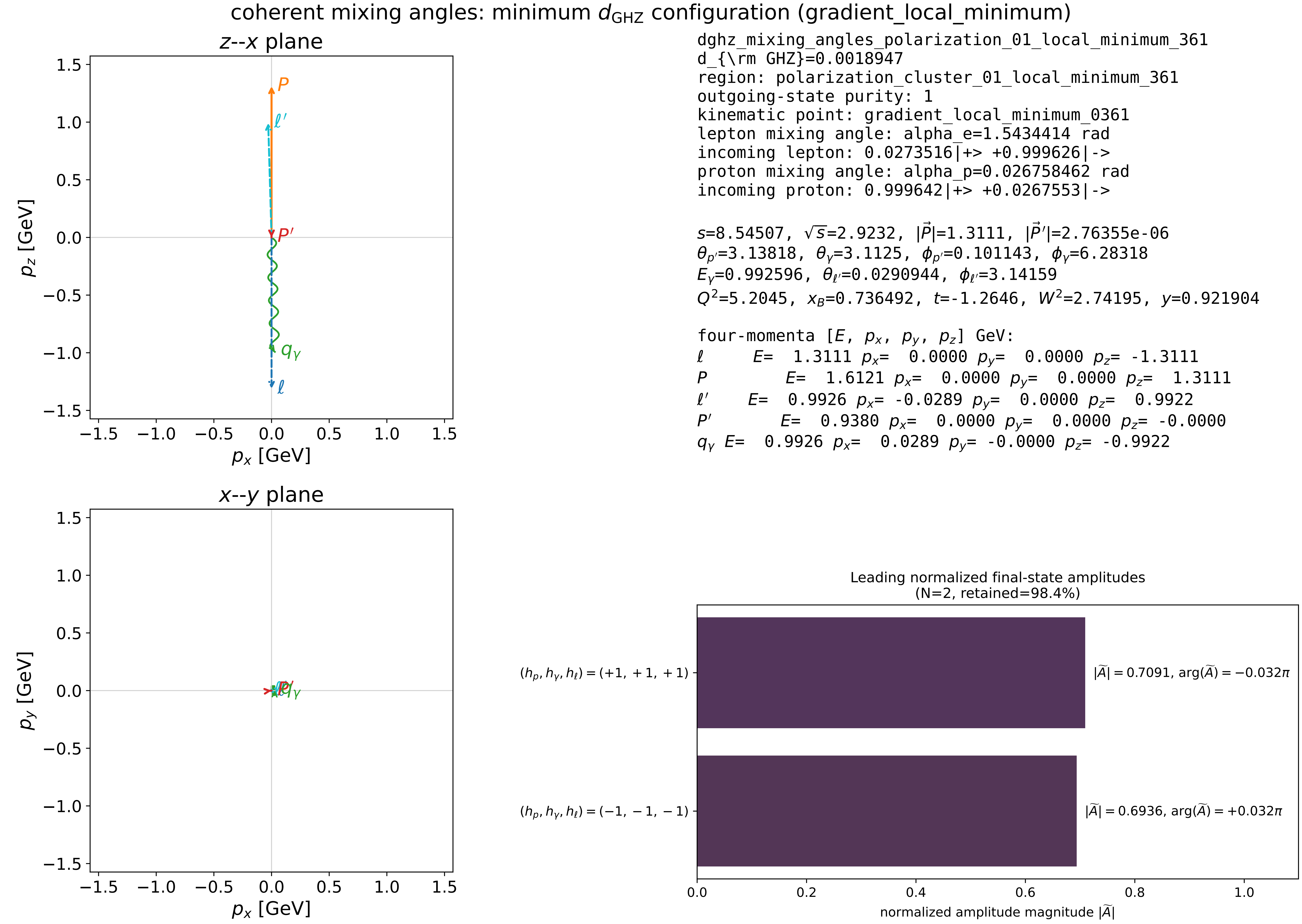




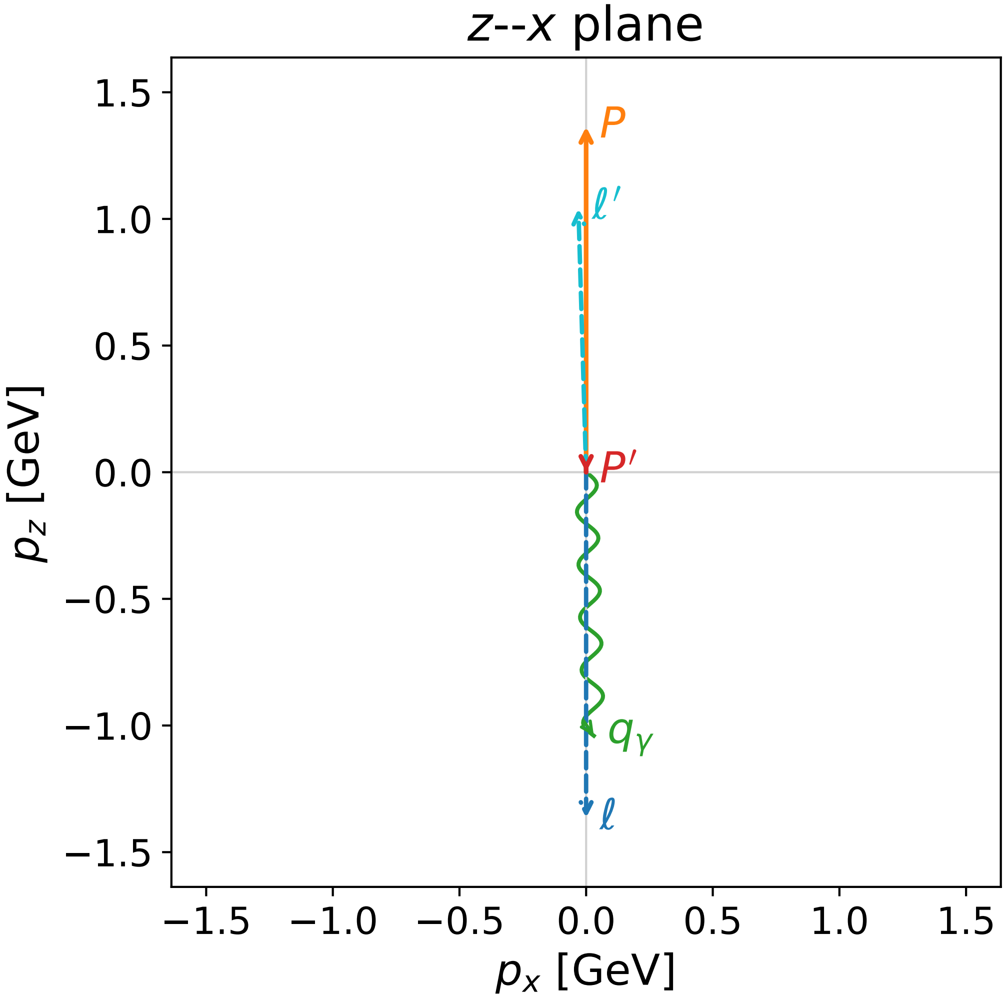

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P
ℓ'
P'
q_γ
ℓ
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_359
d_{\rm GHZ}=0.00157373
region: polarization_cluster_01_local_minimum_359
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0359
lepton mixing angle: alpha_e=1.5434269 rad
incoming lepton: 0.027366|+> +0.999625|->
proton mixing angle: alpha_p=0.026716948 rad
incoming proton: 0.999643|+> +0.0267138|->
s=6.9343, $\sqrt{s}$=2.63331, $|\vec{P}|$=1.14959, $|\vec{P}'|$=3.60035e-07
$\theta_{p'}$=3.14082, θ_γ=3.10976, $\phi_{p'}$=0.000770639, ϕ_γ=6.28315
E_γ=0.847653, $\theta_{\ell'}$=0.0318365, $\phi_{\ell'}$=3.14156
Q^2=3.89684, x_B=0.71019, t=-1.02376, W^2=2.47004, y=0.90628
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.1496 p_x= 0.0000 p_y= 0.0000 p_z= -1.1496
P E= 1.4837 p_x= 0.0000 p_y= 0.0000 p_z= 1.1496
ℓ' E= 0.8477 p_x= -0.0270 p_y= 0.0000 p_z= 0.8472
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.8477 p_x= 0.0270 p_y= -0.0000 p_z= -0.8472
Leading normalized final-state amplitudes
(N=2, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7111$, $\arg(\widetilde{A}) = -0.000\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.6940$, $\arg(\widetilde{A}) = +0.000\pi$
normalized amplitude magnitude $|\widetilde{A}|$



coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_360
d_{\rm GHZ}=0.00187001
region: polarization_cluster_01_local_minimum_360
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0360
lepton mixing angle: alpha_e=1.5430613 rad
incoming lepton: 0.0277314|+> +0.999615|->
proton mixing angle: alpha_p=0.027451861 rad
incoming proton: 0.999623|+> +0.0274484|->
s=5.25221, $\sqrt{s}$=2.29177, $|\vec{P}|$=0.953927, $|\vec{P}'|$=0.00502558
$\theta_{p'}$=3.14128, θ_γ=3.10607, $\phi_{p'}$=5.2495, ϕ_γ=6.28316
E_γ=0.674367, $\theta_{\ell'}$=0.0352562, $\phi_{\ell'}$=3.14151
Q^2=2.59155, x_B=0.673146, t=-0.759728, W^2=2.1382, y=0.880508
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.9539 p_x= 0.0000 p_y= 0.0000 p_z= -0.9539
P E= 1.3378 p_x= 0.0000 p_y= 0.0000 p_z= 0.9539
ℓ' E= 0.6794 p_x= -0.0239 p_y= 0.0000 p_z= 0.6790
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= -0.0050
q_γ E= 0.6744 p_x= 0.0239 p_y= -0.0000 p_z= -0.6739
Leading normalized final-state amplitudes
(N=2, retained=99.2%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7078$, $\arg(\widetilde{A}) = +0.329\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.7005$, $\arg(\widetilde{A}) = -0.329\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
x--y plane
p_y [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_361
d_{\rm GHZ}=0.0018947
region: polarization_cluster_01_local_minimum_361
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0361
lepton mixing angle: alpha_e=1.5434414 rad
incoming lepton: 0.0273516|+> +0.999626|->
proton mixing angle: alpha_p=0.026758462 rad
incoming proton: 0.999642|+> +0.0267553|->
s=8.54507, $\sqrt{s}$=2.9232, $|\vec{P}|$=1.3111, $|\vec{P}'|$=2.76355e-06
$\theta_{p'}$=3.13818, θ_γ=3.1125, $\phi_{p'}$=0.101143, ϕ_γ=6.28318
E_γ=0.992596, $\theta_{\ell'}$=0.0290944, $\phi_{\ell'}$=3.14159
Q^2=5.2045, x_B=0.736492, t=-1.2646, W^2=2.74195, y=0.921904
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 1.3111 p_x= 0.0000 p_y= 0.0000 p_z= -1.3111
P E= 1.6121 p_x= 0.0000 p_y= 0.0000 p_z= 1.3111
ℓ' E= 0.9926 p_x= -0.0289 p_y= 0.0000 p_z= 0.9922
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0000
q_γ E= 0.9926 p_x= 0.0289 p_y= -0.0000 p_z= -0.9922
Leading normalized final-state amplitudes
(N=2, retained=98.4%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
$|\widetilde{A}| = 0.7091$, $\arg(\widetilde{A}) = -0.032\pi$
$(h_p, h_\gamma, h_\ell) = (-1, -1, -1)$
$|\widetilde{A}| = 0.6936$, $\arg(\widetilde{A}) = +0.032\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)




```
dghz_mixing_angles_polarization_01_local_minimum_364
d_{\rm GHZ}=0.00240137
region: polarization_cluster_01_local_minimum_364
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0364
lepton mixing angle: alpha_e=1.5438592 rad
incoming lepton: 0.0269338|+> +0.999637|->
proton mixing angle: alpha_p=0.026354725 rad
incoming proton: 0.999653|+> +0.0263517|->
```

s=9.12098, $\sqrt{s}$=3.0201, $|\vec{P}|$=1.36438, $|\vec{P}'|$=4.15695e-06

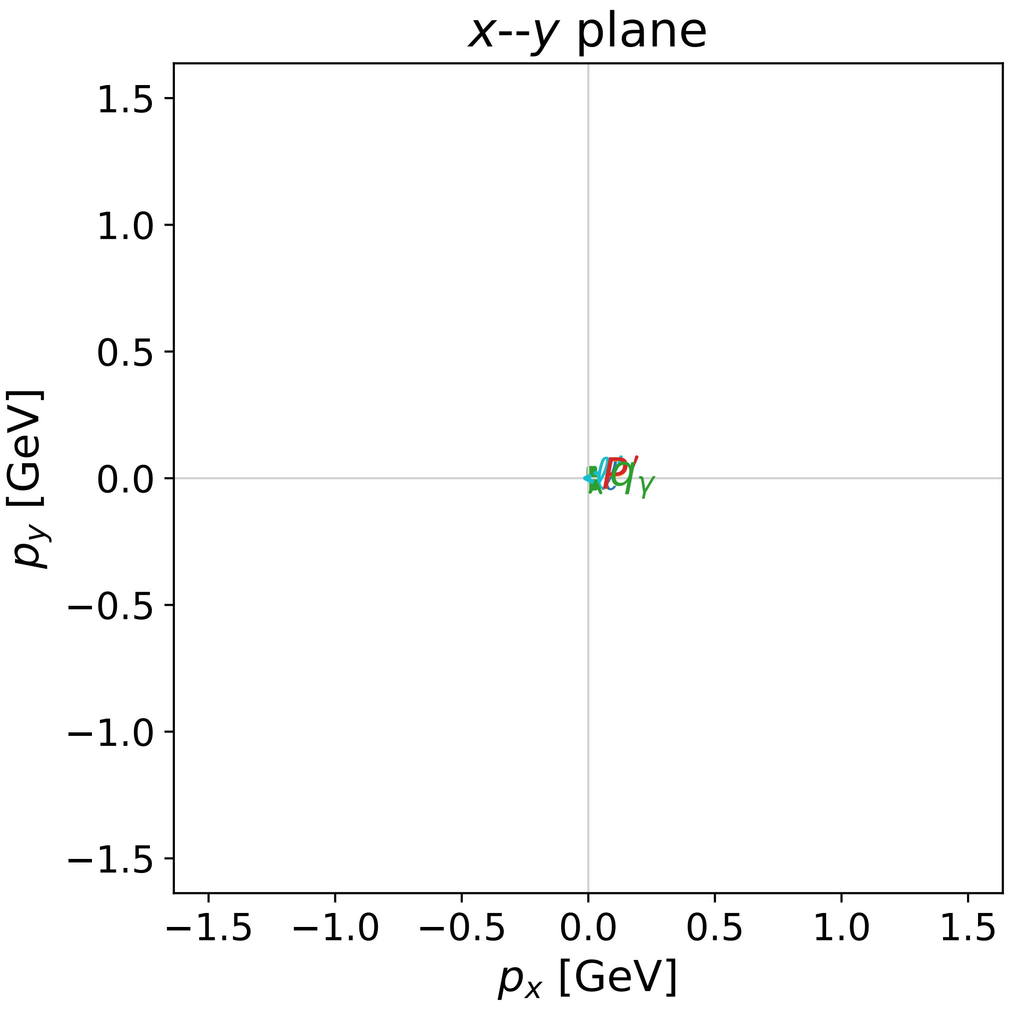
$\theta_{p'}$=3.14159, θ_γ=3.11231, $\phi_{p'}$=0.518887, ϕ_γ=6.28317

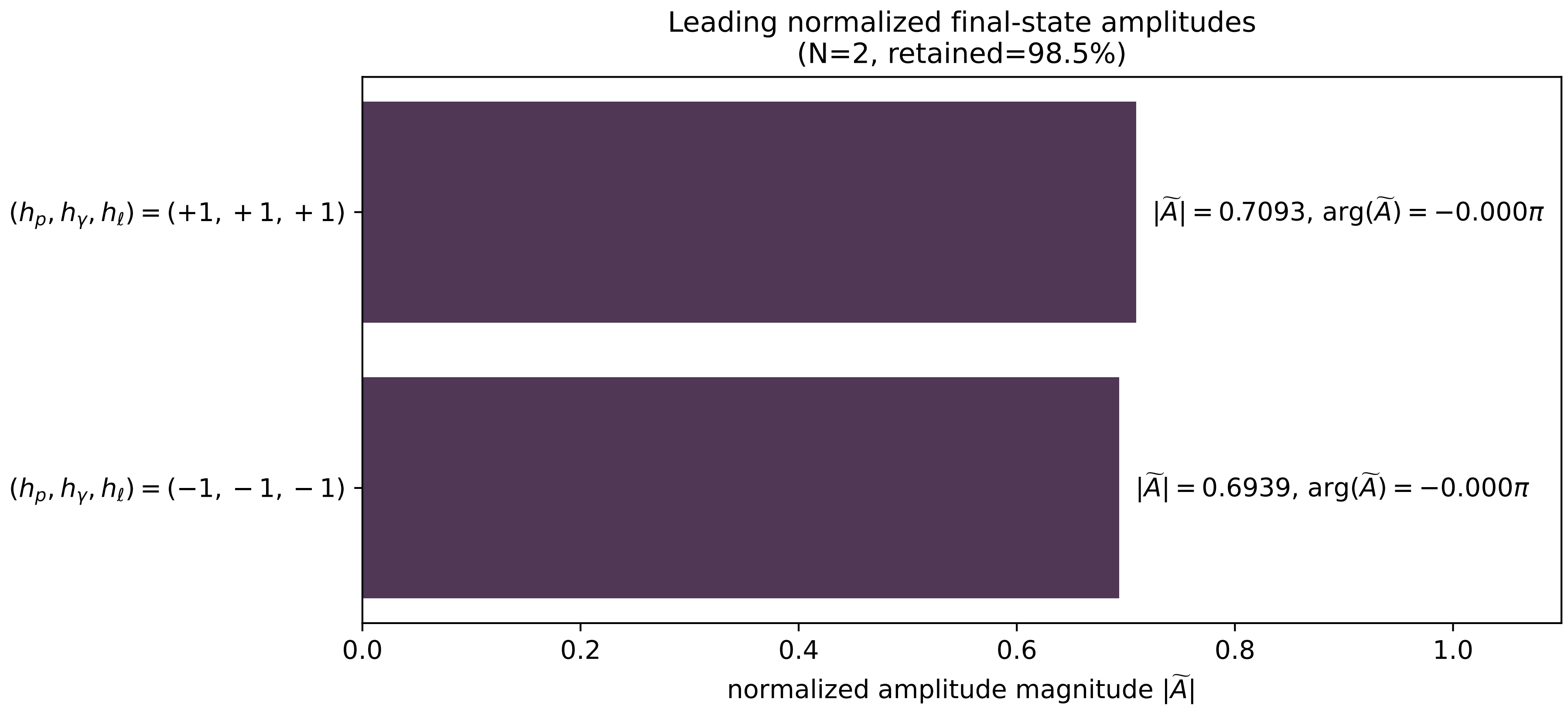
E_γ=1.04105, $\theta_{\ell'}$=0.0292827, $\phi_{\ell'}$=3.14158

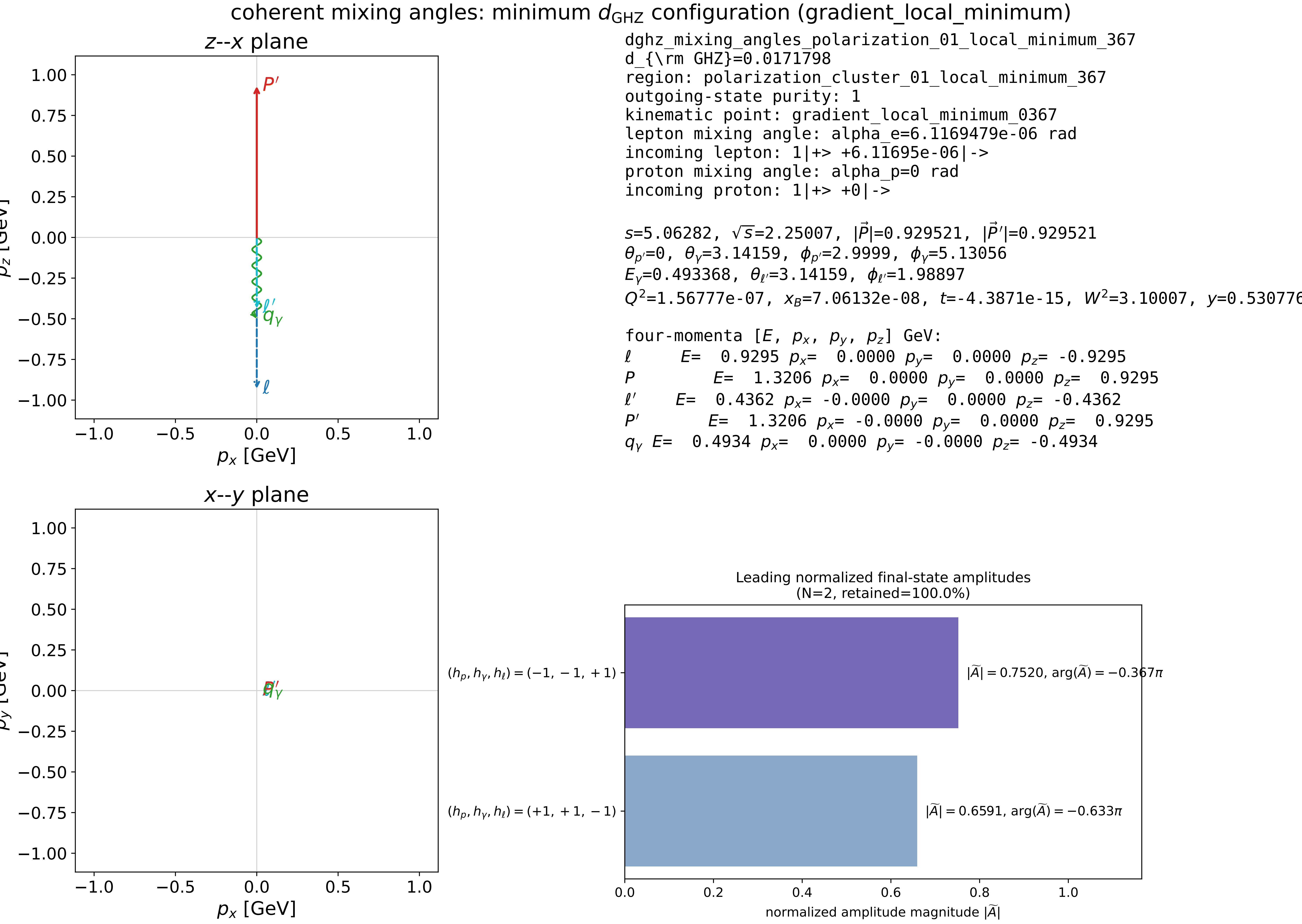
Q^2=5.68034, x_B=0.74415, t=-1.34644, W^2=2.83284, y=0.926248

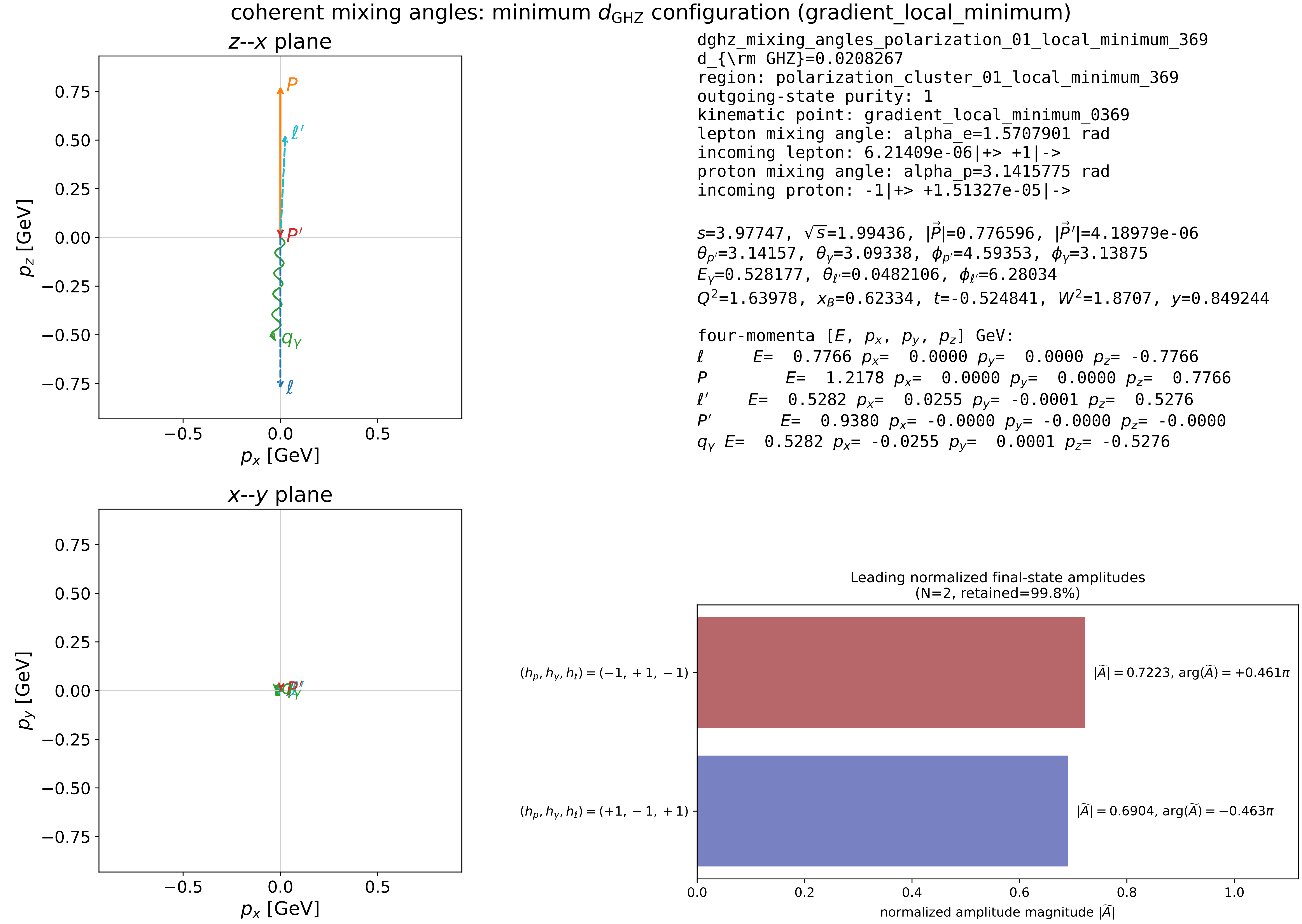
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 1.3644	p_x= 0.0000	p_y= 0.0000	p_z= -1.3644
P	E= 1.6557	p_x= 0.0000	p_y= 0.0000	p_z= 1.3644
ℓ'	E= 1.0410	p_x= -0.0305	p_y= 0.0000	p_z= 1.0406
P'	E= 0.9380	p_x= 0.0000	p_y= 0.0000	p_z= -0.0000
q_γ	E= 1.0410	p_x= 0.0305	p_y= -0.0000	p_z= -1.0406

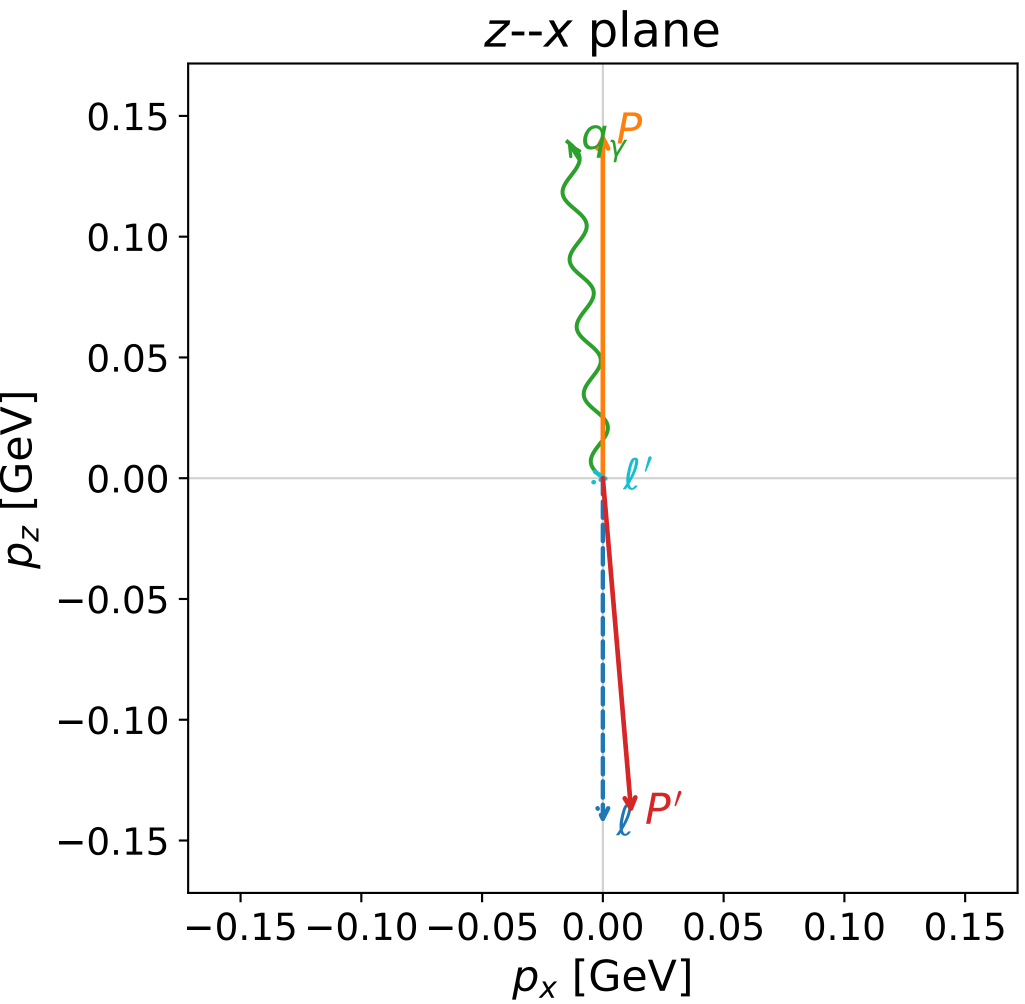





coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)
z--x plane
p_z [GeV]
p_x [GeV]
P'
ℓ'
q_γ
ℓ
dghz_mixing_angles_polarization_01_local_minimum_367
d_{\rm GHZ}=0.0171798
region: polarization_cluster_01_local_minimum_367
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0367
lepton mixing angle: alpha_e=6.1169479e-06 rad
incoming lepton: 1|+> +6.11695e-06|->
proton mixing angle: alpha_p=0 rad
incoming proton: 1|+> +0|->
s=5.06282, $\sqrt{s}$=2.25007, $|\vec{P}|$=0.929521, $|\vec{P}'|$=0.929521
$\theta_{p'}$=0, θ_γ=3.14159, $\phi_{p'}$=2.9999, ϕ_γ=5.13056
E_γ=0.493368, $\theta_{\ell'}$=3.14159, $\phi_{\ell'}$=1.98897
Q^2=1.56777e-07, x_B=7.06132e-08, t=-4.3871e-15, W^2=3.10007, y=0.530776
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.9295 p_x= 0.0000 p_y= 0.0000 p_z= -0.9295
P E= 1.3206 p_x= 0.0000 p_y= 0.0000 p_z= 0.9295
ℓ' E= 0.4362 p_x= -0.0000 p_y= 0.0000 p_z= -0.4362
P' E= 1.3206 p_x= -0.0000 p_y= 0.0000 p_z= 0.9295
q_γ E= 0.4934 p_x= 0.0000 p_y= -0.0000 p_z= -0.4934
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
$|\widetilde{A}| = 0.7520$, $\arg(\widetilde{A}) = -0.367\pi$
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
$|\widetilde{A}| = 0.6591$, $\arg(\widetilde{A}) = -0.633\pi$
normalized amplitude magnitude $|\widetilde{A}|$


coherent mixing angles: minimum $d_{\rm GHZ}$ configuration (gradient_local_minimum)
z--x plane
P
ℓ'
P'
q_γ
ℓ
p_z [GeV]
p_x [GeV]
dghz_mixing_angles_polarization_01_local_minimum_369
d_{\rm GHZ}=0.0208267
region: polarization_cluster_01_local_minimum_369
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0369
lepton mixing angle: alpha_e=1.5707901 rad
incoming lepton: 6.21409e-06|+> +1|->
proton mixing angle: alpha_p=3.1415775 rad
incoming proton: -1|+> +1.51327e-05|->
s=3.97747, $\sqrt{s}$=1.99436, $|\vec{P}|$=0.776596, $|\vec{P}'|$=4.18979e-06
$\theta_{p'}$=3.14157, θ_γ=3.09338, $\phi_{p'}$=4.59353, ϕ_γ=3.13875
E_γ=0.528177, $\theta_{\ell'}$=0.0482106, $\phi_{\ell'}$=6.28034
Q^2=1.63978, x_B=0.62334, t=-0.524841, W^2=1.8707, y=0.849244
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.7766 p_x= 0.0000 p_y= 0.0000 p_z= -0.7766
P E= 1.2178 p_x= 0.0000 p_y= 0.0000 p_z= 0.7766
ℓ' E= 0.5282 p_x= 0.0255 p_y= -0.0001 p_z= 0.5276
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.5282 p_x= -0.0255 p_y= 0.0001 p_z= -0.5276
x--y plane
p_y [GeV]
p_x [GeV]
Leading normalized final-state amplitudes
(N=2, retained=99.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
$|\widetilde{A}| = 0.7223$, $\arg(\widetilde{A}) = +0.461\pi$
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
$|\widetilde{A}| = 0.6904$, $\arg(\widetilde{A}) = -0.463\pi$
normalized amplitude magnitude $|\widetilde{A}|$

coherent mixing angles: minimum d_{GHZ} configuration (gradient_local_minimum)



dghz_mixing_angles_polarization_01_local_minimum_373
d_{\rm GHZ}=0.0300609
region: polarization_cluster_01_local_minimum_373
outgoing-state purity: 1
kinematic point: gradient_local_minimum_0373
lepton mixing angle: alpha_e=0.74214409 rad
incoming lepton: 0.737021|+> +0.67587|->
proton mixing angle: alpha_p=0.90015049 rad
incoming proton: 0.621492|+> +0.78342|->

s=1.19232, $\sqrt{s}$=1.09193, $|\vec{P}|$=0.143083, $|\vec{P}'|$=0.139544
$\theta_{p'}$=3.04448, θ_γ=0.109503, $\phi_{p'}$=0.509518, ϕ_γ=3.44722
E_γ=0.140121, $\theta_{\ell'}$=1.68565, $\phi_{\ell'}$=5.66363
Q^2=0.000885013, x_B=0.00289466, t=-0.0796895, W^2=1.1847, y=0.978442

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1431 p_x= 0.0000 p_y= 0.0000 p_z= -0.1431
P E= 0.9489 p_x= 0.0000 p_y= 0.0000 p_z= 0.1431
ℓ' E= 0.0035 p_x= 0.0028 p_y= -0.0020 p_z= -0.0004
P' E= 0.9483 p_x= 0.0118 p_y= 0.0066 p_z= -0.1389
q_γ E= 0.1401 p_x= -0.0146 p_y= -0.0046 p_z= 0.1393